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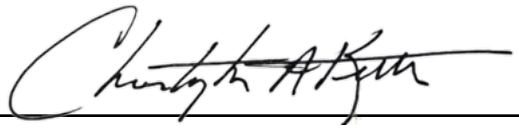
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MODULAR END EFFECTOR SYSTEM FOR A LIGHT INDUSTRIAL ROBOTIC WORKCELL

TO BE ACCEPTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF

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IN
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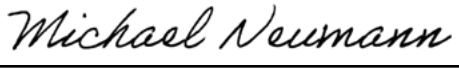
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MODULAR END EFFECTOR SYSTEM FOR A LIGHT INDUSTRIAL ROBOTIC WORKCELL

by

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CAPSTONE PROJECT REPORT

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of

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Master of Science in Mechanical Engineering

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Abstract

A modular interchangeable end effector system was developed for a wide range of light industrial robot arms, with a focus on cost-effectiveness, reconfigurability, and autonomous tool changing. Intended for use in a smart modular workcell, the MATRIX (Modular Adaptive Tooling & Robotic Interface eXtension) system enables a single robotic arm to support multiple manufacturing processes through a universal robot arm mount, integrated electrical and pneumatic passthroughs, and a passive tool depot for automated tool changing. The current end effector tool portfolio includes a force-controlled electrical parallel gripper, a pneumatic suction gripper, and an adhesive applicator under continued development. Initial testing demonstrated autonomous tool swapping in 9 of 10 trials, with only one fault detected during a tool drop-off maneuver. The system also had 100% success rates in picking up a coaster and a pair of electronics tongs over the same number of trials for both objects. The base section and tool depot's low cost of $133 and an average gripper tool cost of $233, coupled with the system's open-source approach, make the design easily reproducible for a variety of users. These results and cost data establish MATRIX as a practical and extensible foundation for modular end effector development in reconfigurable robotic workcells.

Acknowledgements

I would like to use this opportunity to thank the people and organizations involved in this project for both their technical guidance and financial support along the way.

Most importantly, I am grateful to Dr. Christopher Kitts, Dr. Manoj Sharma, and Dr. Michael Neumann for sharing their expertise in the field of mechatronics systems design and for helping guide the development of the project. Of these professors, the former two were key technical advisors throughout the design process, with Dr. Sharma regularly providing detailed advice on the end effector's electromechanical design and Dr. Kitts leading the push to integrate the design into a broader workcell system. Dr. Sharma was also instrumental in providing the vision for this workcell's software architecture and physical layout. Dr. Neumann, meanwhile, was the project's administrative advisor and procurement lead, helping ensure compliance with the university's capstone requirements and ordering any components needed for prototyping.

I would also like to thank my colleague Demilade Kunle-Kunbi (MS '25) for a successful year of collaboration on the project, most notably for his work on the software infrastructure needed to actualize the full potential of the MATRIX end effector system. Through our work together with Dr. Sharma, we were able to set up a room full of robotic arms connected in a unified software environment running on ROS and capable of performing a variety of light industrial tasks via MATRIX.

On an organizational level, I am grateful to Santa Clara University, the SCU School of Engineering (SoE), the SCU Robotic Systems Laboratory (RSL), and Lam Research for their continued financial and logistical backing in this capstone project. More specifically, all of the funding for the in-house portion of the project was provided by SCU and the SoE via the RSL, while Lam Research funded all key equipment required for the adhesive applicator system. The RSL–under the leadership of Dr. Kitts–was particularly crucial in providing a workspace for the development of the end effector system, along with the robot arms and infrastructure equipment used in the creation of the aforementioned workcell room.

Last, but certainly not least, I am also deeply thankful to my family and friends, who have supported me in many ways over the course of this project. My parents in particular have been the bedrock of my educational journey and are an inspiration to me every day. I am truly grateful for their love, support, and understanding throughout this engineering endeavor.

Thank you all for helping bring this vision to life–the success of the project could not have been possible without all of your help, and I am excited to see how far it will go in the future.

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Chapter 1: Introduction

1.1. Project Motivation

As the development of robotics and artificial intelligence (AI) has accelerated over the past decade, it has also become viable to introduce these advances to fields long rooted in conventional industrial paradigms. Robotics researchers are therefore exploring many avenues to extend human capabilities and boost productivity across industrial processes ranging from fabrication to inspection. Hailed as foundations of "Industry 4.0," AI, cyber-physical systems (CPS), and the internet of things (IoT) are quickly being adopted in mainstream industry through off-the-shelf and proprietary systems [1–3]. A common theme in this field is the simplification of complex workflows via interconnected systems that can be centrally managed with minimal user input. One particularly salient example is 3D printing, which in suitable use cases has delivered lead-time reductions of up to ~90% and significant cost savings (e.g., ~41% per-part in a NIST case study), enabling faster production and iteration for complex, low-volume parts [4–5].

Much like how 3D printing has streamlined prototyping and small-batch manufacturing, we believe that a large-scale robotic workcell could deliver similar gains across light-industrial processes by seamlessly automating material handling, assembly, and inspection in an end-to-end system. Robotic workcells are in fact used already in many industrial applications, ranging from automotive welding to electronics assembly, but such systems usually require expensive machinery and long setup times [6–8]. Compared to these industrial-grade workcells, our system is intended to serve as an adaptive, cost-effective solution for high-precision light industrial workflows such as product assembly and inspection. Through a design built for modularity, users of the RSL workcell can set up process chains for countless manufacturing tasks within an hour, and then let the system complete them with full autonomy.

Within this broader system, the Modular Adaptive Tooling & Robotic Interface eXtension (MATRIX) project aims to provide an integrated, modular portfolio of end effectors throughout the range of processes the workcell will be used for. In line with the workcell's goals for adaptability, the base section of the end effector is designed to be near-universal and thus be mounted onto the distal ends of popular robotic arms like the Stäubli TX60, Fanuc LR Mate, and Franka-Emika Panda with minimal modification. After the manual installation of this base–along with a compact tool depot–the design of the end effector tools enables the arm to autonomously swap between them as needed in a task workflow. Through its combination of built-in electrical and pneumatic passthroughs, MATRIX provides each robotic arm with numerous hot-swappable tooling options for executing its tasks with a high degree of precision and speed.

1.2. Stakeholder Needs

The MATRIX project is highly grounded in the needs of its primary backer and customer, the Santa Clara University Robotic Systems Laboratory (RSL). The RSL is led by Dr. Chris Kitts and Dr. Michael Neumann, and operates within the SCU School of Engineering as an interdisciplinary laboratory that brings students together to solve a variety of challenges at the

forefront of modern robotics. As mentioned in the preceding section, the RSL intends to use the MATRIX end effector system as a key part of its modular workcell room. Through continued discussions with both Dr. Kitts and Dr. Neumann, as well as with the lead workcell advisor Dr. Manoj Sharma, a set of design needs for the end effector system was determined. These needs are compiled in Table 1 below, and were used as the basis for the subsequent design process.

Table 1: Preliminary Design Needs Summary

Need	Description	Priority Ranking
Modularity	Design should consist of a universal base across all its functional disciplines that can be set up in a single-install operation and augmented as needed for a wide range of end effector tools	1
Low cost	Should cost significantly less than proprietary counterparts, with a reasonable preliminary target being $800	2
Hot/self-swappability	Robot should be able to regularly swap end effector tools during a process chain without any human intervention	3
Ease of manufacture	All components of the system should be either producible in the RSL's Garage workshop or be available off-the-shelf from US-market retailers	4
High-level control capabilities	End effector tools should contain sensors onboard and provide data for a high-level control system with ROS functionality to manage their task execution within the workcell	5
User-friendliness	The end effector system should adhere to the principles of user-friendly product design, i.e. be easy to install, perform as intended, and look good; design should be open-source	6
Maximize existing infrastructure	Design should interface with the robot arms it is used on in a way that maximizes the built-in power and pneumatic routing within them; the tool depot and any peripheral electronics should use any relevant base mounting points	7

1.3. Background Research

In light of the significant body of work created in applied robotics over the past decade, it is imperative that a survey of this material should precede any technical design development on a project in the field. For the purposes of this specific project, a broader review of the state of Industry 4.0 will first be conducted in order to better understand how new robotics technology has been integrated and where the main needs of manufacturing companies currently lie (Section 1.3.1.). Within this space, more detailed research was also done on workcell systems and robotic arm end effector designs, as summarized in Sections 1.3.2. and 1.3.3., respectively. Further background information for the project was also gathered from the manuals for the arms to be used in the workcell and via conversations with Drs. Sharma and Kitts, and will be presented in Section 1.3.4.

1.3.1. Industry 4.0 Research

First formally defined in the early 2010s, Industry 4.0 encompasses the integration of advanced digital technologies into industrial processes to improve their efficiency and also maximize productivity across "the entire product lifecycle" [9–10]. With the proliferation of high-speed networking infrastructure and artificial intelligence over the past decade, Industry 4.0 has quickly become a part of the modern supply chain in countless applications. In 2024 alone, over 540,000 industrial robot installations were completed globally, continuing an upward trend that has seen the operational stock of such robots rise to over 4.6 million units [11]. Beyond the scale of its applications, Industry 4.0 has also yielded similarly paradigm-shifting results in the field. Per a report by consulting firm McKinsey & Company, Industry 4.0 automation has increased productivity by 15-30% in the surveyed industries, primarily through a reduction in machine downtimes ranging from 30% to 50% [12–13]. Furthermore, the wealth of data made available at the systems level by Industry 4.0 has enabled major improvements in product defect detection and mitigation, as shown by the up-to-45% waste reduction achieved in industrial plants around the world [14].



Figure 1. Industry 4.0 Assembly Line for Solar Panel Fabrication [15]

1.3.2. Workcell Research

One particularly common approach within Industry 4.0 automation setups is the implementation of robotic workcell systems that function as self-contained units for specific process chains. Such workcells are used across countless industries to improve manufacturing productivity, ranging from the automotive sector to consumer electronics [16]. Quantitatively, the electronics industry accounted for 24% of all robotic system installations, followed closely by the automotive industry with 23% and more distantly by the metal and machinery sphere with 16% [11]. Workcells are also particularly useful in the post-production and part inspection stages of the manufacturing chain, helping verify the integrity of both parts and assemblies as they exit the production line [17]. At a systems engineering level, workcells are most commonly designed around a specific type of process chain, with the manipulators being chosen according to the needs of the processes involved. From here, the selected manipulators are organized into functional groupings and high-level control flows are developed to achieve macro-scale functionality within the workcell [18]. As such workcells become increasingly sophisticated and are introduced into more high-precision environments, the introduction of end effector-level sensing has emerged as an important avenue of study [19]. Together with the implementation of new sensing modalities such as tactile and vision feedback, this shift has enabled robotic manipulators to interact more reliably with complex environments and generalize their functionalities within workcell systems [20].

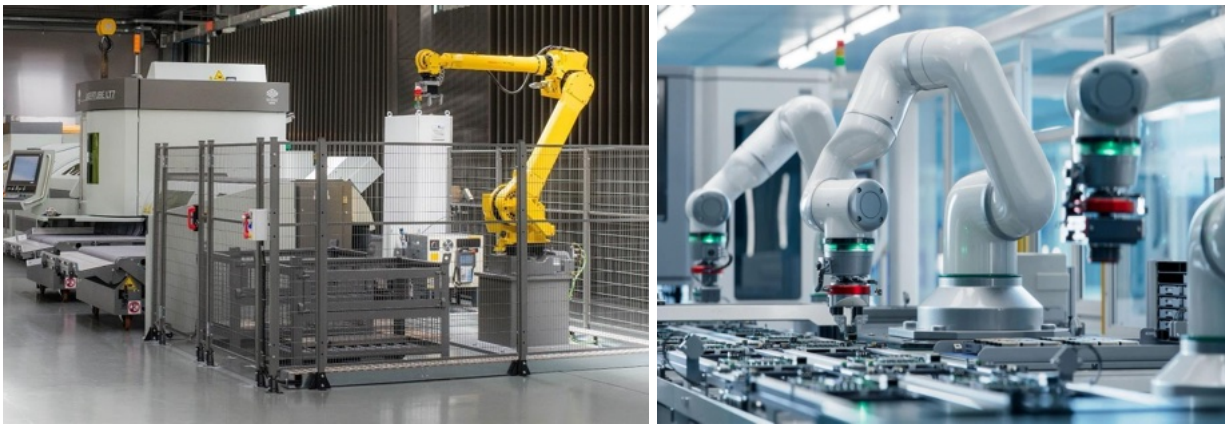


Figure 2. Robotic Workcells Used in a Manufacturing Application [21–22]

1.3.3. End Effector Research

One of the primary areas where hot-swappable, modular end effectors have been particularly effective is in humanoid robotics, where robots often need to switch out their tools in-situ for maximum operational efficiency. In a collaboration between researchers in France and Japan under the CNRS research program, Roux, Izard, et. al. developed such a modular end effector system for the hand, head, and foot sections of their humanoid robots [23]. The design this group created is described as light-weight, requires no embedded actuators, has large

precision tolerance in mounting and dismounting, and has no need for human operator assistance, all of which align with the MATRIX design needs in Table 1.

The research team achieved these functionality targets through a multi-stage end effector module (Figure 3) built around a rotational locking mechanism that joins a static portion mounted at the distal ends of the humanoid's appendages with any relevant tool that the robot might need. Beyond mechanically locking the tool onto the robot, this design also includes electrical passthrough connections via a pair of 24-pin pogo connectors on the opposing sides of the module. The end effectors developed to utilize these electrical lines included, most notably, a gripper for the hands of the humanoid and a USB camera for its head. The external locking bell forms the third portion of the end effector mechanism and helps improve resistance to disengagement under high torque, albeit while adding significant complexity to the design and often requiring human intervention to lock or unlock.

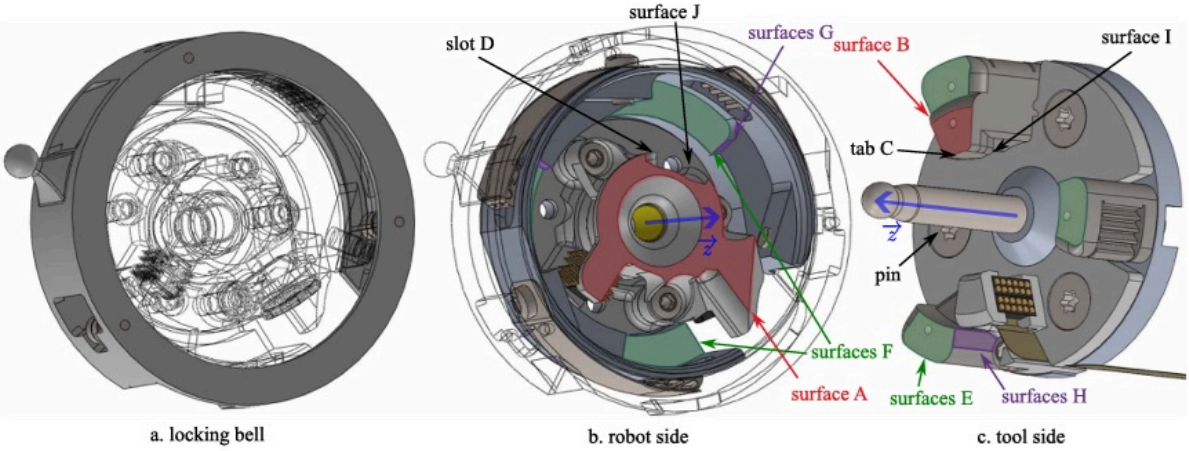


Figure 3. CNRS End Effector Changer (EEC) System CAD Model [23]

A similar rotationally-locking end effector system developed for stationary robotic arms was also identified alongside the CNRS design as a relevant case study during the background research process. Designed by researchers at the Korean Institute of Science and Technology (KIST) and Samsung Research Center, the two-stage "ModMan" system serves as a modular, genderless connector for the various multi-degree-of freedom (MDoF) robotic arms used by the research center for light industrial tasks [24]. The researchers placed great emphasis on the integration of electrical passthroughs into the design, including 115 such connections via a pair of custom parallel, coaxially aligned PCBs. Each connection uses a pogo-type spring pin to both ensure strong electrical contact and provide the needed level of compliance when mating the two sides of the end effector.

Similarly to the CNRS effector system, ModMan also uses a rotating interlock for joining its robot-end and tool-end stages together, as highlighted in Figure 4e. Here as well, this architecture proved to be adept at interfacing the robot arm with a variety of tools, even being used to serially join multiple robot arm links onto the base link. In order to help accommodate this wide array of end effectors and links, the ModMan team also developed its own custom

kinematic modeling algorithm and used the aforementioned electrical connections to transmit all required data across the joint. The innermost pins on the PCBs are particularly important in this regard, as they provide the robot with the offset angle of the otherwise axially symmetric connector modules. Due to these complexities, ModMan's mechanical capabilities were somewhat limited, with the system requiring human intervention to be locked and unlocked. An overview of the final design of ModMan is shown in Figure 4 for reference.

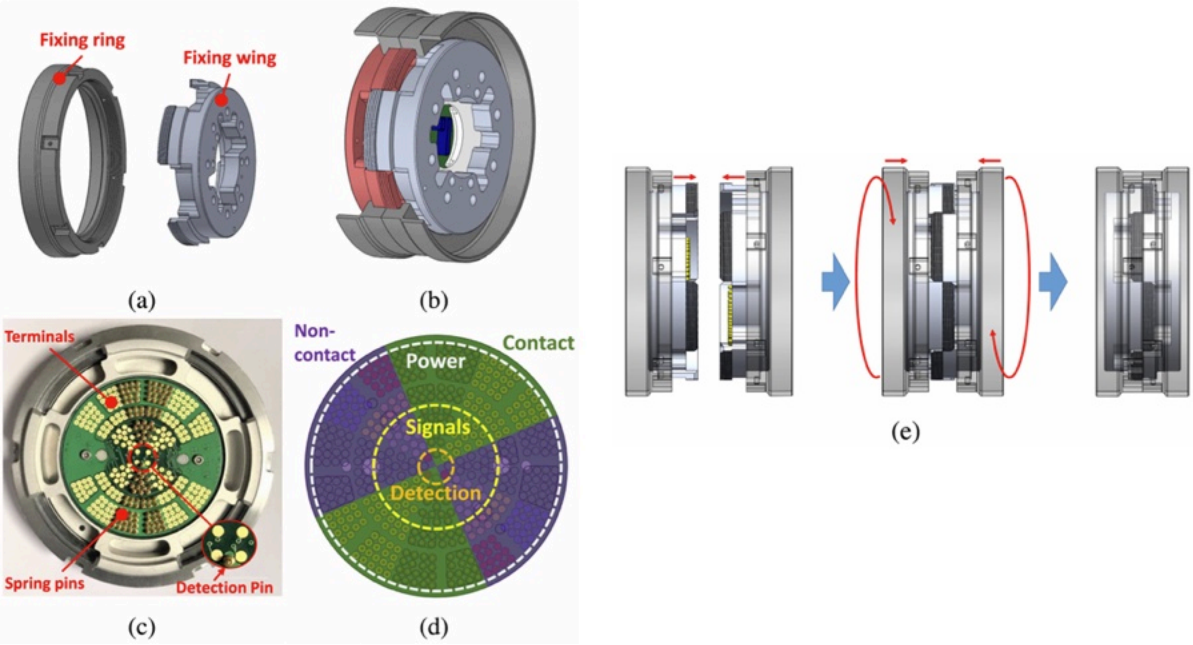


Figure 4. ModMan Modular End Effector System CAD Model [24]

Beyond the rotational locking architecture seen in the CNRS and ModMan systems above, other mechanisms for connecting the different end effector stages were also explored. A prominent example in this regard is the work of Li, Cappelleri, et. al. on their modular end effector system (MEES) in a collaboration between Purdue University and Harvard University [25]. This combined research group sought to develop the MEES for autonomous robot maintenance and repair, especially for use with mobile robots. With the specific goal of streamlining robot maintenance on missions to outer space, it was critical to both make the system robot agnostic and to ensure it required no complex peripherals or proprietary connectors.

Based on these target applications, the researchers included both electrical and pneumatic passthroughs in their design to maximize the system's ability to be recombined across different toolheads. These passthroughs are routed from the distal end of the robot through a standardized robot-side module (RSMS), which passes the two connections onward to the end effector-side module (EEMS) once the latter is attached and locked in place via a solenoid latch. The RSMS contains an internal microcomputer and all of the female connectors required for the 12 electrical lines, along with the inlet and outlet ports for the system's pneumatic line. The EEMS, meanwhile, is designed to serve as a basis for three tools that make use of these passthroughs: a

gripper, a suction cup, and a camera. A complementary tool depot is also part of the MEES design, enabling the robot to self-swap its end effectors via a vision-based alignment approach. While the solenoid locking mechanism does provide a robust connection in this regard, it also requires active power in the EEMS to be actuated, thereby increasing the complexity of the docking process. Figure 5 shows the full MEES assembly and its two constituent subsystems.

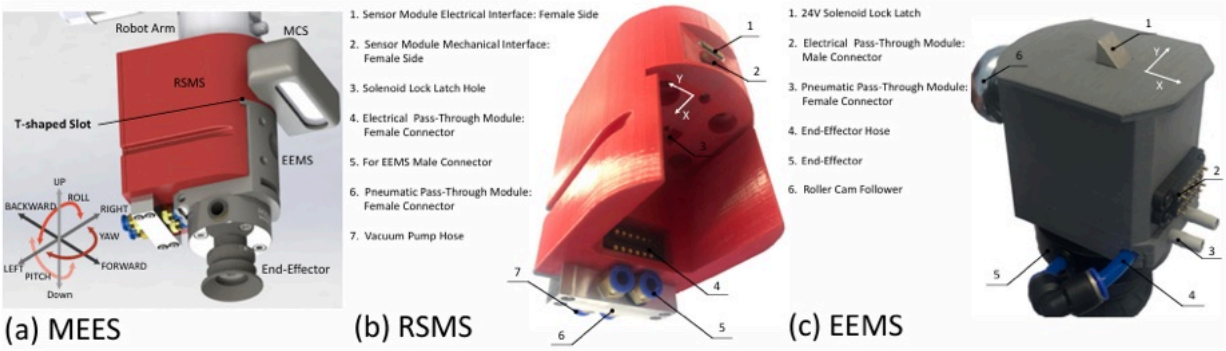


Figure 5. Purdue-Harvard MEES System and its Two Main Subassemblies [25]

The final main design studied as a reference for MATRIX was the ACROBOTER tool changer developed and tested by Gyimothy and Toth at the Budapest University of Technology and Economics. This tool changer consists of a similar rotational architecture to the CNRS and ModMan tool changers covered in the preceding paragraphs, using an axial rotation of the tool-side assembly to connect to the robot-side assembly [26]. In contrast to these tools, however, ACROBOTER uses a set of three spring-loaded pins that are manually pressed in by the operator as they twist the end effector portion, actuating a corresponding set of pins that lock into a slotted internal disk. The ACROBOTER tool changer also includes 12 male-to-female pogo pin connectors spread out along a circular path into three segments of four pins each, making it ideal for accommodating electrical end effector tools that can be used in light industrial applications.

Complementing the mechanical and electrical design aspects of the ACROBOTER project, Gyimothy and Toth also conducted extensive repeatability and structural testing, helping provide a strong foundation to validate the operating theory of rotationally-locking tool changers. Most notably, the researchers identified the primary failure mode of the tool changer in the region around the locking pins and ran a corresponding strength test with the system attached to a robot arm. The specific failure mode tested in this experiment was the shearing of the spring pin retainers' walls when the assembly is subjected to an off-center axial load similar to what it would encounter when picking up an irregular object. This test was used to construct a bending moment diagram of the tool changer assembly under the given load, validating its ability to hold together under loads of up to 184N. The other key testing stage involved a 50-cycle locking and unlocking repeatability test, which used two micrometers to measure the planar displacement of the tool-side section relative to the robot-side section. The data for this experiment showed that the displacement in each Cartesian axis was below the 0.1mm displacement limit for electrical continuity for each of the cycles. Figure 6 shows the three stages of the ACROBOTER assembly.



Figure 6. ACROBOTER Tool Changer System (Left) and Interlock Plate (Right) [26]

In addition to the research sources surveyed above, developments in the industrial sphere can also provide valuable insights into the state of the field in end effector design. While most firms in the field make much higher-end robotic tooling than the focus subspace for MATRIX, a review by the leading end effector company Robotiq provides a cost-agnostic strategic summary of leading practices [27]. Within this review, end effector standardization via universal interfaces and repeatable setup and maintenance processes are discussed extensively as the foundations for streamlining robotic tooling systems. On the former point, the author of the article, Michelle Vallée, highlights the importance of standardizing both the mechanical and electrical connections of end effector systems in order to ensure that they can easily connect between a variety of tools and robots. This "plug and play" class of end effectors then serves as the foundation of the second point, enabling the development of robot-agnostic installation processes and simplified maintenance routines that cumulatively lead to 15-30% cost savings per automation project where they are applied.

1.3.4. Research on Existing Equipment

The primary robotic arm used for the RSL workcell system was the Stäubli TX60, which is classified as a light industrial, high-precision arm. Upon an in-depth study of the user manual provided by the arm's manufacturer, multiple key specifications and reference points were compiled. This information would later be instrumental in the design process for the end effector modules and in maximizing the existing infrastructure (per Table 1) during the integration stage of the project, and is thus summarized in the following two paragraphs for reference. More detailed dimensional information will be presented in later sections as is relevant to specific aspects of the system's design.

The Stäubli TX60 is a 6 degree-of-freedom (6 DoF) serial chain manipulator robotic arm introduced in the 2000s and widely used in industrial automation applications. The TX60 has a maximum load capacity of 3.5 kg at its nominal speed, 4.5 kg at a reduced speed, and 9 kg at its low speed state [28]. The joint velocities corresponding to these states are compiled in Table 2

below, along with the angular position ranges and torque limits of each rotational axis. In the latter category, the most important value to note is that of Axis 6, which defines the upper torque limit of the end effector axis as 2 Nm. This table also highlights the high precision of the arm, with all angular resolutions on the order of 10^{-3}°.

Table 2: Stäubli TX60 Joint Axis Characteristics [28]

Joint Axis	1	2	3	4	5	6
Range (°)	360	255	285	540	255	540
Range distribution (°0)	± 180	± 127.5	± 142.5	± 270	+132.5 −122.5	± 270
Low speed (°/s)	45	45	45	45	45	45
Nominal speed (°/s)	287	287	431	410	320	700
Maximum speed (°/s)	435	410	540	995	1065	1445
Angular resolution (°×10^{-3})	0.057	0.057	0.057	0.114	0.122	0.172
Static torque limit (Nm)	126	71	38	7	8	2

Beyond the kinematic and dynamic properties outlined in Table 2, the Stäubli TX60 also contains a set of built-in pneumatic and electrical passthroughs that are worth mentioning. Regarding the former, the primary pneumatic line of the TX60 is rated for 0.6 MPa (6 bar) of pressurized air, and is accessible via an ⅛” NPT threaded entry port at the base of the robot [28]. The distal outlet of the pneumatic line, meanwhile, is located on the fourth link of the robot through an M5 threaded exit port. The electrical passthrough can similarly be accessed from the base and fourth links of the robot, albeit through a proprietary 19-pin socket pair rather than standard connectors. These electrical lines consist of 2 shielded twisted pairs and 6 unshielded twisted pairs, with the latter requiring a 24V DC input to operate reliably and the former being useful at lower voltages due to its shielding [29]. Together, the two passthrough lines described above provide the end effector with pressurized air and electrical power from the robot’s base without any external cabling or tubing that would cause obstructions in the workspace.

1.4. Objective Statement & Contributions

The goal of the MATRIX project is to develop a low-cost, modular end effector system for use within a light industrial robotic workcell, enabling the robotic arms involved to autonomously swap end effector tools and thus produce a variety of high-quality products.

While a wide variety of robotic end effectors currently exist in both industrial and academic settings, these are generally either proprietary or prohibitively difficult to manufacture in most contexts. Furthermore, the vast majority of these tools require manual intervention to be

swapped, adding friction to the production stream and increasing manufacturing time. Based on these gaps in the current end effector field, MATRIX provides the following contributions to the research space:

1) *A low-cost, open source portfolio of end effector tools:*
   MATRIX encompasses a set of three low-cost, ROS-integrated electropneumatic tools–including two grippers and an adhesive applicator–built around a modular architecture with a common mechanical base and distributed computing capabilities.
2) *Universal mount compatible with popular robot arms:*
   The foundation of MATRIX is a universal mounting section that can be attached to most modern popular robot arms via an integrated single-time mechanical and electrical installation process.
3) *Electrical and pneumatic passthrough integration:*
   The universal passthrough architecture of MATRIX enables five voltage lines of up to 24V each and a 0.6 MPa pneumatic line to be routed from a robot arm to any toolhead in the system.
4) *Fully autonomous tool swapping capability:*
   Tools within the MATRIX system can be hot-swapped at local tool depots by robot arms during a process chain, and are able to self-identify via a serial marker sent through the system's ROS node network.
5) *High-level control from a central computer:*
   A central computer is used to command both tool-level and robot-level actions based on real-time telemetry obtained from onboard sensing systems on each MATRIX toolhead.

Chapter 2: System Definition and Overview

2.1. High-Level System Requirements

Combining the needs of the project's primary stakeholder with the information gathered on the state of the field in the preceding chapter, a high-level set of system requirements were developed to provide more concrete design drivers for MATRIX. These requirements were compiled into Table 3 and grouped according to their corresponding design needs. Each target requirement was also chosen with the intention of being verifiable via either quantitative data, experimental testing, or by inspection, as listed in the aforementioned table.

Table 3. System Requirements Derived from Key Needs

Need	Requirement	Description	Verification
Modularity	Mechanical modularity	End effector system shall have a universal robot-side section and a complementary interlock for toolheads	Demonstrate by mounting universal base on 3 robot arms
	Software modularity	Each tool shall be its own ROS node and connect to the workcell's local wireless network	Check connection of each tool to ROS node network via a simple test command
	Electrical modularity	Computing elements shall be packageable on either the tool section or at the base of the robot	Demonstrate via different computing configurations
Low cost	Cost less than $800	Core end effector portfolio (universal parts and two grippers) shall cost less than $800	Verify via the final assembly's bill of materials (BoM)
Hot/self-swappability	Autonomous tool swapping	A 6 DoF robot arm shall be able to autonomously swap its end effector with over a 90% success rate	Test toolhead pick-up and drop-off via repeated trials and inspect to make sure tool remains intact at least 9 times out of 10

Table 3 (Continued). System Requirements Derived from Key Needs

Need	Requirement	Description	Verification
Ease of manufacture	Use standard components	End effector portfolio shall use only either off-the-shelf components or 3D-printed parts	Source components from reputable sellers during design and verify via BoM
	Easy to assemble	Each section shall be mechanically assembled in under 5 minutes	Assemble each section and use timer to verify
High-level control capabilities	Sensor on each tool	Each tool shall have at least one sensing ability	Demonstrate sensor data streams on laptop
	Sensing informs high-level control	Sensing onboard the toolheads shall inform real-time control at both the tool and robot levels	Verify data being received via ROS nodes and the receipt of commands at toolheads
User-friendliness	Easy to install	End effector system shall be installable in under 5 minutes	Install the system on a robot 5 times and use timer to verify
	Open source design	All design documentation shall be made public	Demonstrate via publication or webpage
Maximize existing infrastructure	Use electrical passthroughs	Universal mount shall connect to robot arm electrical ports	Set up end-to-end electrical passthrough connection and check continuity with a multimeter
	Use pneumatic passthroughs	Universal mount shall connect to robot arm pneumatic ports	Set up end-to-end pneumatic passthrough connection and check continuity by holding paper to tool-end outlet
	Mountable near robot	Tool depot shall be mountable on both robot carts and tables	Install tool depot on one instance of each mounting surface

2.2. Concept of Operations

In order to further develop the design for MATRIX from the above requirements, concepts of operation must also be formulated for both the end effector and broader workcell system. At a high level, the workcell is designed to have each robot arm and conveyor within it be its own ROS node and have all nodes connected over a local network to a central command computer. A combination of overhead cameras and an OptiTrack point tracker array enables the real-time monitoring of the system's operation and crucially provides a platform for further automation through computer vision and object recognition.

Within the workcell, MATRIX implements a similarly modular approach to the usage and swapping of end effector tools, as shown in Figure 7. The diagram below illustrates this operational process broken down into its four main stages: (1) a manual installation of the end effector's base module, (2) the navigation of the arm's distal end to the tool depot, (3) the tool's locking and unlocking process, and (4) the subsequent usage of the tool in the robot's workflow. The first of these steps is only required when the end effector system is set up, after which the robot can cycle through Steps 2-4 as needed during its regular usage for any given process.

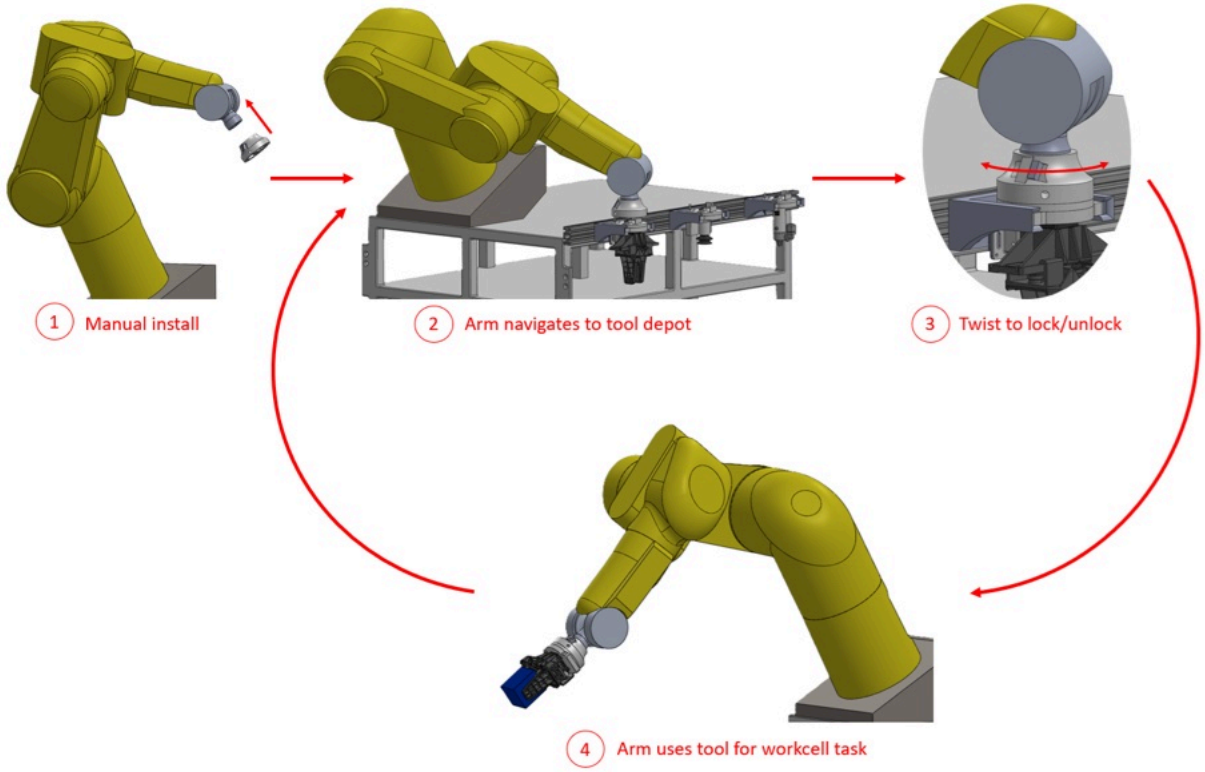


Figure 7. MATRIX Concept of Operations Diagram

Across the stages shown above, each end effector tool functions as a self-contained ROS node, allowing it to be remotely controlled even if it does not have an electrical connection to an arm running ROS. This architecture is advantageous both for simplifying the electrical connections that need to pass through the arm and also for mounting any effector tool on robots

that exist outside of the RSL workcell system. On the former point, each tool can communicate with the command hub computer wirelessly, meaning that only the power and ground wires need to run through the robotic arms. The ability for the tools to be used outside of the workcell, meanwhile, is particularly useful for the RSL given the wide range of terrestrial robots it operates with multiple overlaps in tasks like pick-and-place and sample collection.

2.3. Preliminary Design

Based on the key system requirements and concept of operations outlined above, a preliminary design for the MATRIX end effector system was created. The medium for communicating this design is an overview sketch of the end effector's main assembly, which is shown in Figure 8 below. The focus for this initial design is placed on the universal mount and interlock stages of the design, since these will be consistently used in the workcell regardless of the end effector tool attached to the arm.

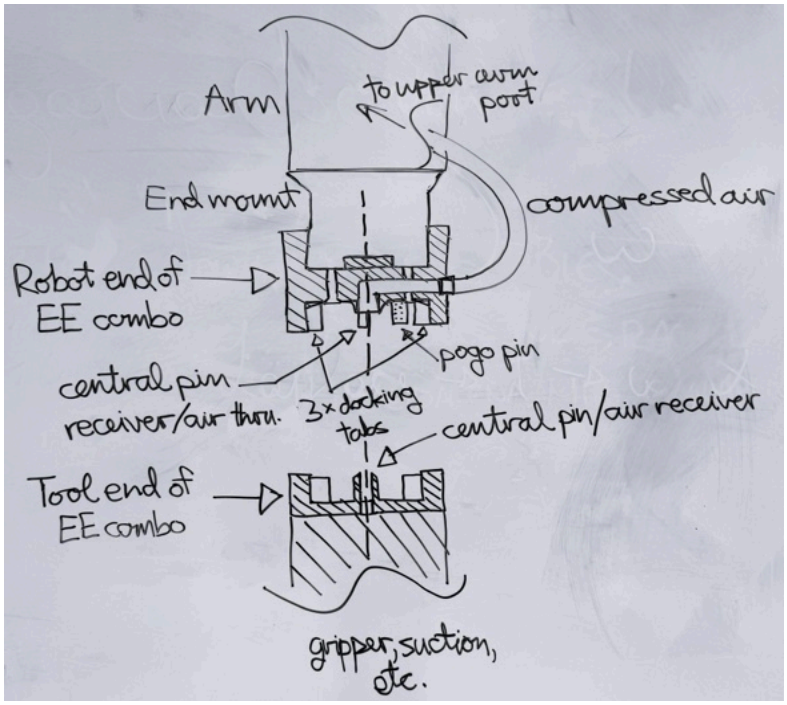


Figure 8. Sketch of the End Effector's Preliminary Design

As shown in Figure 8, the core idea of the MATRIX system is to split the robot's end effectors into two rotationally mateable stages, with the base section always connected to the robot and the tool section being hot-swappable. While the final design did end up splitting the latter portion into two further stages to make the addition of new tools easier, the rotational locking concept proved to be a successful approach. Within this system, three curved prongs evenly spaced along a circular path on the tool plate's top side lock into corresponding slots as the robot rotates its distal link about a central axis, robustly joining the two sections together. The

incorporation of pneumatic and electrical passthroughs via a central air channel and a pair of magnetic pogo pin connectors, respectively, were also key elements introduced in this stage.

2.4. Subsystem Decomposition

In order to further develop the design for MATRIX, the subsystems within it were broken down into their elementary components and the key functional relationships between them were thus identified. This decomposition process first involved creating a high-level component block diagram of the system (Section 2.4.1.), along with an electrical schematic and a software architecture diagram at a similar level of detail (Sections 2.4.2. and 2.4.3.). From here, the system was brought back together through an overview table of the end effector family's key subsystems (Section 2.4.4.) and a CAD visualization of the three existing tools and the universal base sections, showing their interaction with a Stäubli arm (Section 2.4.5.).

2.4.1. Component Block Diagram

The component block diagram of the MATRIX portfolio is intended to help break the full system down into its main constituent parts and define how these are linked together. On the latter point, the component block diagram defines both the spatial groupings of subassemblies and the types of connections that occur between these parts (i.e. electrical, pneumatic, or network-based). Figure 9 shows this block diagram, with the system broken down into subunits for each toolhead–namely, the electrical parallel gripper, pneumatic suction gripper, and adhesive applicator–and for the peripheral components at the robot's base.

2.4.2. System-Level Electrical Schematic

The component block diagram above was used as the basis for a more detailed and formal electrical schematic for the whole system, which can be seen in Figure 10. This schematic provides a detailed overview of all of the electrical components across the MATRIX system and the connections between them, as would be needed to build these circuits in the physical design. Figure 10 maintains the same subsystem-based blocking that is used in Figure 9 in order to reflect the spatial groupings of both the end effector tools and the universal base section of the MATRIX system.

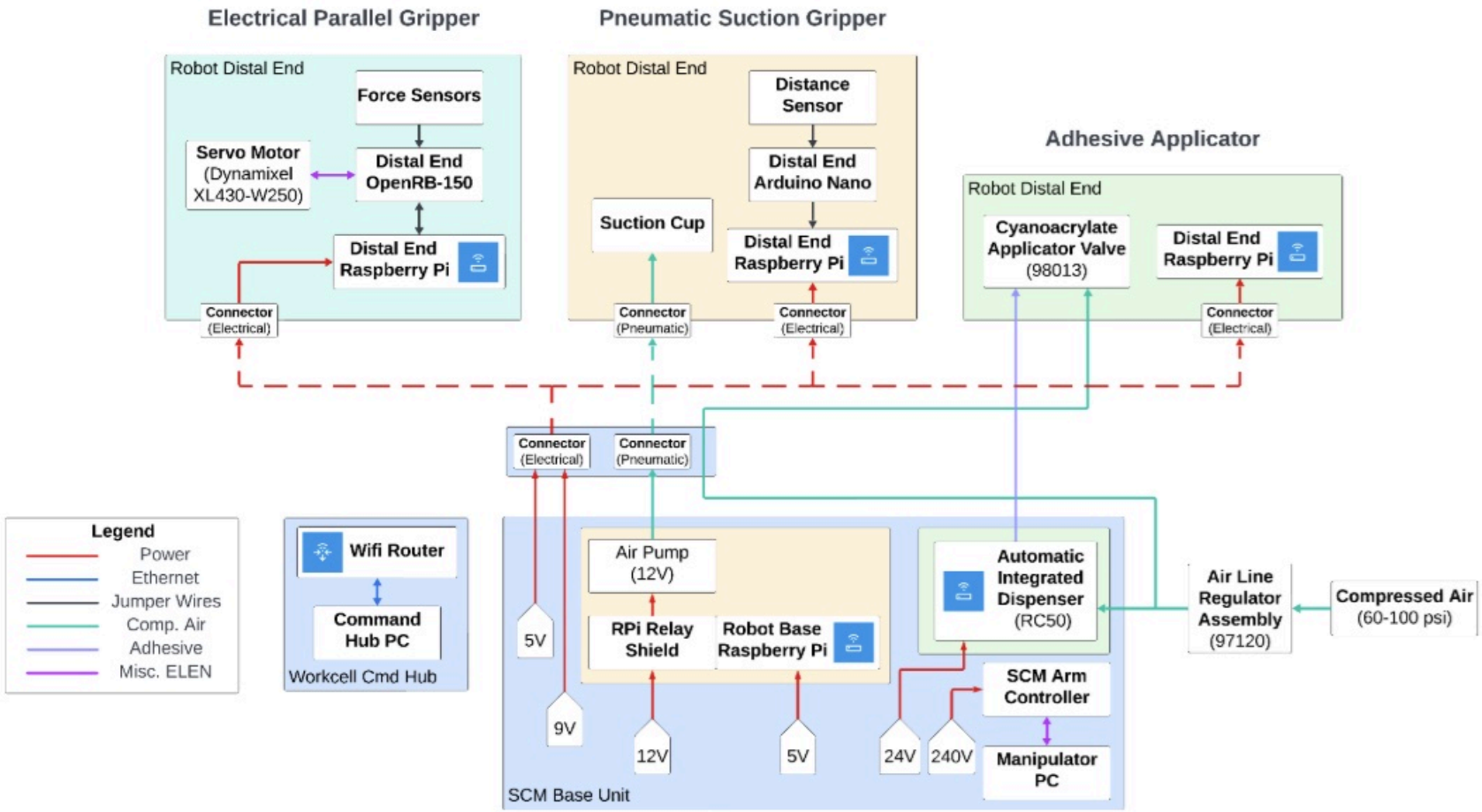


Figure 9. Component Block Diagram of the MATRIX End Effector System

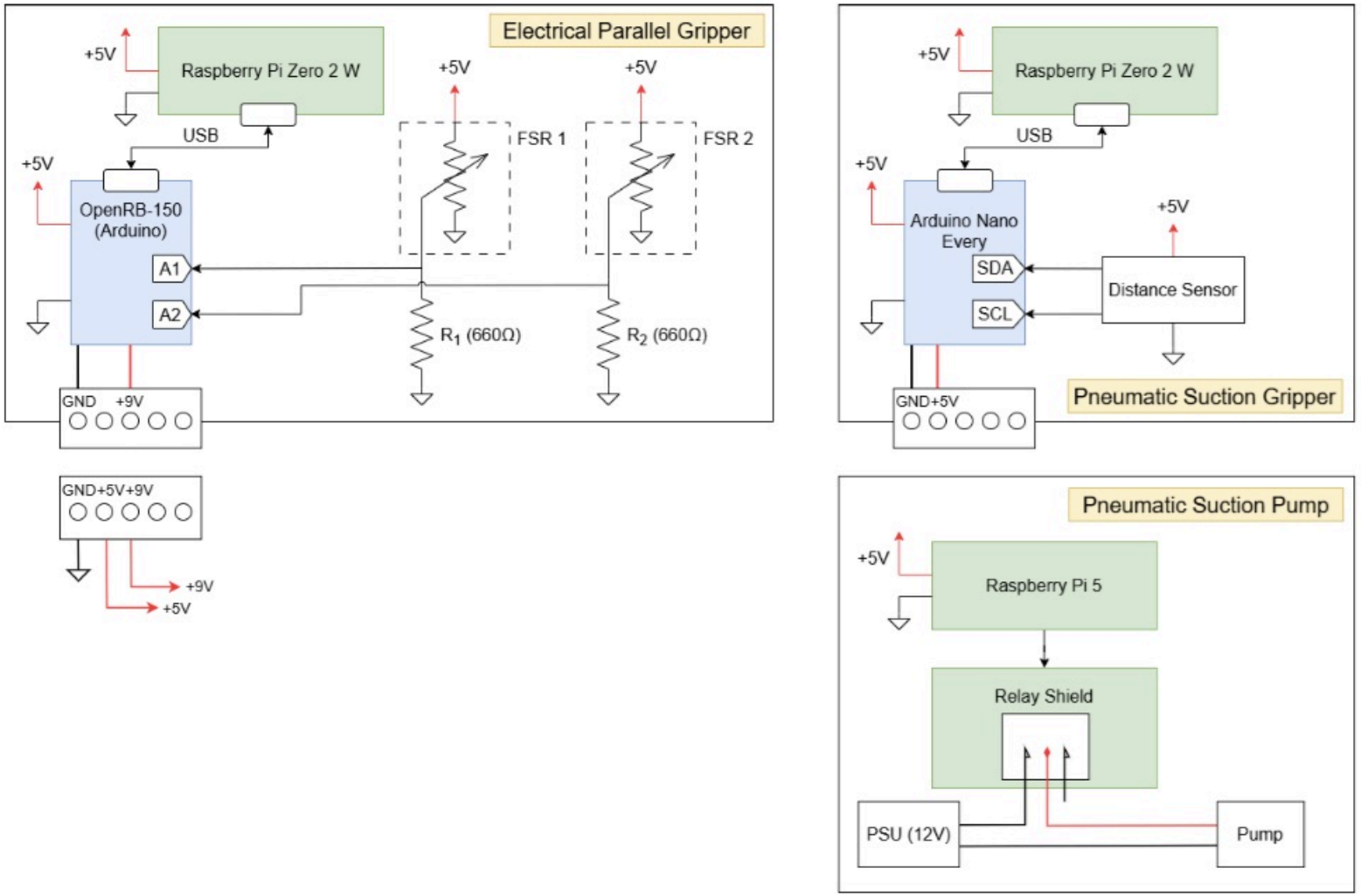


Figure 10. Electrical Schematic of the MATRIX End Effector System

2.4.3. High-Level Software Architecture

In addition to visualizing the physical relationships between components of the modular end effector system, it was also necessary to outline the logic used in their interactions in order to develop a cohesive high-level software architecture. For this purpose, the software diagram in Figure 11 was created, showing the key data lines of the system and the decision-making process they drive in a ROS node network formulation.

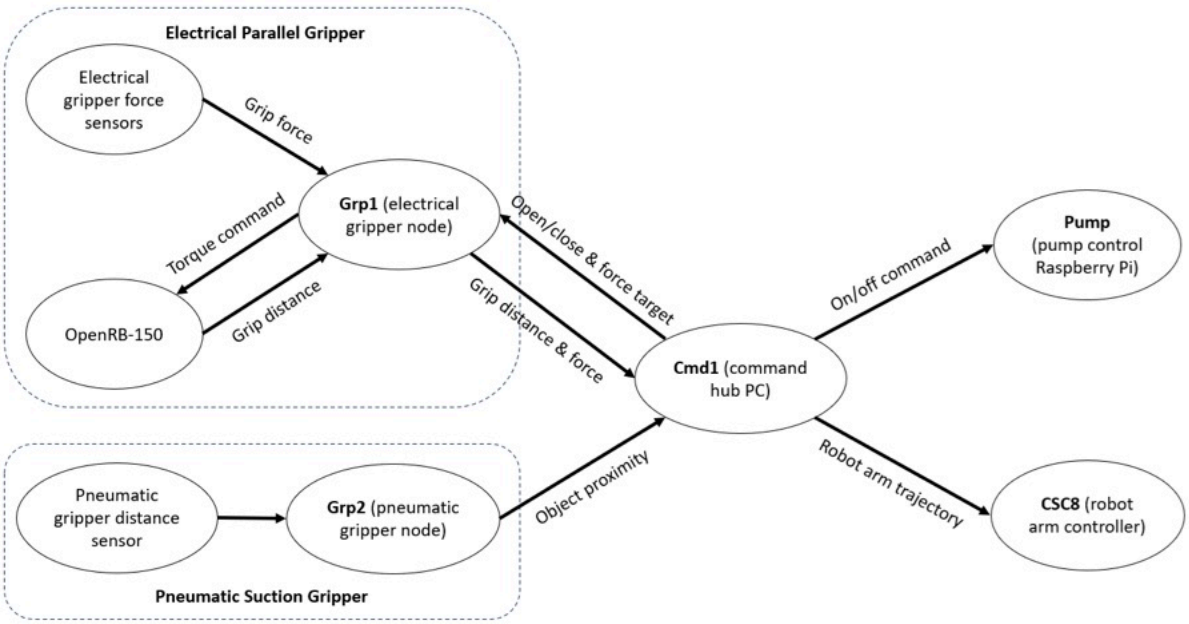


Figure 11. ROS Node Diagram of the MATRIX End Effector System

2.4.4. Full End Effector Portfolio Outline

Based on the subsystem breakdowns of the system in the preceding diagrams, a three-stage outline of the end effector portfolio was also tabulated. As shown in Table 4, the universal subsystems and a preliminary set of relevant end effector tools were developed first in order to validate the modular approach behind MATRIX and demonstrate some of its most pertinent applications, respectively. The future end effector tools listed in the rightmost column will be explored in the continued development of the project and will build on the universal subsystem functionalities to complement the existing tool set.

Table 4. Full End Effector Portfolio Outline

Universal Subsystems	Current End Effector Tools	Future End Effector Tools
Universal mount	Electric parallel gripper	Electric screwdriver/drill
Tool interlock plate	Pneumatic suction gripper	Vacuum cleaner
Tool depot	Adhesive applicator	High-precision probe

2.4.5. Subsystem Overview in CAD

Combining the subsystem-level decompositions above with extensive development based on the preliminary design of Section 2.3, a high-level visualization of the final MATRIX design was created in SolidWorks. This CAD visualization is shown in Figure 12 below and highlights both the universal components of the system and the three primary end effector tools in the current portfolio. The MATRIX system is shown as it relates to the Stäubli TX60 robot arm used for the majority of the project in order to demonstrate how it is mounted during operation.

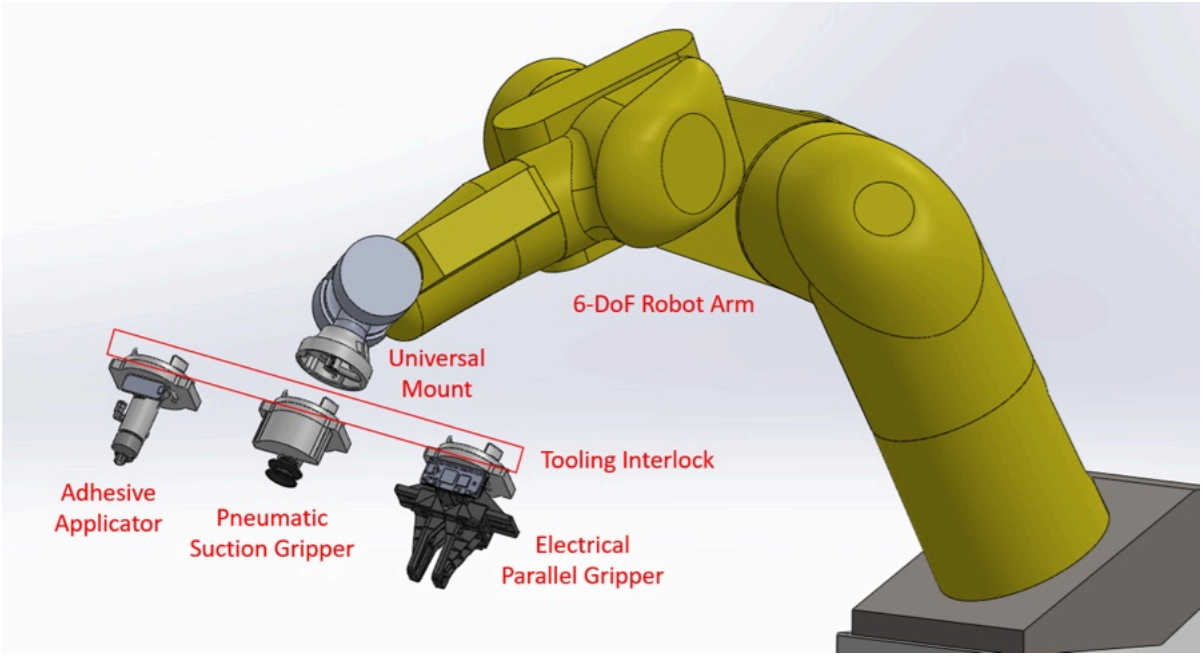


Figure 12. MATRIX End Effector Subsystem Overview in CAD

Chapter 3: Project Management

3.1. Timeline

The MATRIX end effector project was started concurrently with the RSL workcell in October 2024, and thus the development of the two systems was closely interconnected. As noted in the earlier sections, the requirements for the workcell helped drive the design decisions for MATRIX, influencing the end effector system's form factor and integration in particular. The majority of the prototyping for the universal parts of the end effector was done in the winter and spring quarters of 2025, while the end effector tools and their holding depot were built out mostly in the summer and fall of 2025. The latter period also involved the integration of the entire system with the workcell, helping fully connect the two aspects of the overall project. A high-level timeline of the work described above can be seen in Table 5.

Table 5. High-Level Project Timeline

Fall 2024	Winter '25	Spring / Summer '25	Fall '25 / Winter '26
● Project brainstorming ● Early workcell infrastructure installation ● Stäubli arm setup ● Stakeholder interviews ● Literature review	● Design and prototyping for EE base ● Fabrication of OpenManipulator gripper tool ● Continued development of workcell software ● Design of pneumatic tool attachment	● Pneumatic tool prototyping ● Electrical integration of the end effector system ● Early testing on Stäubli arm ● Early development of tool depot ● Start of writing MS thesis ● Fanuc arm setup	● Full stack ROS integration ● Packaging design revisions ● Assembly and early testing of adhesive system ● Final validation testing of all end effector tools ● Completion of MS thesis

3.2. Budget

As one of the major research projects being developed within the RSL from the ground-up over the timeframe listed above, MATRIX significantly benefited from the generosity of the lab via the financial support it received to get off the ground. While there was no fixed budget limit for project-related expenses, the components purchased throughout the development process were all chosen to fit within the broader philosophy of creating a low-cost final product. Appendix D contains an itemized bill of materials used in the completed end effector portfolio, sectioned by subsystem. While the overall project cost of $5039.85 was quite expensive, most of the components that contributed to this total were either shared across multiple projects in the RSL or were purchased by Lam Research for a collaboration between RSL and the company

(detailed in Chapter 9). The latter category in particular made up the majority of the project's cost at a subtotal of $4389.45, primarily due to the high-precision adhesive dispensing equipment required for the production processes at Lam. The following section will further discuss the cost breakdown of the final design, especially as it compares to similar systems within industry.

3.3. Economic Analysis

While the primary stakeholder in this project is also its university backer—and thus does not require a comprehensive market economic analysis like a traditional client—it is nonetheless important to develop the end product in a manner that is economically responsible. To this end, both the material costs and operational expenses of the final product were considered throughout the design process. As mentioned in Section 3.2, Appendix D also contains the total cost of the core MATRIX end effector system, which is $598.32 for the base modules, the tool depot, and the two RSL-only end effector tools. This price is significantly lower than the general market price range of $870 to $9800, determined by a brief survey of end effectors available for retail [30–31]. Given the business-to-business (B2B) sales approach most companies take when selling robotics tooling, pricing data was not available for many of the key manufacturers, and thus the upper end of this price range should be treated as a conservative estimate.

3.4. Safety Considerations and Risk Management

From the outset of this project, safety has been of the utmost priority for both the development of the MATRIX end effector toolset and the operation of the RSL robotic workcell. Risks were identified early on and clear plans to mitigate these issues were developed, as summarized in the following two paragraphs on the project's fabrication and operational risk management, respectively.

Regarding the prototyping and fabrication of the end effector design, great care was taken to ensure safe usage of all the tools and power equipment involved. More specifically, the FDM and SLS 3D printers were both operated in accordance with their recommended operating procedures, with the former requiring caution to avoid contact burns and the latter requiring filtered masks to be worn for the post-processing of printed components. Similarly, the drills and circular saws used during the fabrication process for the tool depot were handled according to their respective procedures so as to avoid any power tool-related injuries.

Throughout the system's regular operation, the focus of the safety procedures shifted to the robot arms and the enforcement of their physical separation from the operator. Safe operation of the workcell is primarily implemented through a set of 3 foot-tall barriers made of T-slot framing with lockable wheels at the base. While the robots are powered on, the barrier is fully extended across the room and nobody is allowed to cross into the broader workspace. If an operator does need to go into the workcell area, all robots are returned to their home positions and powered off remotely. While working on a robot arm, the operator can use a teach pendant to guide its motion in a reduced-speed state, and also has access to emergency stop (E-stop) buttons on both the pendant and on a fixed surface outside of the arm's range (i.e. a table or a robot cart).

Figure 13 shows an overview of the RSL workcell room, with each of these safety features labeled for reference.

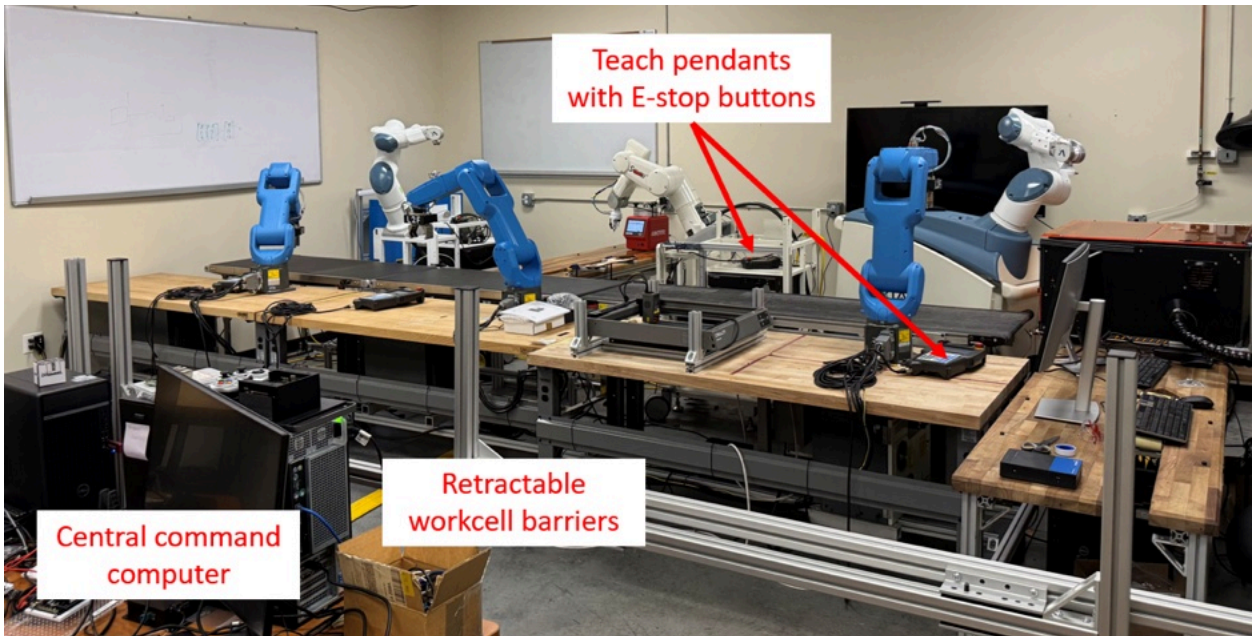


Figure 13. RSL Workcell Room with Key Risk Management Features Labeled

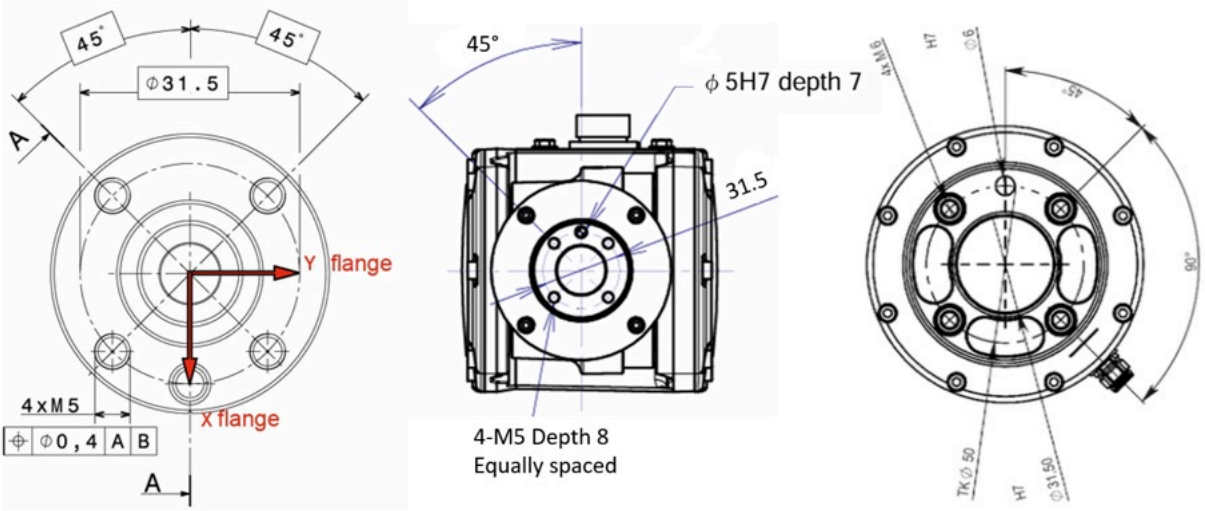
Chapter 4: Universal Mount Subsystem

4.1. Universal Mount Overview

The universal mount subsystem of the end effector is the foundation for connecting all of the tools within MATRIX to a given robot, and as such, was the first section to be designed. The universality of this section goes both ways, enabling both a simple installation on the distal end of any robot in the RSL workcell and the ability for these robots to self-swap all the tools within the MATRIX system. With these requirements in mind, the following sections will cover the electromechanical design and development of the universal mount. The focus here will be how this mount connects to the robots, while Chapter 5 will cover the tool connection in depth.

4.2. Mechanical Design

The overall development approach for the universal mount of the MATRIX system was done from the distal end of the robot arms outward, starting from the mounting hole pattern at the final link to determine the key dimensions. In this regard, all three of the arms that will be used in the RSL workcell feature a cylindrical final link with a central counterbore for alignment purposes. The distal end of each end link has a four-hole mounting design, with tapped metric holes evenly spaced around a circular pattern. The specific dimensions of each end link vary from robot to robot, and are shown in Figure 14 below for reference.



(a) Stäubli TX60 (b) Fanuc LR Mate 200id/7L (c) Franka Emika Panda

Figure 14. Mounting Point Configurations for Three Popular SCM Robot Arms [28, 32–33]

As the three panels of Figure 14 show, the mounting hole arrangements and their dimensions are identical across the three arms. From this common basis, the initial portion of the universal mount was designed as a negative of the typical end link shown in Figure 14. This

initial section in turn serves as the foundation for all of the downstream mechanical and electrical connections to the end effector tools. Harnessing another useful similarity between the three robot arms–their built-in pneumatic passthroughs–the universal mount is designed with a quick-disconnect pneumatic fitting located distally to the robot arm connection point. Figure 15 below shows a view of the universal mount from above, with the screw holes and pneumatic connector both labeled.

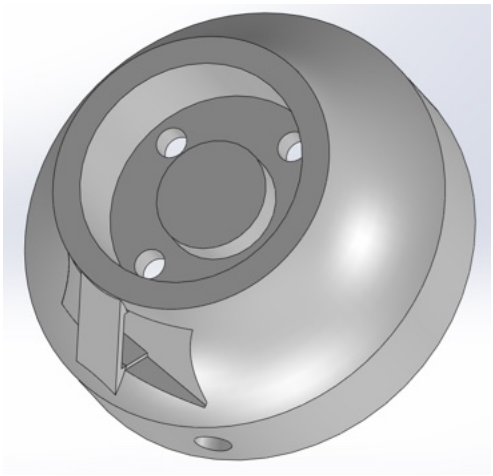


Figure 15. Upper View of the Universal Mounting Piece

The last main consideration in the development of the universal mount was the usage of the space downstream of the robot arm connection. Building on the ideas of Section 2.3, the circular area of the base is fully enclosed by a thin wall, with the slots for the prongs of the tool interlock built into this shell. The interior portion of this enclosed volume contains the pneumatic and electrical passthrough infrastructure, with the former being routed through a radial tunnel to the mount's center. Here, the tunnel takes a 90-degree turn to match the central axis of the mount, leading the pressurized air out distally. This passthrough is shown in Figure 16, while the electrical passthrough system will be discussed in more detail in Section 4.3.

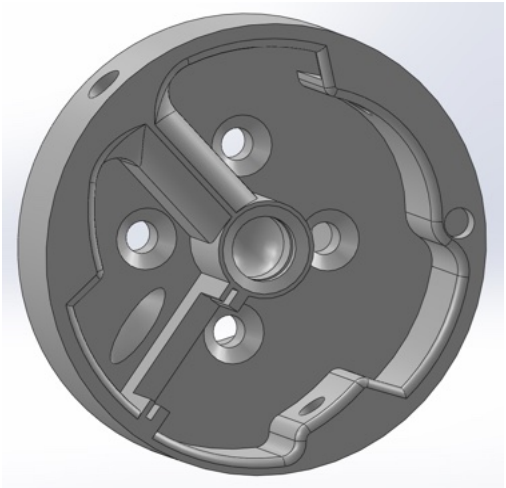


Figure 16. Lower View of the Universal Mounting Piece

4.3. Electrical Design

The universal base outlined in the above section not only serves as the mechanical connection point between the robot and the tool segment of MATRIX, but also as the housing for the electrical components needed to establish a passthrough line towards the distal end of the tool. In this regard, it is important to provide an electrical connection that is both easy to disconnect–in keeping with the hot-swappability requirement of the end effectors–and secure enough to maintain continuity during operation. Based on the research of Roux, Izard, et. al. and the successful implementation of similar components in consumer electronics, a pair of 5-pin male and female magnetic "pogo pin" connectors (Figure 17) were chosen for this objective.

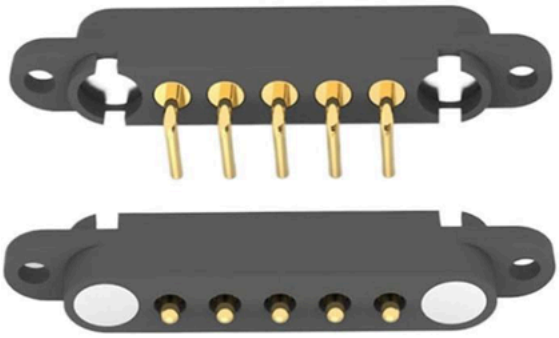


Figure 17. Magnetic Pogo Pin Connectors Used for Electrical Passthroughs [34]

The five spring-loaded pins on the male side of the connector (the bottom panel of Figure 17) help provide contact compliance for the electrical connections, while the magnets that flank both pin arrays help ensure a strong mechanical bond between the two sides of the connector. Upon selection of the final pogo pins connectors, the male component was integrated into the design of the housing shown in Figure 16, being fixed in place on a radially-oriented pedestal via a pair of slots that hold the flanges of the connector in a press fit. As will be further explained in Section 5, the female part of the connector pair is located on the tool interlock plate on a complementary pedestal to align with this male connector.

The other main part of the electrical design for the universal mount involves the connection of the pogo pins to the passthrough access points of the robotic arms in a way that minimizes setup time. For this purpose, the female end of a JST XH was built into the external portion of the universal mount, fitting into a press-fit slot and being connected directly to the pogo pin via a lofted wire channel through the body. The corresponding male end of the JST connector can then be wired to any proprietary connector that a given robot uses as its passthrough outlet without requiring any internal wiring changes to the mount itself. This modular approach to the external connection pipeline is in line with the broader vision of the MATRIX system and enables the universal mount to easily be connected to a variety of robots in the field. The JST XH connector–along with the pogo pin and an O-ring for the pneumatic passthrough–is shown in Figure 18 in its external mounting position on the universal mounting base for reference.

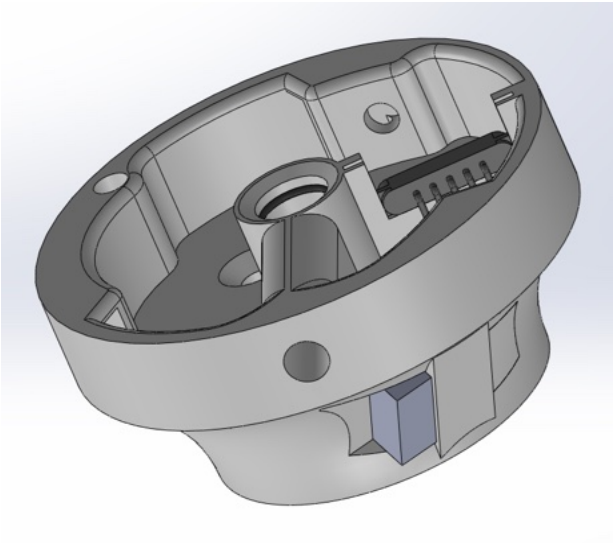


Figure 18. Universal Mounting Piece with Embedded JST XH Connector at the Base

4.4. Manufacturing and Assembly

In keeping with the project's overall focus of producing an affordable end effector system, the majority of the components within it–including the universal mount–were manufactured using fused deposition modeling (FDM) 3D printing. The machine used for this purpose was a Prusa MK4 3D printer, which is one of the most popular printers on the consumer market. FDM printing was used to fabricate the entire body of the universal mount into which the electrical and pneumatic components were subsequently added. Given that the body of the mount is 3D-printed, a threaded heat-set insert was embedded into the end of the pneumatic inlet, thus providing an interface for the M5 quick disconnect to screw into. An O-ring was then placed into the groove at the pneumatic passthrough's opposite end, and the pogo pin was also soldered to the JST connector's female side. This completed assembly of the universal mount is shown in Figure 19.

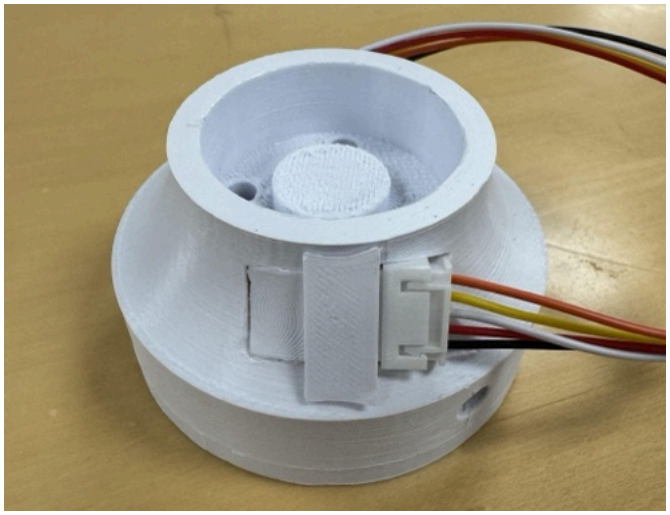
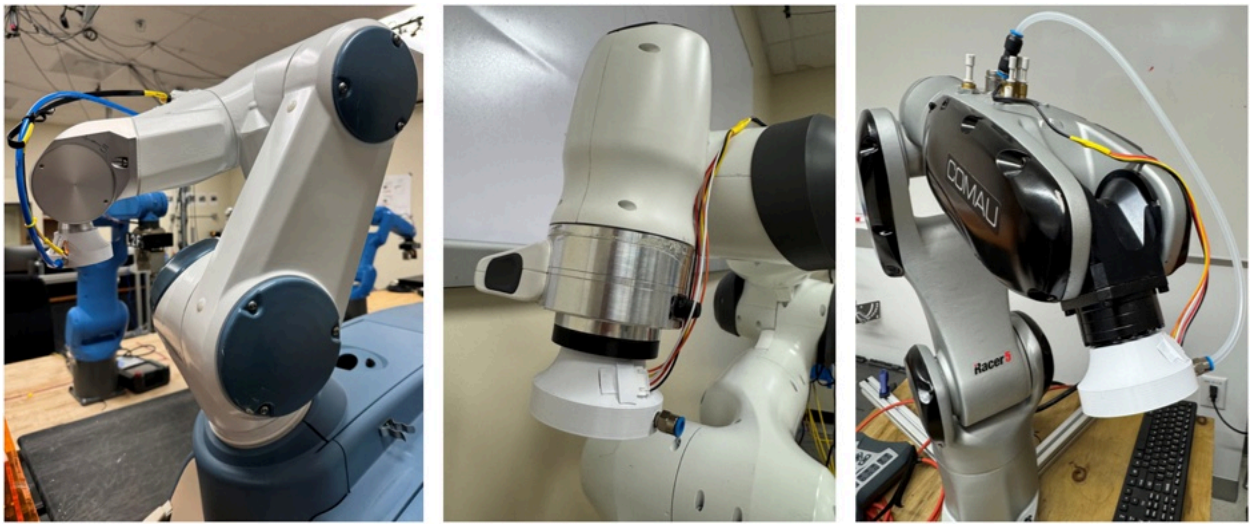


Figure 19. Universal Mounting Piece Final Assembly

Once the mechanical integrity of the universal mount had been verified, it was also mounted onto three available robots in the lab to demonstrate its versatility: the Stäubli TX60, the Franka Emika Panda, and the COMAU Racer-5-0.80. The COMAU arm was substituted in place of the originally-intended Fanuc arm due to the latter being used for another application at the time of testing, with the change being deemed reasonable since the two arms have near-identical distal mounting surfaces. While the Panda and the Racer ended up requiring an additional adapter plate for complete mounting, this adapter was manufacturable in under an hour via the FDM printer, and thus its presence does not detract from the universal mount's ease-of- application. Figure 20 shows the universal mount as it was attached to each of the three robots listed above.



(a) Stäubli TX60 (b) Franka Emika Panda (c) COMAU Racer-5-0.80

Figure 20. Universal Mounting Piece Installed on Three Popular SCM Robot Arms

Chapter 5: Tool Interlock Subsystem

5.1. Tool Interlock Overview

As the direct companion to the universal mount described in the previous section, the tool interlock segment of the MATRIX system is designed as a universal intermediate plate onto which any electro-pneumatic tool assembly can easily be mounted. The tool interlock thus contains both the female counterparts of the pogo pin connectors and the male counterparts of the pneumatic passthrough and rotational locking prongs. The distal surface plane, meanwhile, is intended to provide a variety of mounting solutions for any toolheads to connect to. This section will cover the implementation of all of these requirements, along with a description of the subsequent manufacturing process.

5.2. Mechanical Design

The two main objectives of the tool interlock subsystem were to connect to the universal mount and to extend the passthroughs from it to the toolhead region of the broader assembly. Regarding the former goal, a set of three L-shaped prongs was spaced equally along a circular path such that each prong would slot into a corresponding slot on the base section. The lower surface of each prong was drafted along its arc in order to help ease the rotational locking process. Similarly, the prongs' upper outer edges were filleted to maximize the plate's ability to self-align with the universal mount when docking.

The two passthroughs, meanwhile, were integrated into the design via a pedestal for the female pogo pin for the electronics and a tubular center channel for the pneumatics, respectively. Two sets of four screw holes each were also added to the planar face of the interlock plate, with the countersunk set used as the mounting points for any toolhead downstream of this section. The set of straight through holes was added between design iterations to provide easier access to the screw holes of the motor for the electrical parallel gripper, to be further discussed in Chapter 7. Figure 21 shows the final design of the tool interlock subsystem assembly.

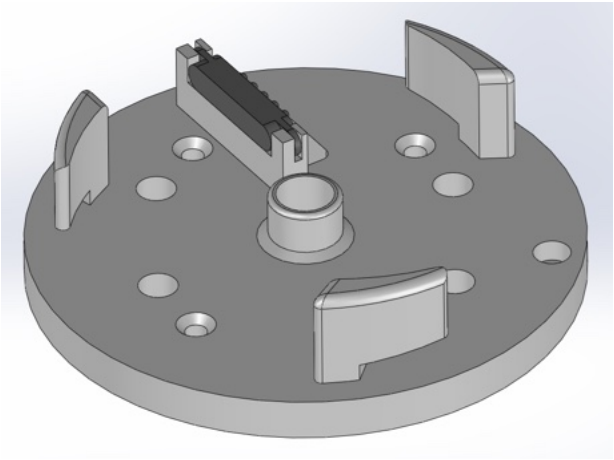


Figure 21. Tool Interlock Plate with Pogo Pin Connector

5.3. Electrical Design

Similarly to its mechanical design, the electrical architecture of the tool interlock plate was largely dictated by the prior decisions made pertaining to the universal mount in Section 4.3. More specifically, the female half of the 5-pin pogo connector is the main electrical component on the interlock segment, providing the distal-end output for the electrical passthroughs supplied from the robot arm. The pogo pin's pedestal is thus designed to vertically align it with its male counterpart when the rotating prongs are fully locked in place. The wires connected to this distal pogo pin are then routed through a slot cutout that in turn serves as the base access point for any tool plate being attached to the interlock plate. The pogo pin connector can be seen mounted on the interlock plate shown in Figure 21 above.

5.4. Manufacturing and Assembly

In keeping with the fabrication approach outlined earlier, the tool interlock plate was also manufactured using an FDM 3D printer. After a brief printing process, the pogo pin was added to the interlock plate and distal outlet wires were soldered to each of its five pins. Figure 22 shows the completed subsystem after being mounted onto one of the toolhead sections.

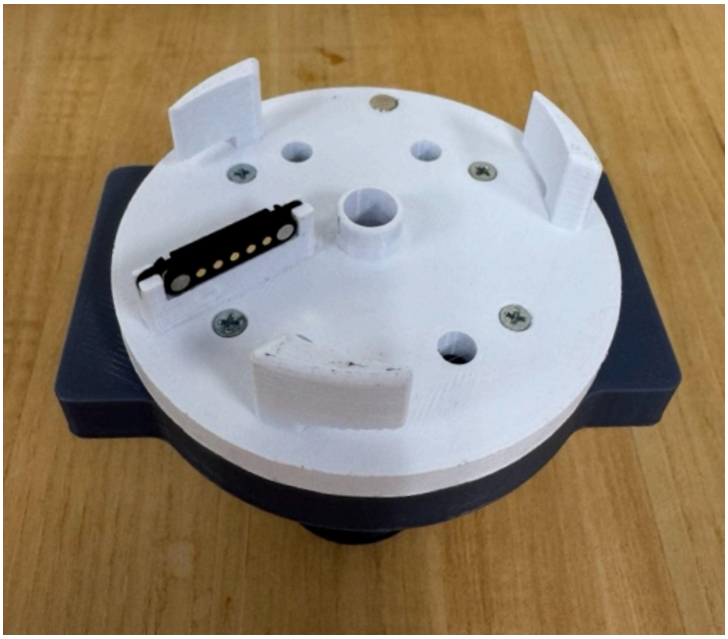


Figure 22. Tool Interlock Plate Assembly

Chapter 6: Tool Depot Subsystem

6.1. Tool Depot Overview

Arguably just as important as the means by which the end effector pieces are connected is the infrastructure that enables the robot to utilize this mechanism with full autonomy. The tool depot subsystem is the primary point of contact between the robot arms and the end effectors when tool swapping occurs, and is thus crucial to ensuring that the whole process is easily repeatable without human intervention. This section will focus on the development of the tool depot, regarding both the standalone system and its integration with the existing robot platforms.

6.2. Mechanical Design

In keeping with design needs in Section 2.1, a focus was placed on making the tool depot system fully passive, building in compliance and self-alignment measures to counteract any alignment issues that may occur during regular use. The general design of the tool depot involves a set of C-shaped 3D-printed guide rails that each provide a support platform for the side flanges present on every tool's interlock plate. The two panels of Figure 23 highlight this functionality, with the left image showing the C rails of an empty tool holder and the right one showing how a tool slots into them via the interlock plate's flanges.

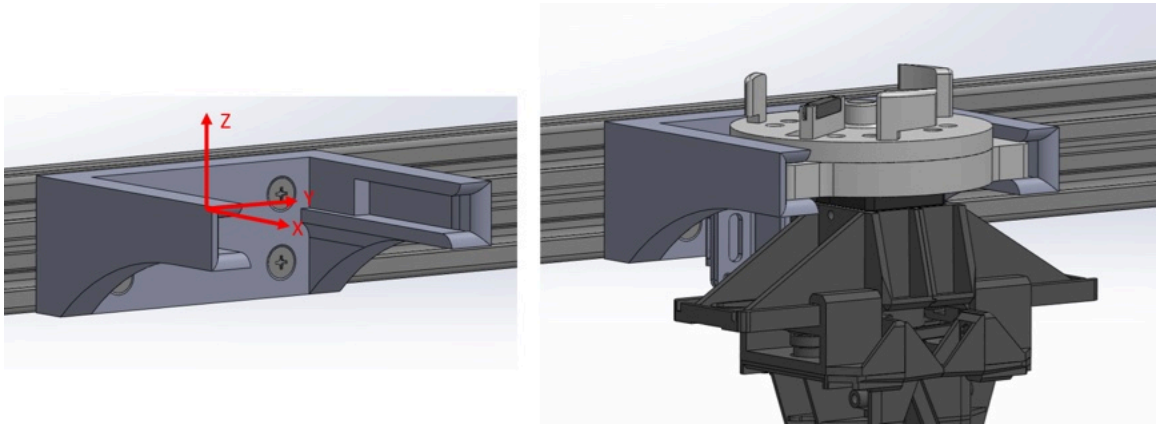


Figure 23. Tool Holder in its Empty (Left) and Occupied (Right) States

One of the key advantages of this configuration is that it leaves the locking prongs and magnetic connector pins exposed for direct docking. When the robot arrives at the depot to pick up a tool, it can simply lower its distal end parallel to the depot's Z-axis onto the interlock section to make a clean contact for the locking rotation. This portion of the operation is also streamlined by the depot's design, with the tool requiring only the most distal axis of the arm to rotate for full engagement. From here, the robot can slide the tool out along the depot's X-axis and then proceed to the general workspace once it is clear of the tool holder. This process is repeated in reverse order when the robot drops off a tool at the depot, with the fillets on the cantilever ends of the tool holder helping the flanges properly align with the C-channels.

For the broader tool depot system, a set of 3D-printed tool holders are mounted onto a T-slot 40mm x 20mm aluminum extrusion, where they can be laterally positioned in accordance with the robot's workspace and then locked in place with a set of metric screws. The extrusion piece unifies the individual tool slots into one assembly which can then be mounted onto any surface near the robot arms, namely the Stäubli carts or the tables where the Fanuc arms are mounted. Figure 24 shows a model of the full assembly prior to being mounted for usage.

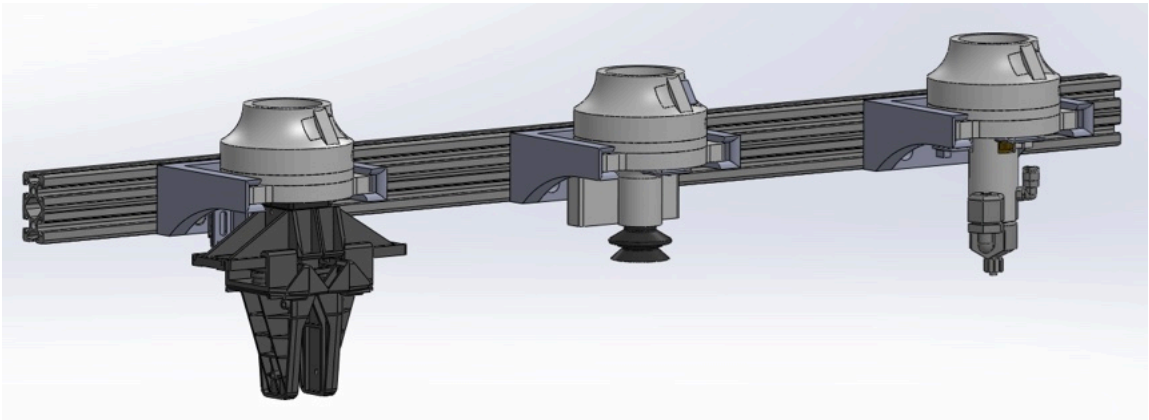


Figure 24. Full Tool Depot Assembly with Primary MATRIX Tools

6.3. Manufacturing and Assembly

The tool holder was manufactured using the FDM 3D printer used for the majority of the MATRIX end effector portfolio. The T-slot extrusion bar was then cut down to length using a circular saw and a set of drop-in nuts were placed into its side rails so that the tool holders could be mounted. Given the primacy of the Stäubli TX60 in the workcell, the tool depot's first version was tested on the associated robot cart. As such, the existing threaded mounting holes on the cart were utilized via an intermediate mounting structure for the tool depot, also made of T-slot extrusions. The completed assembly mounted on the cart is shown in Figure 25, and would later be used as the basis for teaching the Stäubli its tool docking and undocking positions.



Figure 25. Full Tool Depot Assembly Mounted on Robot Cart

Chapter 7: Electrical Parallel Gripper Tool

7.1. Electrical Parallel Gripper Overview

Given the ubiquity and general importance of gripper tools in industrial robotics, the first end effector developed for this project is an electrical parallel gripper. This gripper attachment is intended to be the first part of the electrical toolhead family, demonstrating the viability of MATRIX for both basic actuation and more complex feedback control tasks. For this purpose, the electrical parallel gripper incorporates a smart servo motor to move its two fingers, each of which contain force sensors that are used for a hybrid position-force threshold control scheme. This gripper is particularly useful for pick-and-place operations involving both rigid and soft-bodied non-planar objects in the robot's workspace, ranging from small boxes to foam parts.

7.2. Mechanical Design

In contrast to the rest of the MATRIX project, the mechanical design of the electrical parallel gripper is largely based on an open-source design, namely the gripper attachment of the OpenManipulator robotic arm project created by servo motor manufacturer Robotis. As shown in Figure 26, the OpenManipulator arm consists of a serial chain of Robotis' Dynamixel smart servo motors that form the joints between its aluminum frame links, with the gripper residing on the end of the most distal link. In line with the advanced manufacturing capabilities of a company like Robotis, the gripper's components are designed to be made using injection molding, but can also be fabricated using SLS 3D printing–as will be shown in Section 8.6.

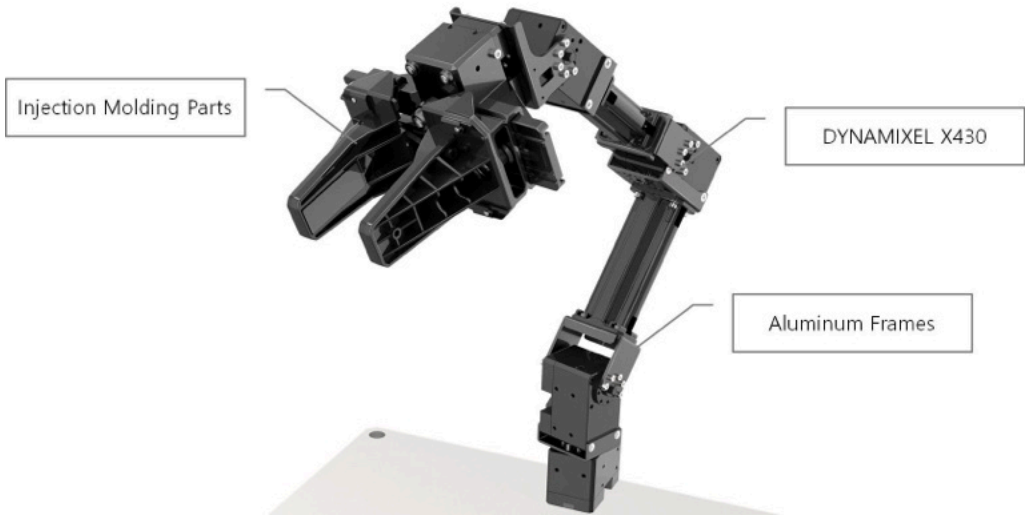


Figure 26. OpenManipulator Gripper on Dynamixel-Based Robot Arm [35]

Studying the design of the gripper more closely, the main sections are the fixed central motor and its housing, the two parallel gripping fingers, and the linkage that connects the motor to these fingers. Per the official Robotis documentation, a Dynamixel X430 series motor is used as the primary actuator for the gripper [35]. While there are a range of motors with varying levels

of capability within the X430 series, all of them share the same form factor and mounting points. For this project's goal of minimizing the system's overall cost–along with its minimal torque and speed requirements–the base-level XL430-W250-T servo was chosen for use in the MATRIX implementation of the gripper. The housing of the gripper assembly is split into two halves, which screw into the sides of the motor and contain the base for the rest of the gripper. The rotating axis of the servo features a circular plate with multiple threaded attachment holes, which is how the motor connects to the linkage shown in Figure 27. This linkage and its operation will be the focus of the following paragraph.

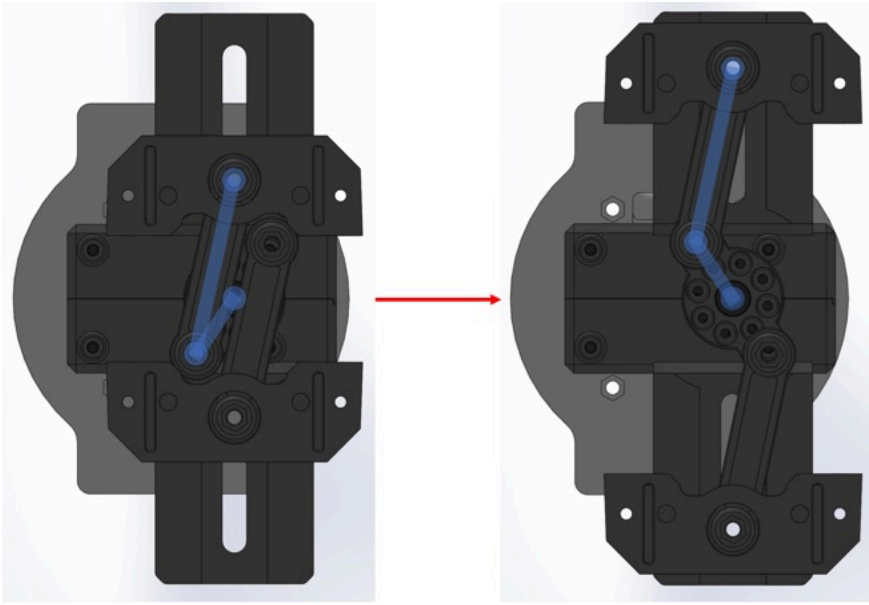


Figure 27. Electrical Parallel Gripper Linkage in its Open and Closed States

As the overhead view in Figure 27 shows, the OpenManipulator gripper design uses a pair of two-bar slider-crank linkage mechanisms to convert the motor's rotational motion into the fingers' linear motion. While the motor can operate in continuous rotation, the linkage uses a linear track at the base of the fingers to constrain their motion within a range of ±10 mm to ±32 mm from the motor's center axis. With the motor simultaneously moving both of the fingers along these tracks, the effective gripping range of the tool therefore becomes 20 mm to 64 mm between the open and closed positions, respectively. Further kinematic analysis of this mechanism is included in Section 7.4, in particular regarding the relationship between the motor's angular position and the gripping distance.

Beyond the core open-source mechanism, the remainder of the electrical parallel gripper attachment mainly involves the base plate that the motor is mounted onto. This plate is designed to include a set of screw holes that match both those in the four corners of the motor's bottom side and the through holes built into the tool interlock plate. Together with corresponding hex nut cutouts at the end of each hole, the gripper's base plate serves to fully join the motor and gripper to the tool interlock plate, creating a unified assembly that the robot can attach at its distal end. Additionally, the base plate contains a lofted cutout for the motor's JST cable to pass through en

route to the pogo pin on the tool interlock segment. The base plate described above is shown in Figure 28, prior to the attachment of any components.

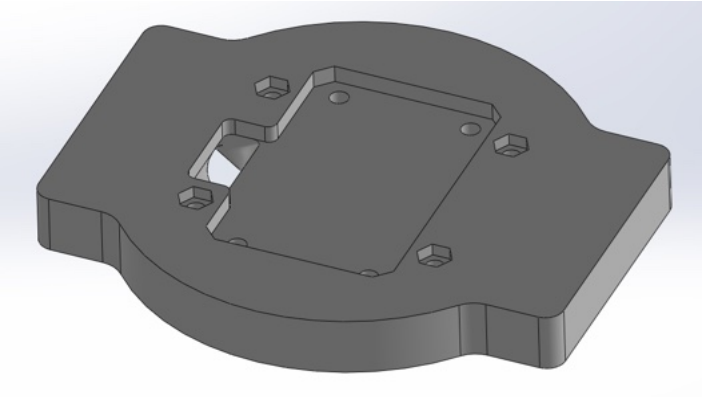


Figure 28. Electrical Parallel Gripper Base Plate

The final part of the electrical parallel gripper's design is focused on the integration of its key electrical components, namely the two force sensors and the onboard microcontrollers. The former of these components are mounted onto foam pads in the tips of the gripper's fingers, which help ensure that the contact force from any object they pick up is uniformly distributed on their surface. The sensors are secured to the fingers using wraparound pieces of tape, and their leads are passed over the tops of the fingers towards the base of the gripper. Regarding the microcontrollers, these components are mounted onto the side of the motor housing via a vertical plate that can be detached as needed for any tests or wiring adjustments. Figure 29 shows the final parallel gripper mechanical assembly in SolidWorks.

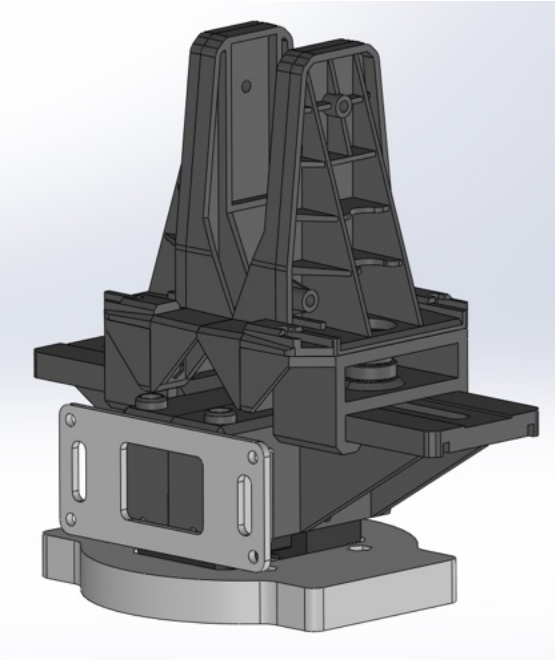


Figure 29. Electrical Parallel Gripper Mechanical Assembly

7.3. Electrical Design

As highlighted in the preceding section, the central component in the electrical parallel gripper tool is the Dynamixel smart servo, which consequently also formed the focus of the electrical design of this subsystem. In order to properly use the Dynamixel servo, a motor control board with Arduino IDE connectivity is required onboard the toolhead. While earlier iterations of the design included a shield board that mounted onto an Arduino Uno or Mega, the final version uses the Robotis OpenRB-150 motor controller shown in Figure 30, which combines an Arduino computing chip into a single PCB with the servo's control circuitry. In order to fit into the low-profile form factor required for it to be mounted on the end effector tool, the header pins and external power connectors of the OpenRB-150 were removed from the board via a soldering iron, after which their leads on the PCB could be directly wired up.



Figure 30. Robotis OpenRB-150 Motor Control Board [36]

In light of the requirement to make each end effector tool its own ROS node while maintaining its compactness, the smallest computer capable of running ROS 2–namely the Raspberry Pi Zero 2 W (Figure 31)–was chosen as the OpenRB-150's complementary board. The Pi Zero is used within the overall workcell to send and receive motor commands and telemetry, respectively, between the gripper tool and the central command computer. This pipeline enables the gripper to be actuated remotely based on the force it encounters when picking up an object.

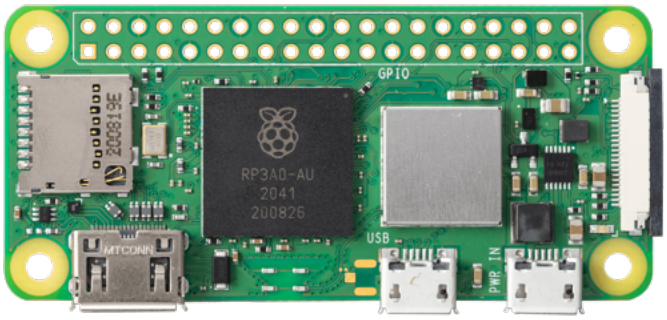


Figure 31. Raspberry Pi Zero 2 W Microcontroller [37]

The final part of the feedback loop, as alluded to above, are the force sensors placed on both of the gripper's fingers. Based on the contact force range the gripper is likely to encounter with its light pick-and-place tasks, the sensor was chosen to be the Adafruit Alpha MF01A, which is a force-sensitive resistor (FSR). In order to ensure that the distribution of the contact across the two fingers does not adversely impact the accuracy of the force detection, an FSR is placed on each finger's gripping plane. The two FSRs are then put into identical voltage divider circuits with a 660Ω resistor in the base position to help maximize the voltage reading resolution over the sensors' available range of 0V to 5V. The outputs of the voltage dividers are then connected to the analog input ports of the OpenRB-150 and thus have their force data sent back to the command computer for control along the ROS pipeline outlined in the previous paragraph.



Figure 32. Adafruit Alpha MF01A Force Sensitive Resistor [38]

Powering all of these functionalities is a universal backend adapter system that provides both 5V and 9V electrical lines along with a common ground. Providing both of these common voltage levels from the robot's base ensures that no major step-down converters are needed at the distal end of the robot and helps maximize the usage of the five electrical lines built into the universal mount. With the internal power lines of the SCM robot arms used being capable of supporting at least 24V, this framework can also easily be utilized for higher-voltage applications as needed. For the electrical parallel gripper, the 9V line is used to power the OpenRB-150, which requires this voltage to run the motor. The OpenRB-150 also contains a step-down converter on its PCB, which is used to provide a 5V power source to both the Raspberry Pi and the FSRs without requiring any peripheral circuitry.

7.4. Kinematic Analysis

Using the dimensions and spatial configuration of the mechanism introduced in Section 7.2, a brief kinematic analysis of the electrical gripper was also conducted in order to determine the relationship between the motor's rotation and the distance between the parallel fingers. This analysis relied on the use of the slider-crank analysis equations, which can be solved to yield the

linear translation of the slider in terms of the crank's rotation and the mechanism's link dimensions. The detailed analysis calculations are included in Appendix A, while the resulting output plot over the gripper's relevant window of operation is shown in Figure 33.

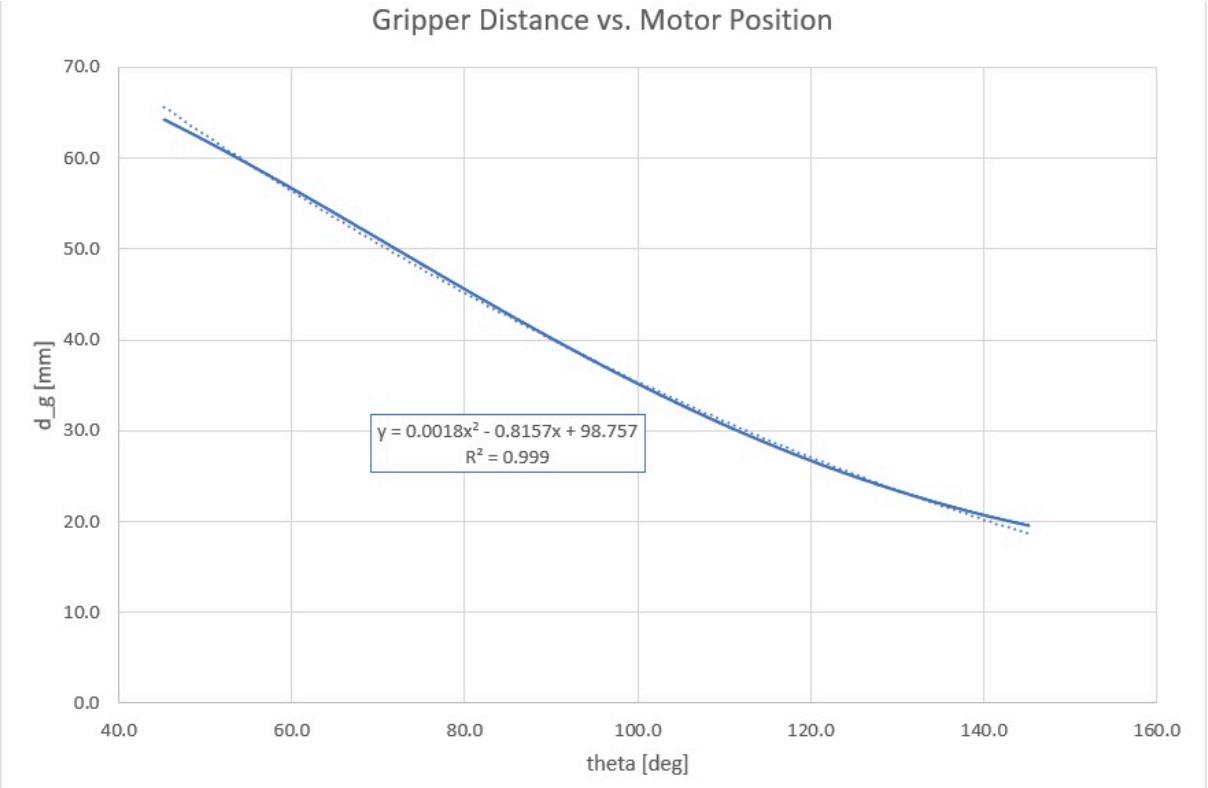


Figure 33. Electrical Parallel Gripper Distance vs. Motor Position Over Open-Close Range

Beyond the analytically-computed distance-rotation curve, this plot also includes a second-order trendline that was fitted through the in order to obtain a more computationally-efficient conversion equation. The corresponding trendline equation does not require the cosine and square root functions of the original result in Appendix A.1 and is thus particularly useful in computing the gripper's finger distance directly from the motor's rotational position data with minimal overhead to the tool's core software.

7.5. Hybrid Control Implementation

Using the understanding of the gripper's kinematics derived above, a proportional control algorithm was developed to provide the mechanism with a hybrid of force and position control. More specifically, a dynamically-adjustable force threshold controller was combined with a basic motor position-based open and closed controller for a flexible object pickup interface. Under this control scheme, the gripper can be commanded to open and close either fully or up to a user-sent force limit that can be updated in real time by sending a new threshold value. The force control law then uses these thresholds to calculate the magnitude of the motor's rotation command,

combining proportional and integral terms to achieve a fast response (τ < 1s) with negligible steady-state force error. Figure 34 below shows the control flow diagram for the controller.

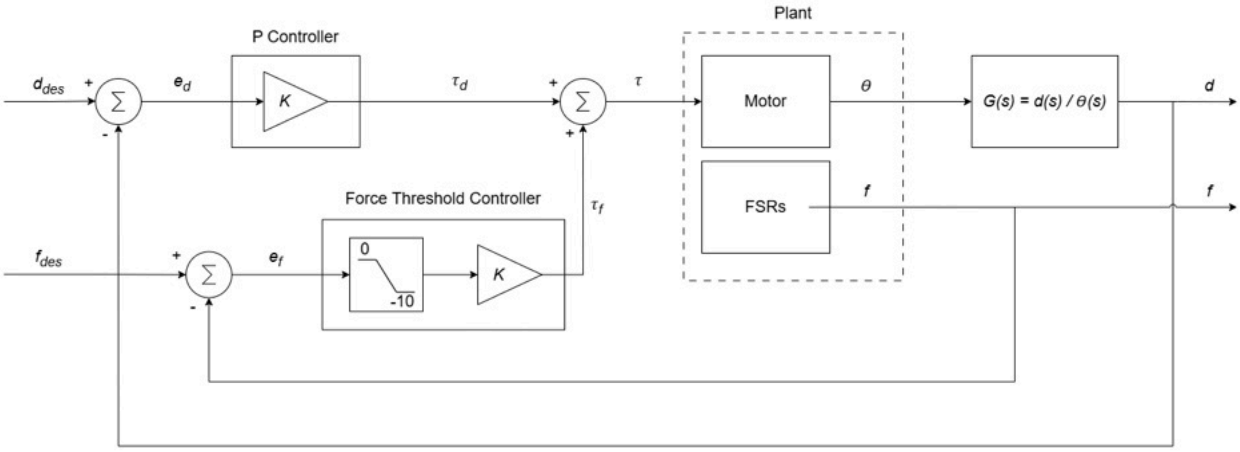


Figure 34. Control Flow Diagram for Electrical Parallel Gripper

In the practical context of the gripper's operation, the control approach described above is implemented using an Arduino script that resides on the OpenRB-150 and is called remotely via a ROS publisher-subscriber pair with one end running on the onboard Raspberry Pi. The first script includes all of the core computational functionality needed to take the FSR readings, filter them, and then calculate the control effort to reach the user-specified setpoint. The tool-end node, meanwhile, both sends key telemetry data to the central command PC and listens to the open, close, and force limit commands that are sent from the corresponding two-way node from either direct user input or a pre-sequenced callback script. The first two scripts are included in Appendices B.1 and B.2, respectively, while their nodal network is shown in Figure 35 below. The final command hub PC script is still in development and is substituted by console commands for testing purposes; thus, it is not included in the report's appendices.

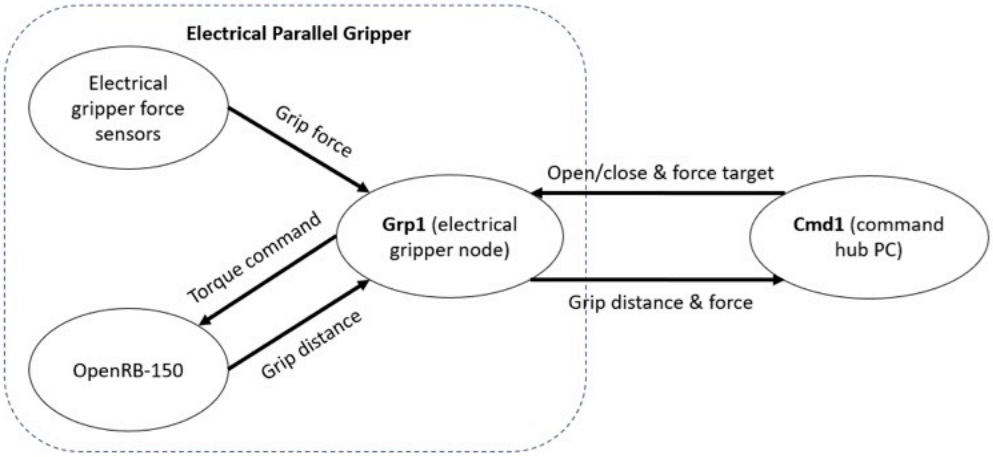


Figure 35. ROS Node Diagram for Electrical Parallel Gripper

7.6. Manufacturing and Assembly

The manufacturing process of the electrical parallel gripper consisted of three distinct stages, namely the 3D printing of the gripper itself, the 3D printing of the corresponding base module, and the assembly of the subsystem. The first of these was the most involved of the three processes due to the complexity of the gripper's ribbed structural design, which made fabrication on a standard FDM 3D printer non-viable. Even with various orientation and support pairings tested, all of these setups resulted in significantly compromised print quality along at least one planar surface set of the gripper. Given the critical importance of the gripper's dimensional accuracy to its proper kinematic functionality, it was decided that its components would be made using an SLS printer with Nylon 12 powder. This manufacturing process, though more expensive than a standard FDM printer, results in a final product with much smoother surfaces and more isotropic material properties, both of which are highly desirable in the gripper. The contrast between the quality of the two printing methods is highlighted in Figure 36 below, with the annotated left panel denoting the key areas of concern.

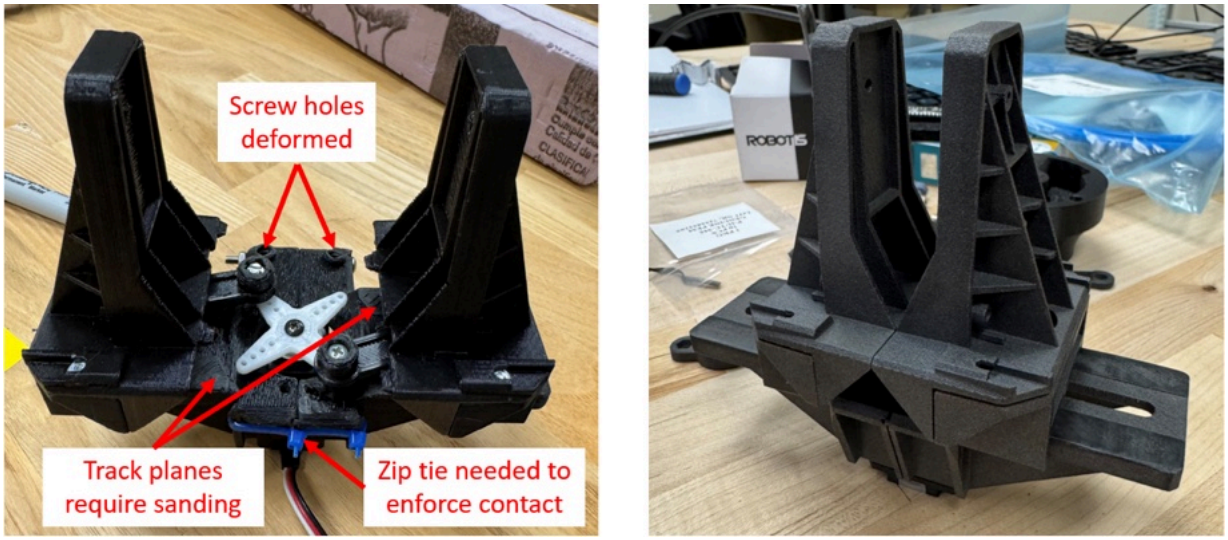


Figure 36. FDM-Printed (Left) and SLS-Printed (Right) Electrical Parallel Gripper Toolheads

In contrast to the main gripper toolhead, the accompanying base section and electronics platform pieces could be fabricated on a standard FDM printer thanks to their simpler geometry. From here, all of the gripper's sub-parts were screwed together around the Dynamixel motor (with its JST cable plugged in) and the gripper itself was screwed onto its base plate. Two adhesive-backed foam pads were then cut using a knife and pasted into their recesses on the gripper's two fingers. The foam pads were then used as the basis for mounting the two FSRs, which had their wiring routed over the tops of the fingers and away from their path of travel. The leads of these sensors–along with the three lines of the motor's JST cable–were then soldered to the Raspberry Pi Zero and OpenRB-150 to be mounted onboard the gripper platform. Once all of the necessary connections were soldered, the boards were zip-tied onto the electronics platform together and screwed onto the side of the gripper, completing the assembly seen in Figure 37.

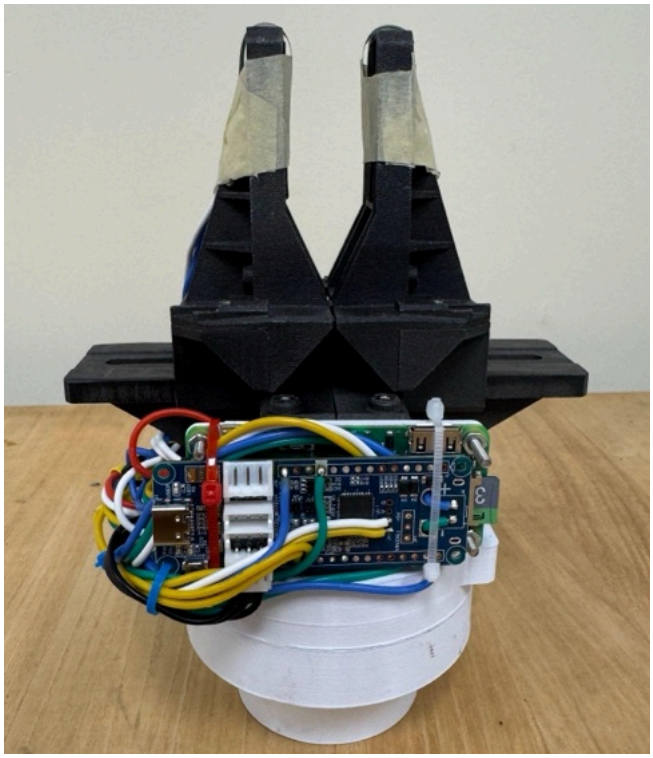


Figure 37. Completed Electrical Parallel Gripper on Universal Mount

Chapter 8: Pneumatic Suction Gripper Tool

8.1. Pneumatic Suction Gripper Overview

While the electrical parallel gripper is the most common choice for picking up objects within the workcell, it is notably limited in its ability to grasp flat, planar objects like sheets of acrylic and coasters. As such, the pneumatic suction gripper is intended to complement the electrical gripper's functionality and thereby provide the MATRIX system with a more well-balanced skillset. Much like how the parallel gripper utilizes the electrical passthroughs of the arm, the suction gripper makes use of the pneumatic passthrough to provide a grasping force for object handling. A basic distance sensing functionality is also incorporated into the toolhead, providing the foundation for a future distance- based control algorithm at the robot arm level.

8.2. Mechanical Design

The main component of the pneumatic gripper's mechanical design is the suction cup used to grasp objects via the vacuum it creates upon direct contact with their surfaces. Knowing the wide range of pick-and-place scenarios the workcell requires throughout its operation, a bellowed suction cup was chosen for the gripping interface, enabling the toolhead to create a vacuum even when approaching an object from a slightly off-perpendicular angle. This 1.42" diameter suction cup, purchased from McMaster-Carr, also features an ⅛" NPT pipe thread and hex profile on its base end for easy attachment to pneumatic lines. Figure 38 shows this suction cup, which in turn would be the basis for the pneumatic gripper's design.



Figure 38. Bellowed Suction Cup Used in the Pneumatic Suction Gripper [39]

The body of the pneumatic gripper toolhead was thus designed with the goal of holding the suction cup in a stable position and ensuring an airtight seal at its threaded base. Additionally, given the goal of having a distance sensor onboard, the tool piece had to be dimensioned such that the minimum distance of the suction cup's distal plane is at least 50mm away from the sensor's mounted position. A hollow pedestal with a hexagonal counterbore at its top was therefore designed as the suction cup's mounting base, while mounting points were also included on either side of the pedestal for the distance sensor and its associated microcontrollers. A lofted electrical passthrough cutout similar to the one designed on the base plate of the electrical gripper was also built into the design to pass power to all of the onboard electronics. The base of

the pneumatic gripper is shown individually in the left panel of Figure 39, while the figure's right panel shows it as a component within the final gripper assembly.

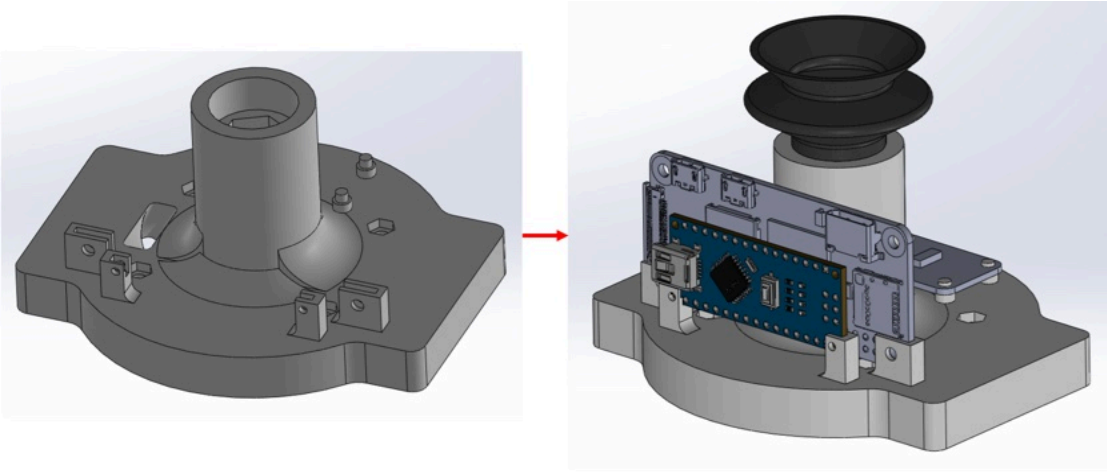


Figure 39. Pneumatic Suction Gripper Prior to and Post-Assembly

8.3. Electrical Design

As mentioned in the previous two subsections, the primary electrical functionalities of the pneumatic gripper involve reading real-time distance data and sending it to the central command computer to determine the toolhead's proximity to the object being picked up. The component chosen for this task is an Adafruit APDS9960 multi-function breakout board (Figure 40), which contains an array of four IR sensors that can be used for both the original goal of near-field distance sensing beyond 50mm and also for alternative sensing modalities like gesture detection. The breakout board is designed to communicate with external devices via the I^2C protocol, and is connected to an onboard Arduino Nano Every microcontroller accordingly. The Arduino Nano handles the distance computations via the calibration curve in Appendix C, and then sends these onward to a Raspberry Pi Zero 2 W (also on the tool). From here, the data is sent to the central command computer and used to control the pump unit outlined in the following section.

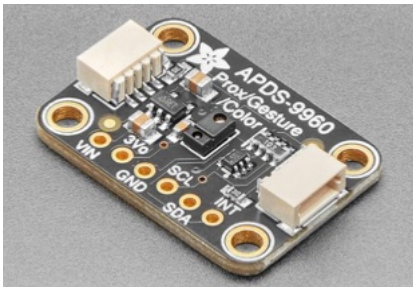


Figure 40. Adafruit APDS9960 Distance Sensor [40]

8.4. Pneumatic Base Unit

As alluded to in the preceding subsections, a peripheral pump assembly is required at the base of the robot arm for the MATRIX pneumatic tools to function properly. This pneumatic base

unit is intended to be a compact package that can easily be set up alongside the universal end mount during the installation process, while also providing expansion points for future additions. As a part of the broader RSL workcell, the base pneumatic unit is designed to fit onto both mobile and stationary robot platforms and thereby make the system's reconfiguration process as frictionless as possible. The pneumatic unit's two main components are an air pump and its corresponding relay, each of which also has its own power adapter. For the first of these, a 12V air pump from Sparkfun was chosen for its ability to provide a useful level of vacuum for light pickup tasks (>47 kPa) at an affordable price and a reasonable voltage. As the calculations in Appendix A.2 show, this pump allows the pneumatic suction gripper to lift objects up to 5 kg under standard operating conditions, covering the majority of the use cases for MATRIX.

Based on the system requirement for high-level control, it was decided to make the base hub unit its own ROS node in order to maximize its placement flexibility and eliminate the need for a connection to the suction gripper tool through the robotic arm. As such, the relay that controls the flow of power to the pump was chosen to be a Raspberry Pi shield, and is mounted onto a Pi 5 microcontroller running ROS 2. Through this pipeline, the central computer for the workcell can send binary commands to the pump over the local wireless network, turning it on and off as needed for the suction gripper to pick up and release objects within the robot's workspace. A ROS node diagram for the system's operation is shown in Figure 41 below, with the corresponding gripper script included in Appendix B.3. The other two nodes are currently in development and are substituted with manual commands; thus, they are not included in the appendices of the report.

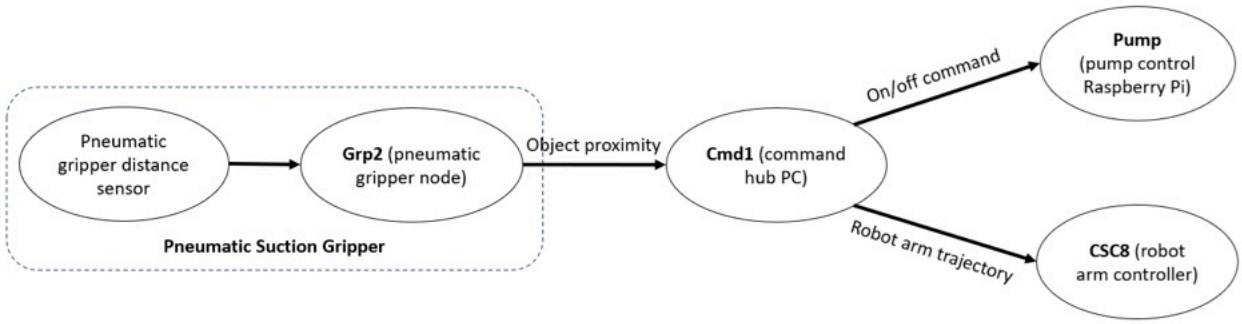


Figure 41. ROS Node Diagram for Pneumatic Suction Gripper

The pneumatic support system is contained on a single substrate board laser-cut from a ¼" sheet of acrylic with outer dimensions of 8" by 6". The cutout pattern, used across many RSL applications, contains an array of #4 through holes that can be used for either screw or zip tie-based attachments. Each of the board's four corners also has a larger primary mounting hole for easy attachment of the board onto any surface. Regarding the components' spatial arrangement, the air pump is laid on its side to make the overall assembly as flat as possible, while the Raspberry Pi-based relay assembly is placed next to it for ease of wiring. The adapters for these two parts are plugged into an outlet close to the board's mounting position and their cables are routed through the main board afterwards. Figure 42 shows a Raspberry Pi with the relay shield mounted on top of it in the same configuration as was used for the pump unit.



Figure 42. Raspberry Pi 5 with Relay Shield on Top [41]

8.5. Manufacturing and Assembly

Similarly to the previous subsystems, the manufacturing of the pneumatic suction gripper was centered around the FDM 3D printing of the main custom piece, with the mechanical and electrical integration of OEM components being done afterward. Once the toolhead's main piece had been printed, the suction cup was inserted into the hexagonal cutout at the top of its pedestal and hot-glued around its base for a strong pneumatic seal. The distance sensor and the two microcontrollers were also soldered to each other per the layout described in Section 9.3, with the power from the pogo pin being sent to the Raspberry Pi and distributed through the system from there. The electronics were lastly zip-tied together and attached to their respective mounting points, and the whole toolhead was screwed onto a tool interlock plate. Figure 43 shows the completed pneumatic suction gripper assembly.

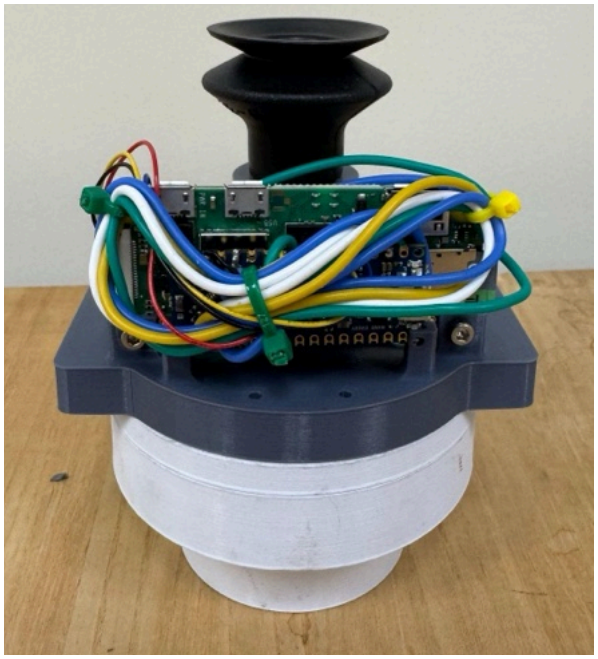


Figure 43. Completed Pneumatic Suction Gripper on Universal Mount

The pneumatic base unit, meanwhile, was cut out of its stock acrylic sheet via a laser cutter, and the pump and relay were mounted as described in the preceding section using zip ties to secure them in place. The power lines of the pump and Raspberry Pi were then connected to their respective 12V and 5V adapter sources, with the relay placed inline with the former to open and close the circuit as intended. The whole assembly was then zip tied to the robot cart of the Stäubli arm being used for end effector testing. Lastly, all of the power adapters were plugged into a nearby outlet and connected to the base of the robot via its proprietary 19-pin connector. Figure 44 shows an overview of the base hub unit in this application environment.

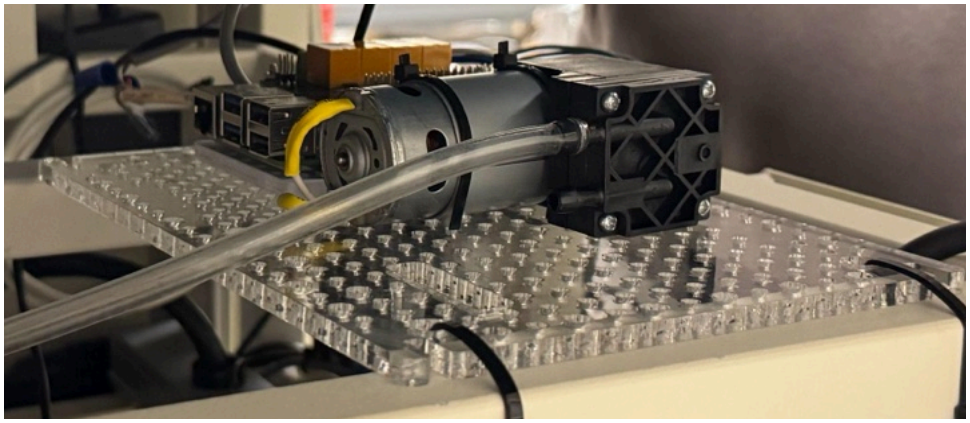


Figure 44. Pneumatic Base Unit Mounted on Robot Cart

Chapter 9: Adhesive Applicator Tool

9.1. Adhesive Applicator Overview

In addition to the two primary end effector tools designed for use primarily within the RSL, a third tool–an adhesive applicator–is also currently in development for Lam Research, a major local firm in the semiconductor manufacturing supply chain. Given the company's extensive robotics development and manufacturing operations, there is a present need for precision adhesive applications on various components in their systems. After an initial consultation with the engineers at Lam and a review of the documentation they provided, the application of a fast-curing adhesive onto a gasket pathway on a robot arm panel was chosen as the initial target task. The development of the adhesive applicator thus far is outlined in the following sections.

9.2. Equipment Selection

The main design drivers in the development of the adhesive applicator toolhead and its companion control system have so far been the type of adhesive required and the need for full autonomy in the dispensing process. On the former point, the adhesive process requires Loctite 380 Black Max single-part cyanoacrylate adhesive, which is commonly used to bond metals, plastics, and rubbers in industrial applications. The adhesive's fast curing upon exposure to air means that its corresponding applicator system must be airtight, most critically at the valve that dictates the dispensing stage. In order to maximize the existing workcell infrastructure for the automation aspect, the dispensing system had to be remotely controllable via an ethernet or wireless connection and preferably have a ROS-callable command structure.

Based on these requirements, the Loctite product catalog was thoroughly reviewed and various adhesive dispensing product combinations were compared to find the most cost-effective solution. The optimal system that emerged from this process is centered around the Loctite RC50 Automatic Integrated Dispenser, which is a digitally-integrated adhesive reservoir and controller that can be commanded over either ethernet or WiFi. The RC50 has a large internal reservoir for housing adhesive containers with up to 2.2 kg of adhesive in them, making use of an external pressurized air line to push the adhesive out at a user-specified rate. Given the sensitivity of the adhesive to external contaminants, the RC50 also requires an inline air filter upstream of its air supply. This filter was chosen to be the Loctite Air Line Regulator, a device capable of filtering a pressurized air supply for particles up to 5 microns and regulating its pressure up to 150 psi. The final component of the adhesive system is the Loctite Cyanoacrylate Adhesive Valve, which is specifically designed for handling fast curing adhesives. The valve assembly's operating principle involves the use of a diaphragm valve driven by pressurized air to push out the exact amount of adhesive required for a given application and preventing any excess from leaking out or being exposed to air. The three components described here are shown in the panels of Figure 45 for reference.



Figure 45. Loctite Air Line Regulator, RC50 Dispenser, and Cyanoacrylate Valve [42]

Given the status of the adhesive applicator system as an externally-funded addition to the internal RSL MATRIX portfolio, the bill of materials for the components is split into a separate table following the primary BoM in Appendix D. The total cost of the system–including a 1-lb container of the 380 Black Max adhesive–is $4389.45, all of which was paid for by Lam Research. The full set of components was delivered to the RSL within a month of the order being placed, and their basic individual functionalities were verified shortly after. Lam Research also provided a sample of the robot arm panel that the adhesive will be applied to and shared the associated CAD files for subsequent engineering work.

9.3. Mechanical Design

The mechanical design of the adhesive applicator subsystem can be split into two primary sections, namely the end effector tool and the fixturing for the panel to which the adhesive will be applied. The design of the toolhead uses a hollow cylindrical shell to hold the applicator valve in a close fit, while a screw hole aligned with a threaded hole on the valve's side provides a point at which the two can be fully secured to each other. The baseplate portion of the toolhead also includes a lofted pneumatic passthrough cavity that exits next to the valve holder via a pedestal for an M5 threaded insert. This insert is then used as the mounting point for quick disconnect that routes an intermediate pneumatic tube onward to the pneumatic inlet port of the valve assembly. Similarly, an electrical passthrough cutout is included on the other side of the valve holder, leading to a slot that can accommodate a variety of electrical platforms depending on which microcontroller is ultimately chosen for the toolhead. The toolhead's structural design is shown in the left panel of Figure 46, while the completed assembly in CAD is shown in the right panel.

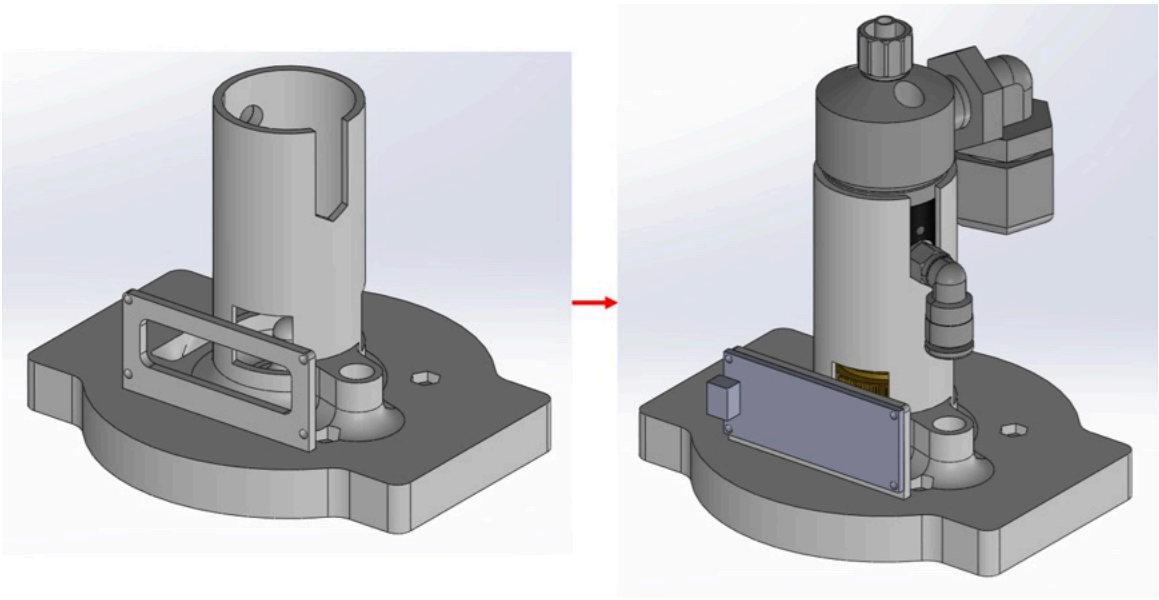


Figure 46. Adhesive Applicator Tool Prior to and Post-Assembly

The fixturing subassembly for the adhesive application process was designed around the CAD model of the arm panel provided by the engineers at Lam Research, focusing on the key mounting holes located on the part's planar surface. A set of key holes was identified on each end of the panel, and a pair of corresponding pin plates were designed to temporarily fix the panel during the adhesive application. These plates were pocketed due to their minimal load requirements, making full use of the design flexibility afforded by the project's additive manufacturing approach. A T-slot aluminum frame was then designed for the mounting plates to be affixed to, creating a fully-contained fixture that can be positioned on either a robot cart or the side of a table. Figure 47 shows this fixture mounted on the Stäubli TX60's robot cart.

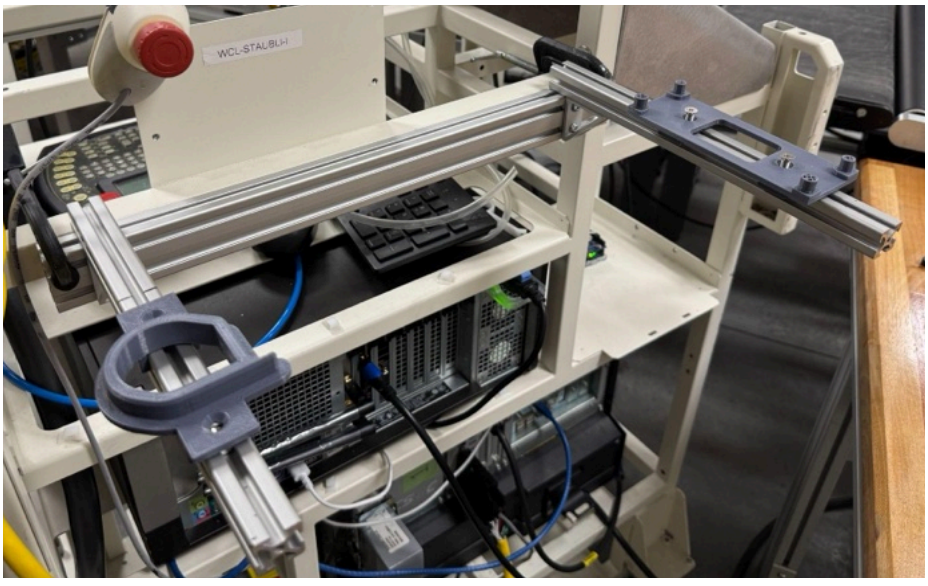


Figure 47. Robot Arm Panel Fixture Prototype on Robot Cart

9.4. Electrical Design

Seeing as the main electrical systems for the adhesive applicator reside within the RC50 dispensing control module, the minimal remaining electronic components are largely peripheral. As mentioned in the previous section, the mechanical base for the OEM components also includes a modular slot for an electronics holder that can be configured for any compact microcontroller such as an Arduino Nano or Raspberry Pi Zero. While no decisions have been made thus far on which microcontroller will be used, a follow-up discussion with the engineers at Lam is imminent to determine their needs for this aspect of the design. Any subsequent integration will follow the same electrical approach as that of the other two end effectors, albeit with a modified software framework to accommodate the application needs of the company.

9.5. Manufacturing and Assembly

At the time of writing, a mechanical prototype of the adhesive applicator toolhead has been completed, with all of the custom parts printed via an FDM printer and the OEM components mounted onto them in a full assembly (Figure 48). The mechanical fitment of these parts has thus been verified and the corresponding pneumatic tubing has been fitted to them in preparation for the assembly of the full system. The inline air filter and regulator has also been connected to the built-in compressed air line that is accessible near the Stäubli TX60 robot arms in the workcell room. A brief test was conducted to verify pneumatic continuity through the filtration system, during which the filter regulator was able to successfully output the desired air pressure without any observable losses.



Figure 48. Mechanical Prototype Assembly of Adhesive Applicator Toolhead

A similar prototype has also been manufactured for the fixture to hold the arm panel during the adhesive application process. The custom components for this section were also manufactured using the FDM 3D printer and then mounted onto a frame assembled from T-slot aluminum beams cut in-house at the RSL. This subassembly was also mounted onto a table near the TX60 to visualize its position within the arm's workspace and to help verify its structural integrity using a mockup of the arm panel. The fixture and the mockup part on it are shown in Figure 49, as they were mounted on the table.



Figure 49. Robot Arm Panel Fixture Prototype on Table with Arm Panel Mockup

Chapter 10: System Integration & Testing

10.1. System Integration Objectives

Once all of the primary subsystems had been assembled, the process to integrate them into a cohesive system began. The main objectives of this stage were the verification of key functionalities between various subsystem combinations and also the physical integration of the MATRIX system with the relevant robotic arms and their peripheral infrastructure. As such, the core intra-subsystem functionalities were integrated and tested in order of primacy to the overall system, starting with the electrical and pneumatic interlock connections and ending with the robots' interactions with MATRIX. The stages of this integration process will thus be covered part by part in the following subsections.

10.2. Electrical and Pneumatic Interlock Connections

As one of the defining mechanical challenges of the modular end effector design, the continuity of the electrical and pneumatic passthroughs across the tool interlock was consequently the first subsystem interaction to be tested. For this purpose, a pared-down version of the base hub unit was assembled and connected directly to the universal mount section to provide the core passthrough functionalities. This testing-focused version of the hub consisted of the 12V air pump, its corresponding relay board, and a 12V battery and a portable charging bank to power these two subunits, respectively. Since this set of tests was conducted at a relatively early iteration of the design's computational architecture, the relay board was used with an Arduino Uno rather than the final Raspberry Pi 5. Figure 50 below shows the base hub unit as it was assembled for this stage of integration testing.

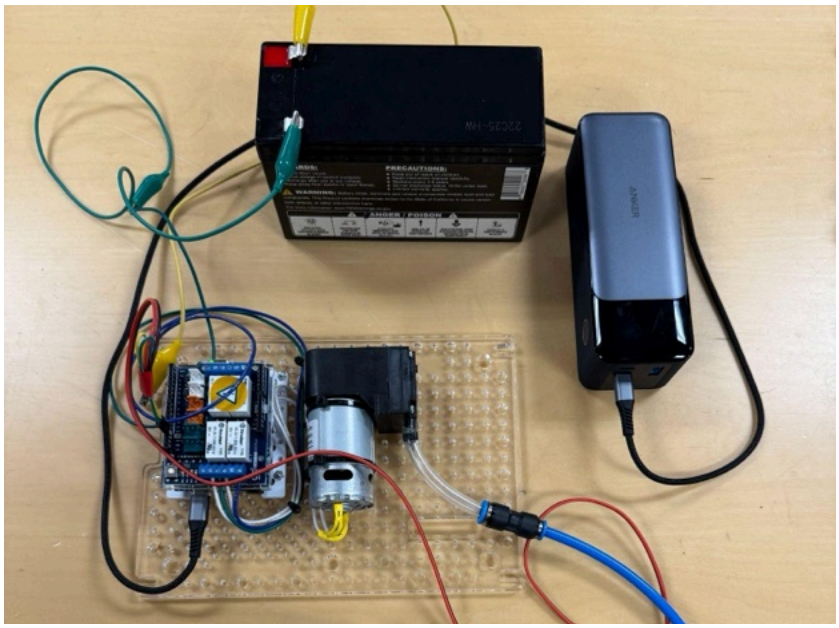


Figure 50. Base Hub Pump Unit Used for Interlock Continuity Testing

In order to evaluate the functionality of the two passthroughs built into the end effector system, the two representative end effector tools were both attached to the universal mount and tested via a set of basic commands sent via the Arduino IDE. As the first of the toolheads to be tested, the electrical parallel gripper's testing process included an initial check of continuity across the five pogo pin output wires via a multimeter prior to the main test. Once the tool's electrical continuity had been verified, the gripper was tested with an open-close command sequence sent to it directly over a wired connection, and demonstrated this functionality successfully. Afterwards, the pneumatic suction gripper was attached to the universal base, and the corresponding passthrough's continuity was verified by turning on the pump and having the tool pick up a large (6" x 10" x ¼") piece of acrylic. These two tests are shown in Figure 51 below, and were instrumental in validating the passthrough integrity of the end effector design.

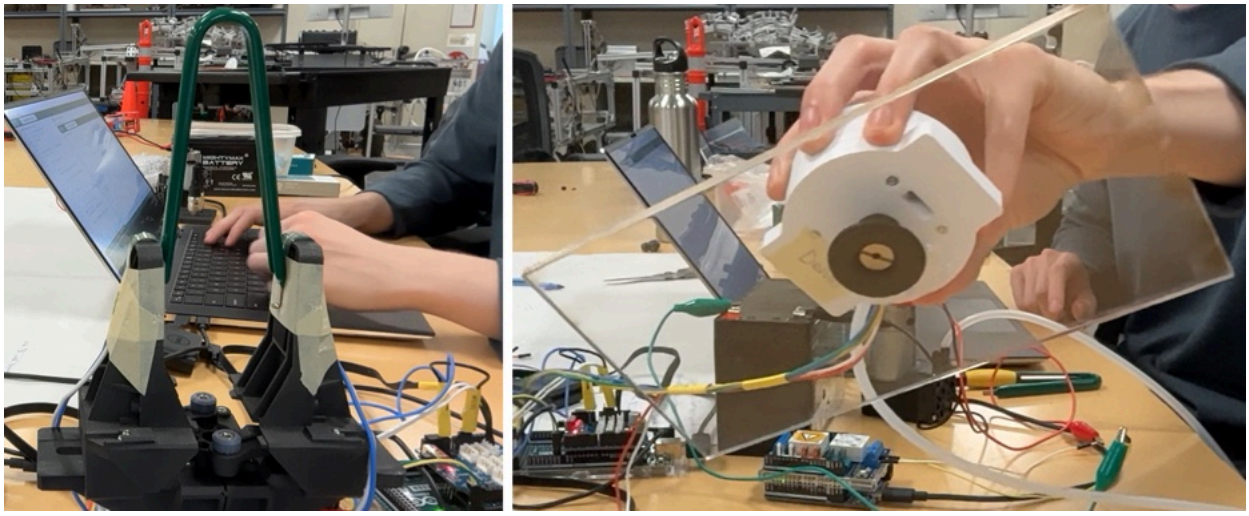


Figure 51. Continuity Testing with Electrical Gripper (Left) and Pneumatic Gripper (Right)

10.3. Remote Gripper Tool Actuation

Since the initial integration tests of the preceding section used a local wired connection to send and receive data across the end effector's two halves, an additional level of testing was later required to ensure that the tools could be controlled remotely via ROS as intended. For this stage, the Arduino computing units of the preceding section were all either replaced or supplemented by the Raspberry Pi Zeros per the final end effector tool designs of Sections 7.3 and 8.3. A laptop computer was also set up with the corresponding ROS nodes to stand in for the role of the workcell's central computer. Once this electrical and software setup had been implemented, the two end effectors were tested to verify that their gripping and data transmission functionalities can be replicated over the ROS pipeline. The command structure from the previous section was used as the basis for the ROS commands used here, which were called from the Linux terminal on the testing laptop. Figure 52 shows the setup for this integration stage, with the end effector tools in the foreground and the laptop behind them in both of the two panels. After some troubleshooting of the serial data lines, these integration tests ultimately proved successful in validating the ROS-based task execution process.

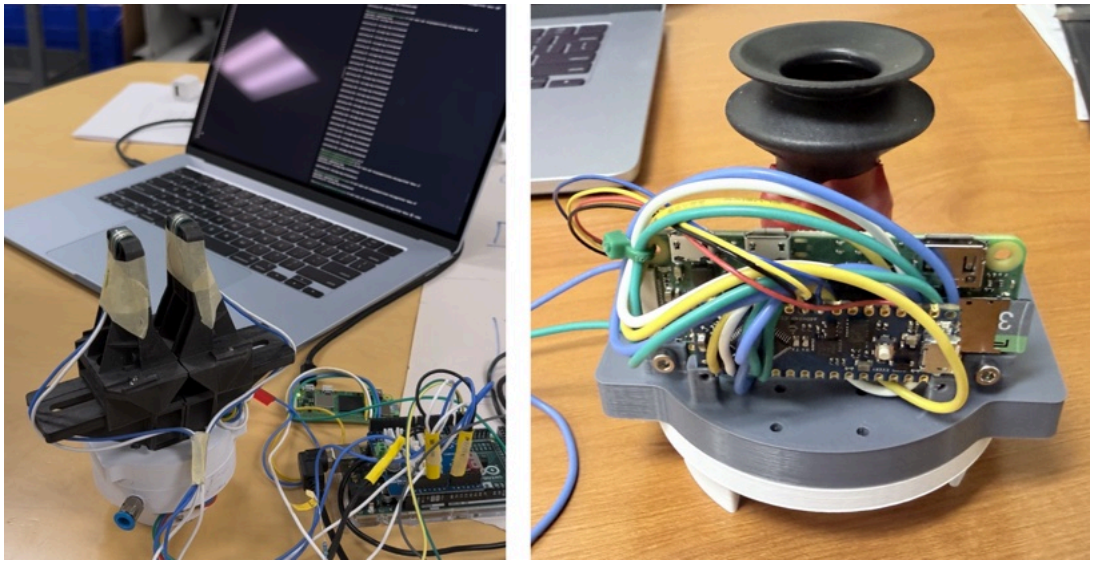


Figure 52. Remote Tool Testing with Electrical Gripper (Left) and Pneumatic Gripper (Right)

10.4. Tool Retrieval Position Training

In parallel with the end effector tool verification process, the integration of the tool depot with the Stäubli TX60 and its robot cart was also implemented. More specifically, the depot was installed on the robot cart and the arm's teach pendant was used to determine the target position waypoints required to complete the tool retrieval process. For this process, one researcher operated the teach pendant to jog the arm's joint positions such that the universal mount at the distal end would align with the tool depot as required at the various sub-steps of the tool-change process. The other researcher, meanwhile, provided visual confirmation of the positional and rotational alignment and logged the angular positions of the arm's six joints at each stage. The teach pendant of the TX60 is shown in Figure 53 for reference, and is typical of the pendants used by modern robot arms for position training.



Figure 53. Stäubli TX60 CS8C Teach Pendant [42]

Once all of the relevant end effector positions had been trained and their corresponding joint positions were logged, the latter set of values was compiled into a script for a tool pickup control sequence. This script–included in Appendices B.4 and B.5–was written in Python and is designed to interface with the TX60's controller via a combined ROS and RoboDK middleware package. The compiled code spanning these levels of the workcell system was then run remotely from the central command computer, with one of the researchers observing the robot from a safe distance to verify the accuracy of the coded positions and also ensure that the resulting trajectories do not cause any collisions. Upon verification of these criteria, the order of the sequence was reversed and the resulting tool drop-off sequence was also validated.

10.5. Robot Arm Passthrough Utilization

The final major integration stage of the MATRIX project involved the fitment of the universal mount to the distal end of the Stäubli TX60 and the verification of the robot arm's internal passthroughs as they interact with the rest of the end effector system. As mentioned in Section 4.3, the universal mount uses a female JST connector as the interface between the MATRIX end effectors and the proprietary electrical connectors used by arms like the TX60. Thus, an adapter cable was manufactured in house using a male JST connector on one end and a male 19-pin proprietary Stäubli connector on the other. A similar cable was also made for the arm's base electrical inlet, completing the five electrical passthrough lines from the power supply to the end effector. Given the use of standard metric threaded holes at the pneumatic ports of the arm, the only steps required to complete the air line were the installation of the appropriate quick-disconnect fittings and the insertion of the corresponding tubing segments. The resulting testing setup with the pneumatic suction gripper on the TX60 is shown in Figure 54, as it was used to verify both passthroughs via the same command sequence as in Section 10.3.

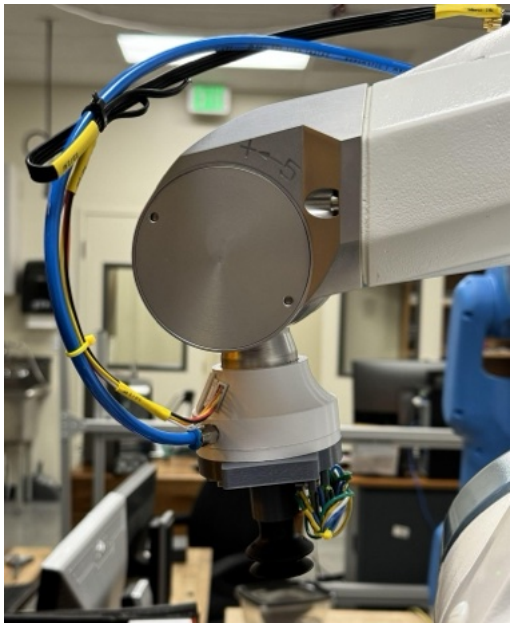


Figure 54. Stäubli TX60 with Passthroughs Connected and Suction Gripper Installed

Chapter 11: System Testing and Discussion

11.1. Testing and Verification Objectives

The final step of the design, manufacturing, and integration testing of the preceding sections involved a series of full system tests that closely mirrored the tasks that MATRIX will be used for in the RSL workcell. This task set was split into the tool changing and object pick-up sub-tasks in order to segment the testing process and thus ensure the direct verification of each key functionality. The following sections will cover these three testing tasks, describing both the procedures and the quantitative results obtained through each test cycle.

11.2. Tool Changing Verification

In order for any end effector tool in the MATRIX system to be functional, it is critical for the mechanical and electrical connections between the robot-side and tool-side sections to first be successfully locked together. Similarly, the tool drop-off sequence must be robust enough for the robot arm involved to remove any given toolhead with full autonomy. As such, these two substeps were tested together prior to any toolhead functionality tests. Based on the tool depot position training done with the Stäubli TX60 in Section 10.4, the tool pick-up sequence was split into four key stages, each with its own waypoint: (1) navigation of the arm to the tool slot, (2) positional alignment of the robot-side section and the toolhead, (3) rotational alignment to lock in the toolhead, and (4) exit from the tool depot. The tool drop-off sequence aligns directly with these four steps as well, albeit with the rotation of the third step being reversed to unlock the toolhead from the robot-side assembly. Figure 55 shows the TX60 approaching the tool depot to pick up a toolhead.

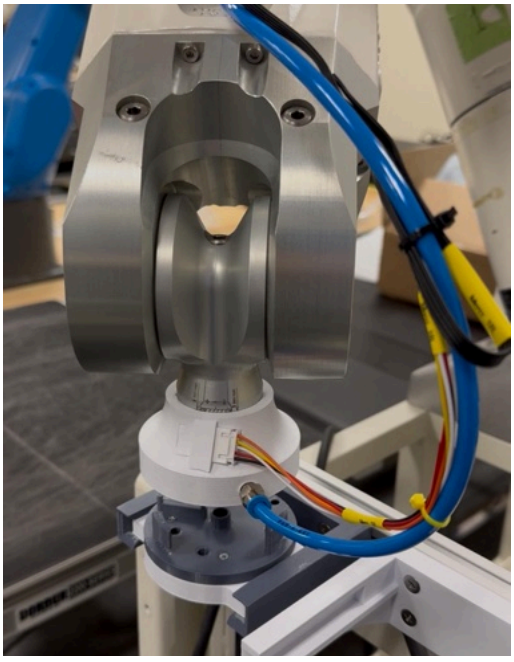


Figure 55. Stäubli TX60 Approaching Depot for Toolhead Pick-Up

The testing methodology for the four stages of the tool changing cycle involved a series of 10 tests for both the pick-up and drop-off operations, as defined via the chain of substeps within them. More specifically, a successful tool change was defined as the robot being able to navigate its distal end to the tool depot, positionally align the tool assembly, lock or unlock the toolhead, and exit the tool depot, all without damaging any involved components. The end effector assembly was observed by one of the operators throughout the entire testing process, and thus any signs of damage were recorded when they first occurred. Table 6 shows the compiled results of this testing series and is followed by a brief analysis.

Table 6. Tool Changing Operation Validation Results

	Success Percentage (out of 10 trials)			
Phase	**Navigation**	**Pos. Alignment**	**Lock/Unlock**	**Exit**
Tool Pick-up	100%	100%	100%	100%
Tool Drop-off	100%	90%	100%	100%

Overall, the results across the testing cycles for both the tool pick-up and tool drop-off sequences were highly successful, with only one error present across the whole test process. In this instance, the flange of the toolhead got stuck on a small ridge of the 3D-printed tool depot, forcing the toolhead to twist in an unnatural manner and consequently snapping off one of the internal prongs. Nonetheless, the tool depot assembly had enough mechanical compliance within its design to tolerate this extra force and eventually allow the flange to snap past the ridge as the arm kept moving into its alignment position. From here, the robot was also able to complete the rest of the tool drop-off process without any interference from the snapped pin, highlighting the overall robustness of the MATRIX system to any potential damage during tool changing. All other tests within the broader tool change test cycle were completed successfully without any damage, providing a strong body of proof for the functionality at hand.

11.3. Tool Usage Verification

Upon the validation of the tool changing functionality in the preceding section, the second half of the full-system testing–namely, tool usage verification–was started. This stage of experiments involved mounting the two gripper tools onto the Stäubli TX60 via the universal mount and replicating the same core functionalities that were tested in Section 10.3. The first of the two grippers to be tested was the pneumatic suction gripper, which was used in a series of tests as a part of the workcell's laser engraving workflow. Within this process, the gripper had to pick up a cork coaster from a stack, place it into the laser cutting enclosure, pick it up again after the engraving was complete, and then move it to a nearby conveyor belt. This cycle was repeated five times, resulting in a total of 10 pick-and-place operations tested with the suction gripper. All

of these operations were successfully completed, helping validate the end-to-end functionality of the suction gripper tool for a relevant workcell task. An example of the TX60 holding a coaster during this testing stage can be seen in Figure 56.

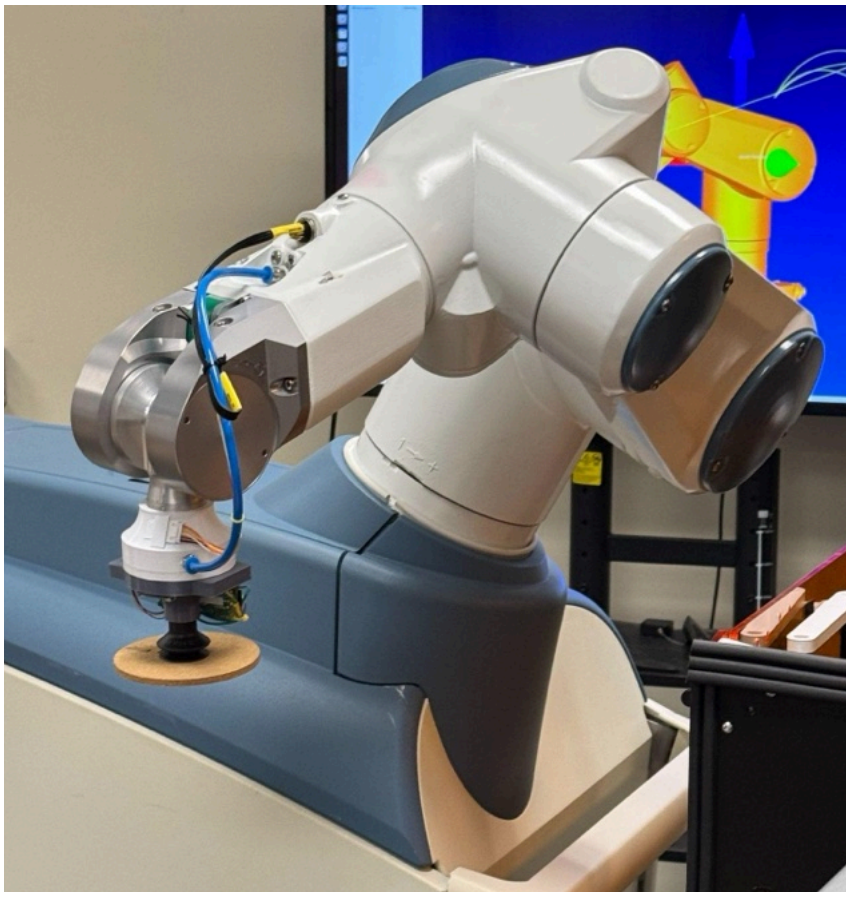


Figure 56. Stäubli TX60 Holding Coaster via Pneumatic Suction Gripper

On the other hand, the electrical parallel gripper was used to pick up a flexible pair of electronics tongs, which were particularly useful since they could accommodate a wide range of gripping force thresholds being tested. Unfortunately, due to power level stability issues with the step-down converter on the OpenRB-150, the electrical parallel gripper was not able to maintain a stable connection to the workcell's local network long enough for in-situ testing on the robot arms. While the source of this issue was eventually identified, the electronics were already fully packaged on the toolhead and thus could not be reconfigured by the time of writing. Thus, for the purpose of this testing stage, the intended functionality was tested with the full system laid out on a table rather than mounted onto the robot. This testing cycle was similarly successful to the preceding ones, with the parallel gripper capable of reaching force thresholds within the range of 1 N to 10 N with a maximum error of 10% relative to any given target value. Figure 57 shows the electrical parallel gripper holding the tongs to a nominal force threshold as described in the testing process above.

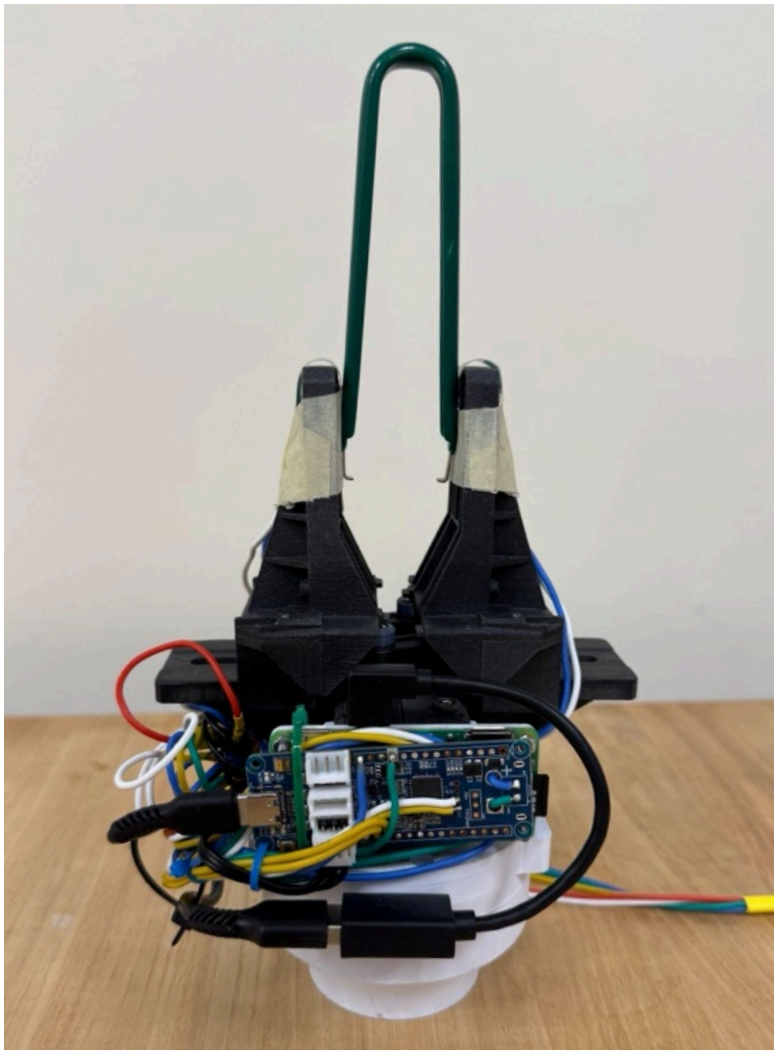


Figure 57. Electrical Parallel Gripper Holding a Pair of Electronics Tongs

11.4. Summary and Future Testing

Recalling the system requirements outlined in Table 3 near the beginning of the project, the completion status of these can now be evaluated objectively. As Table 7 below shows, the vast majority of the requirements have been fulfilled and validated either via the testing above or through other quantitative means. Nonetheless, it is worth noting that the testing conducted thus far provides a preliminary sample space rather than a comprehensive body of proof. As such, there are near-term plans to: (1) test the electrical parallel gripper mounted on the TX60 arm, (2) finalize the ROS node network for fully-automated high-level control, and (3) expand all test segments in the preceding sections to 50 trials each. This additional round of testing will be undertaken in the following academic quarter in preparation for an academic journal publication on the MATRIX end effector portfolio, and will serve to both verify features still currently being revised and expand the validation data for the systems tested thus far.

Table 7. Summary of System Requirement Testing and Outcomes

Requirement	Verification Method	Outcome	Status
Mechanical modularity	Demonstrate by mounting universal base on 3 robot arms	Mounted universal base on 3 robot arms	Completed
Software modularity	Check connection of each tool to ROS node network via a simple test command	Both gripper tools successfully connected; adhesive system not yet connected	Partially completed
Electrical modularity	Demonstrate via different computing configurations	Distal-end and base-end computing modules verified	Completed
Cost less than $800	Verify via the final assembly's BoM	Core MATRIX system costs $598	Completed
Autonomous tool swapping	Test toolhead pick-up and drop-off via repeated trials and inspect to make sure tool remains intact at least 9 times out of 10	Toolhead pick-up verified with 100% success rate; drop-off verified with 90% success rate	Completed
Use standard components	Source components from reputable sellers during design and verify via BoM	All components used are commercially available from major distributors	Completed
Easy to assemble	Assemble each section and use timer to verify	Average section assembly time verified to be around 4 minutes	Completed
Sensor on each tool	Demonstrate sensor data streams on laptop	Force and distance sensor data streams verified	Completed
Sensing informs high-level control	Verify data being received via ROS nodes and the receipt of commands at toolheads	Data and commands sent via ROS nodes, but not yet in a closed high-level control loop	Partially completed

Table 7 (Continued). Summary of System Requirement Testing and Outcomes

Requirement	Verification Method	Outcome	Status
Easy to install	Install the system on a robot 5 times and use timer to verify	Average installation time verified to be around 4 minutes, 30 seconds	Completed
Open source design	Demonstrate via publication or webpage	Publication planned by the end of 2026	Partially completed
Use electrical passthroughs	Set up end-to-end electrical passthrough connection and check continuity with a multimeter	Stäubli TX60 electrical passthrough continuity utilized across 9V, 5V, and ground lines with end-to-end continuity	Completed
Use pneumatic passthroughs	Set up end-to-end pneumatic passthrough connection and check continuity by holding paper to tool-end outlet	Stäubli TX60 pneumatic passthrough used to pick up paper via vacuum pump connected at base	Completed
Mountable near robot	Install tool depot on one instance of each mounting surface	Tool depot installed on robot cart and nearby table	Completed

Chapter 12: Conclusion & Outlook

12.1. Summary

This report is the culmination of over a year's worth of research on modular end effectors and thus reflects all of the engineering that went in developing the final MATRIX end effector system. The project began with a combination of state-of-the-field research and stakeholder discussions with the leadership of the RSL, developed through an iterative design process, and ultimately led to the realization of a modular end effector portfolio that meets most of the key requirements established at its inception. Several advanced features were demonstrated and verified through testing, while the software loops still in development are expected to be completed in the near future. Throughout the project, the interactions between the mechanical, electrical, and software aspects of MATRIX regularly informed interdisciplinary design decisions, unifying an array of OEM components into a set of low-cost, compact modules that can be recombined for a wide range of light industrial task workflows. The development of MATRIX provided numerous insights into the mechatronics systems design process and also served as the basis for a collaboration with a major local industry partner, both of which represent valuable educational opportunities. Overall, the MATRIX project has been successful thus far, becoming a key part of the broader RSL workcell effort and providing a practical set of end effector tools and an extensible architecture for continued research and development.

12.2. Ethical and Societal Implications

Throughout both the MATRIX project's conception and throughout its development, the ethical and societal implications of the proposed design were thoroughly considered. The primary concern in this regard was the potential for the modular end effector system to contribute to the displacement of manufacturing workers through its usage in workcell systems similar to the one it is incorporated into at the RSL. As such, the specific applications of the end effector portfolio were chosen to either be tasks that are already commonly automated or ones that are not optimal for workers to spend their time on. These applications in turn drove the selection of the tools to be designed, with the two grippers being developed for standard pick-and-place tasks and the adhesive applicator for high-precision complex curve tracing. While the self-swapping capability of the MATRIX system does eliminate the need for manual tool changing operations, the automation of the workcell simply shifts the workload to a more centralized interface via its high-level ROS control script.

Beyond the potential drawbacks discussed above, the positive impacts of the MATRIX end effector portfolio were also explored during the project's development. Most notably, the fully 3D-printable design of the system enables it to be manufactured with a significantly lower-waste workflow compared to the traditional subtractive techniques used to fabricate most end effector tools. This manufacturing approach is particularly powerful when paired with the decision to open-source the MATRIX design, enabling everyone from hobbyists to researchers to both design their own end effector tools and manufacture them in-house alongside the universal

parts of the assembly. Through this synergy, members of the mechatronics community around the world can access a robust platform for end effector development and can also further contribute to a growing set of tools that exist within the MATRIX ecosystem.

In the spirit of Santa Clara University's focus on engaging external partners with the school, the MATRIX project also involves an ongoing collaboration with nearby company Lam Research. The adhesive applicator system being developed by the RSL for Lam helps showcase the many benefits of this approach to the local engineering environment, solving a relevant need in industry while getting graduate students involved in the applied engineering process. Upon its completion, the adhesive applicator tool will help boost productivity in Lam Research's robotics manufacturing operations, helping deliver a high-quality product while allowing workers to shift their focus to more human-optimal tasks.

12.3. Outlook

Looking at the future of the MATRIX project, there are numerous innovative additions planned for continued development, and thus many reasons for optimism about the project's staying power as a flagship of the RSL. The current team involved in MATRIX and the broader RSL workcell intends to continue work at the university for at least the next academic quarter as the end effector portfolio is further expanded and a journal paper is written. One of the main goals of such a publication is to help increase the reach of the project and thereby inspire others outside of SCU to experiment with or adopt its open-source architecture. During this upcoming quarter, extended testing will also be conducted on the MATRIX system in order to more accurately quantify its capabilities and identify any potential points for improvement. The industry collaboration with Lam Research will also be continued over this timeframe, with a two-part adhesive dispensing system already in early development to complement the single-part applicator tooling built thus far.

Beyond the current academic year, new student members of the RSL are planned to join the project and continue expanding the available end effector portfolio as a part of their own capstone projects. The modular design of MATRIX is particularly powerful in this regard, enabling students to easily undertake individual projects that also contribute to a broader integrated system with a long-term development plan. Similarly, there is an initiative within the RSL to introduce a new manufacturing engineering course where students will work hands-on with the workcell and MATRIX ecosystem to develop a variety of efficient light industrial workflows. Through these plans, the next few years of research could see the MATRIX end effector portfolio grow to over 15 hot-swappable toolheads and serve both the RSL and broader SCU engineering community as a touchstone for teaching, in-house product development, and local industry collaborations.

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Appendix A: Key Calculations and Derivations

A.1. Electrical Parallel Gripper Kinematics

The mechanism moving the two fingers of the electrical parallel gripper was split into two halves, both of which were modeled as identical slider-crank linkages. Due to the mechanical configuration of the gripper, the general linkage kinematic structure was simplified down such that the slider axis is horizontally coincident with the crank's pivot point. Euler's formula was applied to the geometric equation of the linkage, which was in turn rearranged to express the slider's position in terms of the crank's pivot angle and the lengths of the two non-zero linkages. These parameters were then plugged into the kinematic equation, and the slider distance was offset based on the geometry of the gripper. The offset distance was then multiplied by 2 to reflect the mirrored displacement of the two halves of the gripper. Figure A1.1 shows the kinematic model and the associated derivation described above.

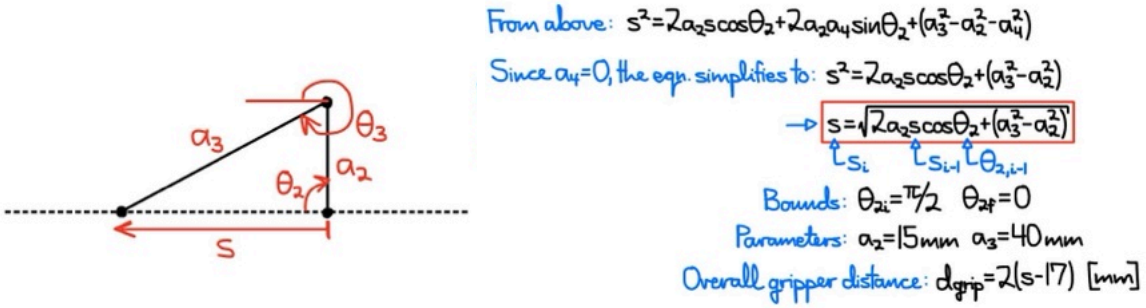


Figure A.1.1. Electrical Parallel Gripper Distance Kinematic Derivation

The resulting equation from this derivation is shown below, and was used as the basis of the rotation-to-gripper distance conversion plot in Figure 33.

$$d_{grip} = 2 * [\sqrt{2a_2 s * cos(\theta_2) + (a_3^2 - a_2^2)} - d_{offset}]$$

A.2. Pneumatic Suction Gripper Force Limit

A simple pressure calculation was performed using the known dimensions of the suction cup and the available pressure from the pump unit. The diameter of the suction cup was first used to calculate the surface area of its distal contact plane, which was then divided by the maximum vacuum pressure the pump can provide to get the maximum suction force–and thus the maximum liftable mass–of the suction gripper, assuming no losses.

$$A_{contact} = \pi r^2 = \pi * (0.036\,m/2)^2 = 0.001018\,m^2$$

$$F_{max} = P_{v,max} * A_{contact} = (47\,kPa) * (0.001018\,m^2) = 47.84\,N$$

$$m_{max} = F_{max}/g = (47.84\,N)/(9.81\,m/s^2) = 4.87\,kg \approx 5\,kg$$

Appendix B: Arduino and ROS Codes

B.1. Electrical Parallel Gripper Arduino Code

```
#include <math.h>
#include <Dynamixel2Arduino.h>

// Please modify it to suit your hardware.
#if defined(ARDUINO_AVR_UNO) || defined(ARDUINO_AVR_MEGA2560) // When using
DynamixelShield
  #include <SoftwareSerial.h>
  SoftwareSerial soft_serial(7, 8); // DYNAMIXELShield UART RX/TX
  #define DXL_SERIAL   Serial
  #define DEBUG_SERIAL Serial1
  const int DXL_DIR_PIN = 2; // DYNAMIXEL Shield DIR PIN
#elif defined(ARDUINO_OpenRB)  // When using OpenRB-150
  #define DXL_SERIAL   Serial1
  #define DEBUG_SERIAL Serial
  const int DXL_DIR_PIN = -1;
#else // Other boards when using DynamixelShield
  #define DXL_SERIAL   Serial2
  #define DEBUG_SERIAL Serial
  const int DXL_DIR_PIN = 2;
#endif

// Dynamixel motor & protocol version defines
const uint8_t DXL_ID = 1;
const float DXL_PROTOCOL_VERSION = 2.0;
Dynamixel2Arduino dxl(DXL_SERIAL, DXL_DIR_PIN);

// This namespace is required to use Control table item names
using namespace ControlTableItem;

// FSR pins (use A0/A1)
int fsrPin1 = A0;
int fsrPin2 = A1;

// PWM and position defines
float pwm = 0;
int count = 0;
float posDeg_raw = 0;

// Offset and error defines
float offset = 0.0f;   // <-- IMPORTANT: float offset to avoid 0/1 jitter from
truncation
int e = 0;
```

```
// Open/close (cmd_oc) & force limit (cmd_force) defines
// Set cmd_oc=2 to idle on startup (no motion) until a command arrives.
int cmd_oc = 2;   // 1=open, 0=close, 2=idle/stop
int cmd_force = 0;

// ---- FSR read struct ----
struct FSRReadings {
  int r1;
  int r2;
  int avg;
};

FSRReadings get_fsr_values() {
  FSRReadings data;
  data.r1 = analogRead(fsrPin1);
  data.r2 = analogRead(fsrPin2);
  data.avg = (data.r1 + data.r2) / 2;
  return data;
}

// Get relative position (rounded consistently)
int get_rel_pos()
{
  float rel = dxl.getPresentPosition(DXL_ID, UNIT_DEGREE) - offset;
  return (int)lroundf(rel); // rounded, not truncated
}

// Gripper command function (split free-close vs force-control, includes hard-stop
stall protect)
void gripper_cmd(int x = 1, int gripping_f = 0, int fsr_avg = 0)
{
  // Close-mode state (latched)
  static int in_force_control = 0;
  static int fsr_baseline = 0;
  static int baseline_set = 0;

  // Stall detection memory (for "no object, hard stop" case)
  static int last_pos = 0;
  static int stall_count = 0;

  int pos = get_rel_pos();

  if (x == 1) // open
  {
    // Reset close-mode state when opening
    in_force_control = 0;
```

```
    baseline_set = 0;
    stall_count = 0;
    last_pos = pos;

    if (pos <= 100)  // opens up to 100 degrees
    {
      e = gripping_f - fsr_avg;
      pwm = 30;
      dxl.setGoalPWM(DXL_ID, pwm, UNIT_PERCENT);
    }
    else
    {
      dxl.setGoalPWM(DXL_ID, 0, UNIT_PERCENT);
      pwm = 0;
    }
  }

  else if (x == 0) // close
  {
    // HARD SAFETY: treat <=1 deg as "closed" to kill the startup closure issue from
0/1 jitter
    if (pos <= 1)
    {
      dxl.setGoalPWM(DXL_ID, 0, UNIT_PERCENT);
      pwm = 0;
      e = 0;

      in_force_control = 0;
      baseline_set = 0;
      stall_count = 0;
      last_pos = pos;
      return;
    }

    // Convert force command to a usable PWM scale (ensure not too weak)
    int x_cmd = int(float(gripping_f) / 10.0);
    if (x_cmd < 10) x_cmd = 10;
    if (x_cmd > 80) x_cmd = 80;

    // Baseline once per closing episode (helps avoid false contact due to preload)
    if (!baseline_set)
    {
      fsr_baseline = fsr_avg;
      baseline_set = 1;
      in_force_control = 0;

      last_pos = pos;
```

```
      stall_count = 0;
    }

    // Contact detection derived from command magnitude
    int contact_thresh = fsr_baseline + ((x_cmd > 15) ? x_cmd : 15);
    if (fsr_avg >= contact_thresh) in_force_control = 1;

    // If no force target, treat as pure close
    if (gripping_f <= 0) in_force_control = 0;

    int err = gripping_f - fsr_avg;

    if (!in_force_control)
    {
      // Free-close: push closed decisively
      pwm = -max(30, x_cmd);

      // Stall protect: while closing, pos should DECREASE.
      // If it stops decreasing for a short time and still no contact, assume hard
stop => stop PWM.
      const int STALL_CYCLES = 25; // ~250ms at 10ms loop
      if (pos >= last_pos) stall_count++;
      else stall_count = 0;

      if (stall_count >= STALL_CYCLES)
      {
        pwm = 0;
        in_force_control = 0;
        baseline_set = 0; // next close re-baselines
        stall_count = 0;
      }
    }
    else
    {
      // Force mode: regulate about setpoint; loosens if setpoint is lowered below
current force
      int deadband = (x_cmd > 10) ? x_cmd : 10;

      if (abs(err) <= deadband)
      {
        pwm = 0;
      }
      else
      {
        // err>0 => close (negative), err<0 => open (positive)
        pwm = -(x_cmd + (err / 2));
      }
```

```
      // Clamp to allowed PWM range
      if (pwm >  100) pwm =  100;
      if (pwm < -100) pwm = -100;

      stall_count = 0; // ignore stall logic in force mode
    }

    e = err;
    dxl.setGoalPWM(DXL_ID, pwm, UNIT_PERCENT);
    last_pos = pos;
  }

  else // idle/stop
  {
    dxl.setGoalPWM(DXL_ID, 0, UNIT_PERCENT);
    pwm = 0;
    e = 0;
  }
}

void setup()
{
  // Debug serial (optional) - OpenRB-150 uses Serial over USB to PC
  DEBUG_SERIAL.begin(115200);
  while(!DEBUG_SERIAL) {}

  // Dynamixel setup
  dxl.begin(57600);                        // <-- set this to match Wizard if needed
  dxl.setPortProtocolVersion(DXL_PROTOCOL_VERSION);
  dxl.ping(DXL_ID);

  dxl.torqueOff(DXL_ID);
  dxl.setOperatingMode(DXL_ID, OP_PWM);
  dxl.torqueOn(DXL_ID);

  // Stream/command channel: OpenRB-150 does NOT have Serial3; use USB Serial instead.
  // (Bare-minimum change: replace Serial3 with DEBUG_SERIAL.)
  // Capture initial offset (float)
  offset = dxl.getPresentPosition(DXL_ID, UNIT_DEGREE);
}

void loop()
{
  // Read command if available (format: cmd_oc,cmd_force like "0,400")
  if (DEBUG_SERIAL.available())
  {
```

```
    cmd_oc = DEBUG_SERIAL.parseInt();
    cmd_force = DEBUG_SERIAL.parseInt();
  }

  // Sense once
  FSRReadings fsr = get_fsr_values();
  int rel_pos = get_rel_pos();
  posDeg_raw = dxl.getPresentPosition(DXL_ID, UNIT_DEGREE);

  // Actuate
  gripper_cmd(cmd_oc, cmd_force, fsr.avg);

  // Print telemetry
  DEBUG_SERIAL.print("Count: ");
  DEBUG_SERIAL.print(count);

  DEBUG_SERIAL.print("| Raw Position: ");
  DEBUG_SERIAL.print(rel_pos);

  DEBUG_SERIAL.print("| FSR 1: ");
  DEBUG_SERIAL.print(fsr.r1);
  DEBUG_SERIAL.print("| FSR 2: ");
  DEBUG_SERIAL.print(fsr.r2);
  DEBUG_SERIAL.print("| Avg:");
  DEBUG_SERIAL.print(fsr.avg);

  DEBUG_SERIAL.print("| cmd_oc:");
  DEBUG_SERIAL.print(cmd_oc);
  DEBUG_SERIAL.print("| cmd_force:");
  DEBUG_SERIAL.print(cmd_force);

  DEBUG_SERIAL.print("| PWM: ");
  DEBUG_SERIAL.print(pwm);
  DEBUG_SERIAL.print("| Error: ");
  DEBUG_SERIAL.print(e);
  DEBUG_SERIAL.print("\n");

  delay(10);
  count = count + 1;
}
```

B.2. Electrical Parallel Gripper ROS Tool-Side Code

```
import rclpy
from rclpy.node import Node

from std_msgs.msg import Int16
```

```
import serial
arduino_obj = serial.Serial(
port='/dev/ttyS0',
baudrate=57600,
parity=serial.PARITY_NONE,
stopbits=serial.STOPBITS_ONE,
bytesize=serial.EIGHTBITS,
timeout=1)

class motor_driver(Node):

    def __init__(self):
        super().__init__('gripper_node')

        self.grp_state = 7
        self.grp_des_force = 0

        self.subscription = self.create_subscription(
            Int16,
            'ee/gripper1/state',
            self.cmd_callback,
            5)
        self.subscription  # prevent unused variable warning

        self.sub_grp_force = self.create_subscription(
            Int16,
            'ee/gripper1/desiered_force',
            self.cmd_callback_des_force,
            5)
        self.sub_grp_force  # prevent unused variable warning

    def move_motors(self):
        payload1 = str(int(self.grp_state)) + "," + str(int(self.grp_des_force))
        #print(payload1)
        arduino_obj.write(str.encode(payload1))

    def cmd_callback(self, msg):    # master serial_write
        self.grp_state = msg.data
        self.move_motors()

    def cmd_callback_des_force(self, msg):
        self.grp_des_force = msg.data

def main(args=None):
    rclpy.init(args=args)
```

```
    gripper_subscriber = motor_driver()
    rclpy.spin(gripper_subscriber)
    gripper_subscriber.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```

B.3. Pneumatic Suction Gripper ROS Tool-Side Code

```
import rclpy
from rclpy.node import Node
from std_msgs.msg import Int16, Float32
from rclpy.qos import QoSProfile, QoSReliabilityPolicy, QoSHistoryPolicy
import serial
import re

_num = re.compile(r'^[-+]?\d+(\.\d+)?$')

class motor_driver(Node):
    def __init__(self):
        super().__init__('ee_pneumatic_node')

        self.grp_state = 7
        self.grp_des_force = 0

        # Move serial object into class
        try:
            self.arduino_obj = serial.Serial(
                port='/dev/ttyUSB0',
                baudrate=57600,
                parity=serial.PARITY_NONE,
                stopbits=serial.STOPBITS_ONE,
                bytesize=serial.EIGHTBITS,
                timeout=1
            )
        except Exception as e:
            self.get_logger().error(f"Failed to open serial: {e}")
            self.arduino_obj = None

        qos_profile = QoSProfile(
            reliability=QoSReliabilityPolicy.BEST_EFFORT,
            history=QoSHistoryPolicy.KEEP_LAST,
            depth=1
        )

        self.subscription = self.create_subscription(
```

```
        Int16,
        'ee/pne/gripper1/state',
        self.cmd_callback,
        5)

    self.distance_pub = self.create_publisher(
        Float32,
        'ee/pne/ee_distance',
        qos_profile)

    self.timer = self.create_timer(0.12, self.read_distance)

def read_distance(self):
    if self.arduino_obj is None:
        return

    try:
        waiting = self.arduino_obj.in_waiting

        if waiting > 100:
            self.arduino_obj.reset_input_buffer()
            return

        if waiting > 0:
            raw = self.arduino_obj.readline()
            text = raw.decode('utf-8', errors='ignore').strip()

            if not text:
                return

            # FIX: Only return if invalid, otherwise continue to publish
            if not _num.match(text):
                self.get_logger().warn(f"Invalid serial data: {text!r}")
                return  # ← Only return if invalid

            # Now this code will execute for valid data
            distance = float(text)

            msg1 = Float32()
            msg1.data = round(distance/100, 2)
            self.distance_pub.publish(msg1)

    except (OSError, IOError) as e:
        self.get_logger().error(f"Serial I/O error: {e}")
        self.arduino_obj = None
    except ValueError as e:
        self.get_logger().warn(f"Value conversion error: {e}")
```

```
        except Exception as e:
            self.get_logger().error(f"Unexpected error: {e}")

    def move_motors(self):
        if self.arduino_obj is None:
            return

        try:
            payload1 = str(int(self.grp_state)) + ",\n"
            print(f"Sending: {payload1}")
            self.arduino_obj.write(str.encode(payload1))
        except Exception as e:
            self.get_logger().error(f"Write error: {e}")

    def cmd_callback(self, msg):
        self.grp_state = msg.data
        self.move_motors()

    def cmd_callback_des_force(self, msg):
        self.grp_des_force = msg.data

def main(args=None):
    rclpy.init(args=args)
    gripper_subscriber = motor_driver()

    try:
        rclpy.spin(gripper_subscriber)
    except KeyboardInterrupt:
        pass
    finally:
        gripper_subscriber.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()
```

B.4. Tool Pick-Up Operation Code

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from geometry_msgs.msg import PoseStamped
from std_msgs.msg import Float64, String, Bool
from zed_aruco.msg import ArucoPoseArray
import numpy as np
import math
```

```
def make_pose(x, y, z, r, p, ya):
    pose = PoseStamped()
    pose.header.frame_id = "base_link"
    pose.pose.position.x = x
    pose.pose.position.y = y
    pose.pose.position.z = z
    pose.pose.orientation.x = r
    pose.pose.orientation.y = p
    pose.pose.orientation.z = ya
    pose.pose.orientation.w = ya
    return pose

def pose_check(goal_pose, current_pose, pos_tolerance=0.02, ang_tolerance=0.02):
    gx, gy, gz, gr, gp, gya = goal_pose.position.x, goal_pose.position.y,
goal_pose.position.z, goal_pose.orientation.x, goal_pose.orientation.y,
goal_pose.orientation.z
    cx, cy, cz, cr, cp, cya = current_pose.position.x, current_pose.position.y,
current_pose.position.z, current_pose.orientation.x, current_pose.orientation.y,
current_pose.orientation.z
    pose_diff = np.linalg.norm([gx - cx, gy - cy, gz - cz])  # calculate difference
    return pose_diff <= pos_tolerance

class Staub1Ctrl(Node):
    def __init__(self):
        super().__init__('staub1ctrl_node')
        self.ctrl_pub = self.create_publisher(PoseStamped, "/pose_publish_topic", 10)
        self.grip_cmd = self.create_publisher(Bool, "grip_cmd", 10)
        # self.subscription = self.create_subscription(ArucoPoseArray, "/pose_data",
self.marker_callback, 10)
        self.robot_pose = None
        self.stage = 'initial'
        self.init_pose_sub = self.create_subscription(
            PoseStamped,
            '/pose_recived_topic',
            self.pose_callback,
            10
        )
        self.home = make_pose(353.836, 0.547, 143.00, -179.264, 0.677, -43.383)
        #self.pick = make_pose(550.734, 20.253, -196.528, -179.66, 31.872, -46.010)
        self.place = make_pose(17.707, 434.738, -386.606, 179.136, 29.876, -155.542)
        self.pick = make_pose(2.850, 435.128, -360.746, 179.136, 29.876, -154.706)
        #self.placeh = make_pose(463.408, -217.027, 186.273, 174.652, 22.828, -16.727)
        self.homeh = make_pose(17.707, 434.738, -386.606, 179.136, 29.876, -133.100)
        self.retra = make_pose(18.273, 491.543, -387.135, 179.137, 29.876, -132.355)
        self.mov = make_pose(18.273, 491.543, -360.746, 179.137, 29.876, -132.355)
        #self.place = make_pose(516.052, 237.310, 173.111, -166.368, 53.451, -53.943)
```

```
    def pose_callback(self, msg):
        self.robot_pose = msg.pose
        grip_msg = Bool()

        if self.stage == 'initial':
            self.ctrl_pub.publish(self.home)
            self.get_logger().info("Moving to home.")
            self.stage = 'moving_to_home'

        elif self.stage == 'moving_to_home' and pose_check(self.home.pose,
self.robot_pose):
            self.get_logger().info("Reached home. Moving to pick.")
            self.ctrl_pub.publish(self.pick)
            self.stage = 'moving_to_zone1'

      #  elif self.stage == 'moving_to_pick' and pose_check(self.pick.pose,
self.robot_pose):
      #      self.get_logger().info("Reached pick position. moving to home")
      #      self.ctrl_pub.publish(self.home)

      #      self.stage = 'moving_to_place'

        elif self.stage == 'moving_to_zone1' and pose_check(self.pick.pose,
self.robot_pose):
            #grip_msg.data = True
            #self.grip_cmd.publish(grip_msg)
            #self.get_logger().info("moving to safe.")
            self.ctrl_pub.publish(self.place)
            self.stage = 'moving_to_zone2'
        elif self.stage == 'moving_to_zone2' and pose_check(self.place.pose,
self.robot_pose):
        #    grip_msg.data = True
        #    self.grip_cmd.publish(grip_msg)
        #    self.get_logger().info("moving to place.")
            self.ctrl_pub.publish(self.homeh)
            self.stage = 'done_one'

        elif self.stage == 'done_one' and pose_check(self.homeh.pose,
self.robot_pose):
        #    grip_msg.data = False
            self.grip_cmd.publish(grip_msg)
            self.ctrl_pub.publish(self.retra)
            self.stage = 'done'

        elif self.stage == 'done' and pose_check(self.retra.pose, self.robot_pose):
            grip_msg.data = False
            self.grip_cmd.publish(grip_msg)
```

```
            self.ctrl_pub.publish(self.mov)
            self.stage = 'initial'

def main(args=None):
    rclpy.init(args=args)
    node = Staub1Ctrl()
    rclpy.spin(node)
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```

B.5. Tool Drop-Off Operation Code

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from geometry_msgs.msg import PoseStamped
from std_msgs.msg import Float64, String, Bool
from zed_aruco.msg import ArucoPoseArray
import numpy as np
import math

def make_pose(x, y, z, r, p, ya):
    pose = PoseStamped()
    pose.header.frame_id = "base_link"
    pose.pose.position.x = x
    pose.pose.position.y = y
    pose.pose.position.z = z
    pose.pose.orientation.x = r
    pose.pose.orientation.y = p
    pose.pose.orientation.z = ya
    pose.pose.orientation.w = ya
    return pose

def pose_check(goal_pose, current_pose, pos_tolerance=0.02, ang_tolerance=0.02):
    gx, gy, gz, gr, gp, gya = goal_pose.position.x, goal_pose.position.y,
goal_pose.position.z, goal_pose.orientation.x, goal_pose.orientation.y,
goal_pose.orientation.z
    cx, cy, cz, cr, cp, cya = current_pose.position.x, current_pose.position.y,
current_pose.position.z, current_pose.orientation.x, current_pose.orientation.y,
current_pose.orientation.z
    pose_diff = np.linalg.norm([gx - cx, gy - cy, gz - cz])  # calculate difference
    return pose_diff <= pos_tolerance

class Staub1Ctrl(Node):
    def __init__(self):
```

```
        super().__init__('staub1ctrl_node')
        self.ctrl_pub = self.create_publisher(PoseStamped, "/pose_publish_topic", 10)
        self.grip_cmd = self.create_publisher(Bool, "grip_cmd", 10)
        # self.subscription = self.create_subscription(ArucoPoseArray, "/pose_data",
self.marker_callback, 10)
        self.robot_pose = None
        self.stage = 'initial'
        self.init_pose_sub = self.create_subscription(
            PoseStamped,
            '/pose_recived_topic',
            self.pose_callback,
            10
        )
        self.home = make_pose(353.836, 0.547, 143.00, -179.264, 0.677, -43.383)
        #self.pick = make_pose(550.734, 20.253, -196.528, -179.66, 31.872, -46.010)
        self.placeh = make_pose(-0.527, 435.218, -354.871, 179.136, 29.876, -158.200)
        self.place = make_pose(17.707, 434.738, -386.606, 179.136, 29.876, -155.542)

        self.pick = make_pose(2.850, 435.128, -360.746, 179.136, 29.876, -154.706)
        #self.placeh = make_pose(463.408, -217.027, 186.273, 174.652, 22.828, -16.727)
        self.homeh = make_pose(17.707, 434.738, -384.606, 179.136, 29.876, -137.100)
        self.retra = make_pose(17.271, 491.570, -384.865, 179.137, 29.876, -137.000)
        self.mov = make_pose(18.273, 491.543, -360.746, 179.137, 29.876, -132.355)
        #self.place = make_pose(516.052, 237.310, 173.111, -166.368, 53.451, -53.943)

    def pose_callback(self, msg):
        self.robot_pose = msg.pose
        grip_msg = Bool()

        if self.stage == 'initial':
            self.ctrl_pub.publish(self.home)
            self.get_logger().info("Moving to home.")
            self.stage = 'moving_to_home'

        elif self.stage == 'moving_to_home' and pose_check(self.home.pose,
self.robot_pose):
            self.get_logger().info("Reached home. Moving to mov.")
            self.ctrl_pub.publish(self.mov)
            self.stage = 'moving_to_mov'



      #  elif self.stage == 'moving_to_pick' and pose_check(self.pick.pose,
self.robot_pose):
      #      self.get_logger().info("Reached pick position. moving to home")
      #      self.ctrl_pub.publish(self.home)
```

```
      #      self.stage = 'moving_to_place'

        elif self.stage == 'moving_to_mov' and pose_check(self.mov.pose,
self.robot_pose):
            #grip_msg.data = True
            #self.grip_cmd.publish(grip_msg)
            self.get_logger().info("moving to retra.")
            self.ctrl_pub.publish(self.retra)
            self.stage = 'moving_to_depot'
        elif self.stage == 'moving_to_depot' and pose_check(self.retra.pose,
self.robot_pose):
        #    grip_msg.data = True
        #    self.grip_cmd.publish(grip_msg)
            self.get_logger().info("moving to depot.")
            self.ctrl_pub.publish(self.homeh)
            self.stage = 'done_one'

        elif self.stage == 'done_one' and pose_check(self.homeh.pose,
self.robot_pose):
        #    grip_msg.data = False
            #self.grip_cmd.publish(grip_msg)
            self.get_logger().info("moving to place.")
            self.ctrl_pub.publish(self.place)
            self.stage = 'done'

        elif self.stage == 'done' and pose_check(self.place.pose, self.robot_pose):
            grip_msg.data = False
            self.grip_cmd.publish(grip_msg)
            self.get_logger().info("moving to dep.")
            self.ctrl_pub.publish(self.placeh)
            self.stage = 'up'

        elif self.stage == 'up' and pose_check(self.placeh.pose, self.robot_pose):
            self.stage = 'initial'

def main(args=None):
    rclpy.init(args=args)
    node = Staub1Ctrl()
    rclpy.spin(node)
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```

Appendix C: Component Technical Documentation

C.1. Dynamixel XL430-W250-T Documentation

The three most pertinent datasheets for the Dynamixel XL430-W250-T motor used in the electrical parallel gripper, namely the table of specifications, the isometric design drawings and the performance curves, are shown in Table C.1.1 and Figures C.1.2-C.1.3, respectively.

Table C.1.1. Dynamixel XL430-W250-T Key Specifications [43]

Item	Specifications
MCU	ARM CORTEX-M3 (72 [MHz], 32Bit)
Position Sensor	Contactless absolute encoder (12Bit, 360 [°]) Maker : ams(www.ams.com), Part No : AS5601
Motor	Cored
Baud Rate	9,600 [bps] ~ 4.5 [Mbps]
Control Algorithm	PID control
Resolution	4096 [pulse/rev]
Operating Modes	Velocity Control Mode Position Control Mode (0 ~ 360 [°]) Extended Position Control Mode (Multi-turn) PWM Control Mode (Voltage Control Mode)
Weight	57.2 [g]
Dimensions (W x H x D)	28.5 x 46.5 x 34 [mm]
Gear Ratio	258.5 : 1
Stall Torque	1.0 [N.m] (at 9.0 [V], 1.0 [A], 1.000 [Nm/A]) **1.4 [N.m] (at 11.1 [V], 1.3 [A], 1.077 [Nm/A])** 1.5 [N.m] (at 12.0 [V], 1.4 [A], 1.071 [Nm/A])
No Load Speed	47 [rev/min] (at 9.0 [V]) **57 [rev/min] (at 11.1 [V])** 61 [rev/min] (at 12.0 [V])
Operating Temperature	-5 ~ +72 [°C]
Input Voltage	6.5 ~ 12.0 [V] (**Recommended : 11.1 [V]**) (lithium-ion battery 3 Cell)
Command Signal	Digital Packet
Physical Connection	TTL Multidrop Bus (5V Logic) TTL Half Duplex Asynchronous Serial Communication (8bit, 1stop, No Parity)
ID	253 ID (0 ~ 252)
Feedback	Position, Velocity, Load, Realtime tick, Trajectory, Temperature, Input Voltage, etc
Case Material	Engineering Plastic
Gear Material	Engineering Plastic
Standby Current	52 [mA]

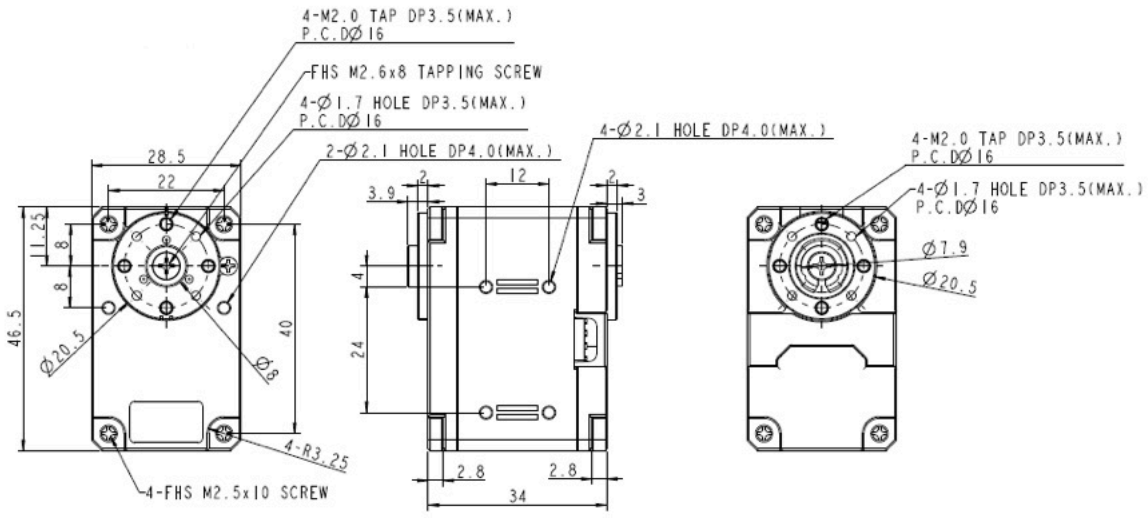


Figure C.1.1. Dynamixel XL430-W250-T Design Drawing [43]

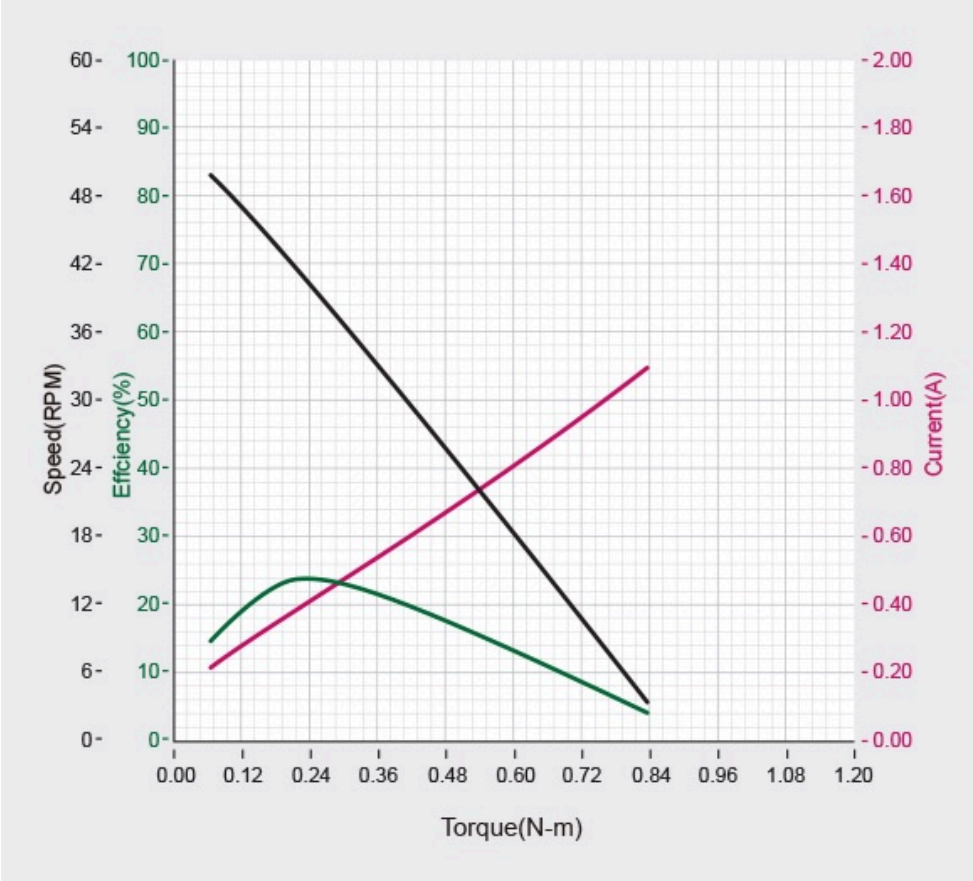


Figure C.1.2. Dynamixel XL430-W250-T Performance Curves [43]

C.2. Robotis OpenRB-150 Pinout

Figure C.2.1 shows the electrical pinout for the OpenRB-150 motor control board with Arduino compatibility, which is used to control the Dynamixel motor as a part of the electrical parallel gripper assembly.

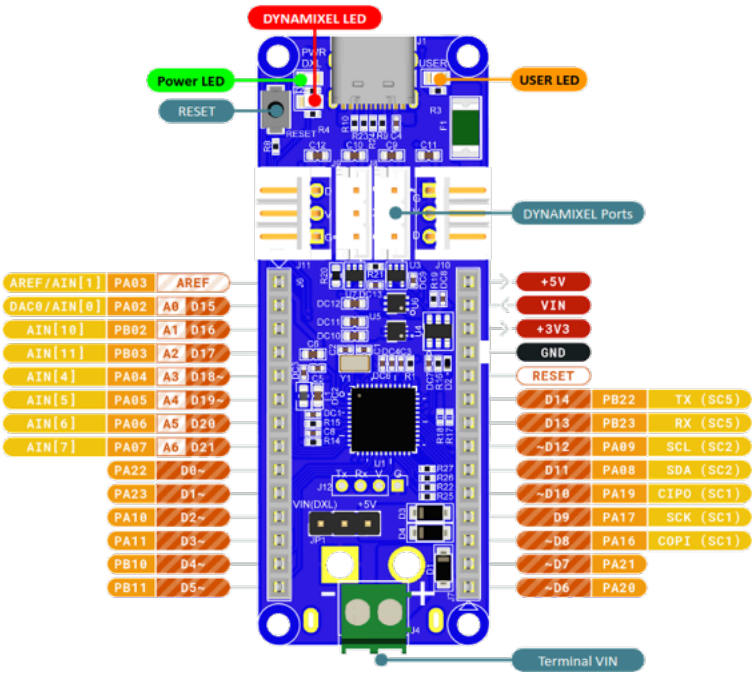


Figure C.2.1. OpenRB-150 Motor Control Board Pinout [36]

C.3. Raspberry Pi Zero 2 W Pinout

Figure C.3.1 shows the electrical pinout for the Raspberry Pi Zero 2 W microcontroller, which is used as the onboard ROS node for both of the gripper toolheads.

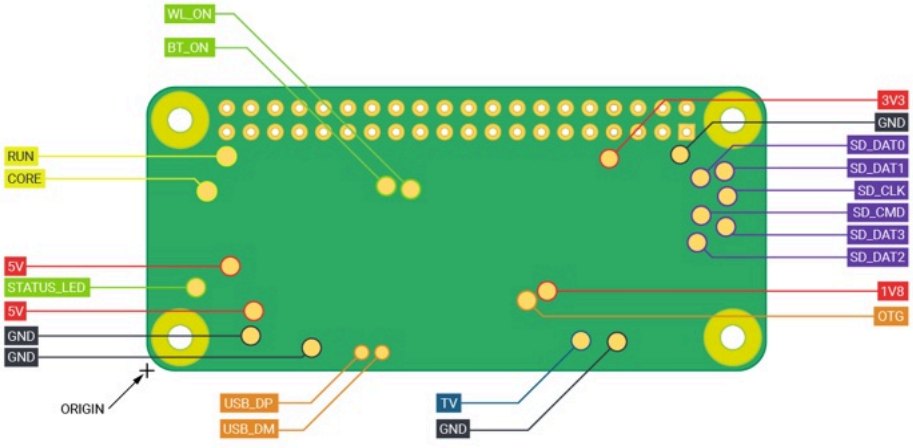


Figure C.3.1. Raspberry Pi Zero 2 W Pinout [37]

C.4. Adafruit 166 FSR Documentation

Table C.4.1 shows the key specifications for the Adafruit 166 force sensitive resistor (FSR) that is used on the fingertips of the electrical parallel gripper, while Figure C.4.1 shows the experimentally-obtained calibration curve with hysteresis included.

Table C.4.1. Adafruit 166 FSR Key Specifications [38]

INSPECTION ITEM	SPECIFICATION
FORCE SENSITIVITY RANGE	30g～1000g (0.3N～9.8N)
FORCE RESOLUTION	Continuous(Analog)
STAND-OFF RESISTANCE (UNLOADED)	>20MΩ
FORCE REPEATABILITY (SINGLE PART)	2%
FORCE REPEATABILITY (PART TO PART)	5%
RESPONSE TIME	<1ms
LONG TERM DRIFT	<10% log10(time)
HYSTERESIS	10%
LIFE CYCLE (WITHOUT FAILURE)	>10 Million
OPERATING TEMPERATURE	-30°C to +85°C
HIGH HUMIDITY STORAGE	+70°C, 85%RH x 96hrs
HIGH TEMPERATURE STORAGE	+85°C x 120hrs
LOW TEMPERATURE STORAGE	-30°C x 120hrs

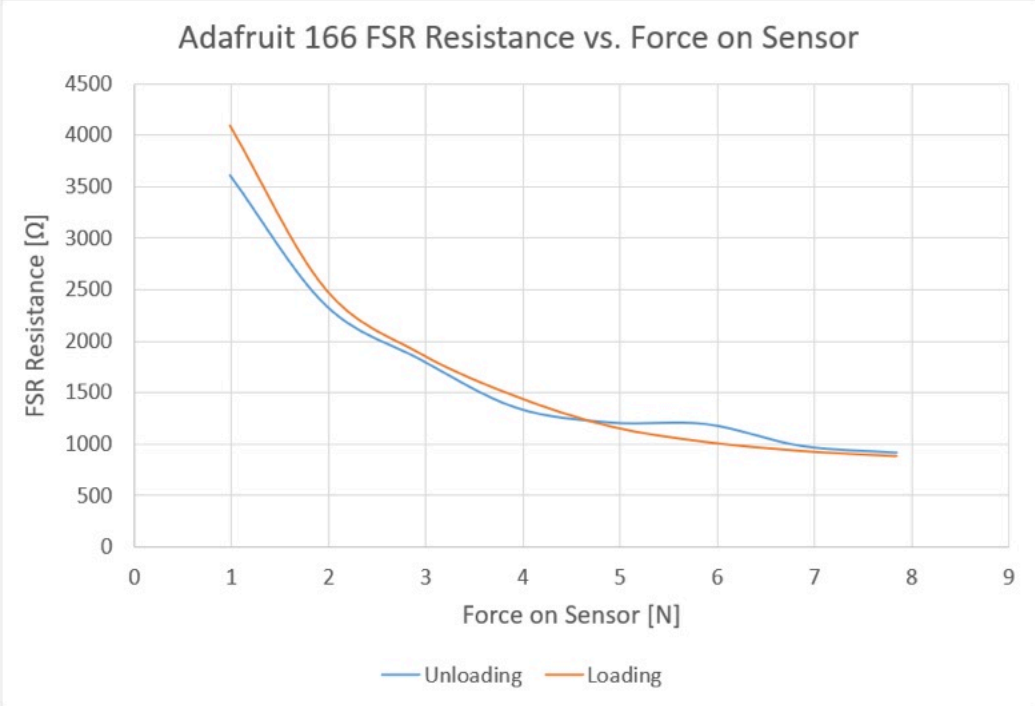


Figure C.4.1. Adafruit 166 FSR Calibration Curve

C.5. Suction Cup Design Drawing

Figure C.5.1 shows the isometric design drawings for the vacuum suction cup used as the contact interface of the pneumatic gripper toolhead.

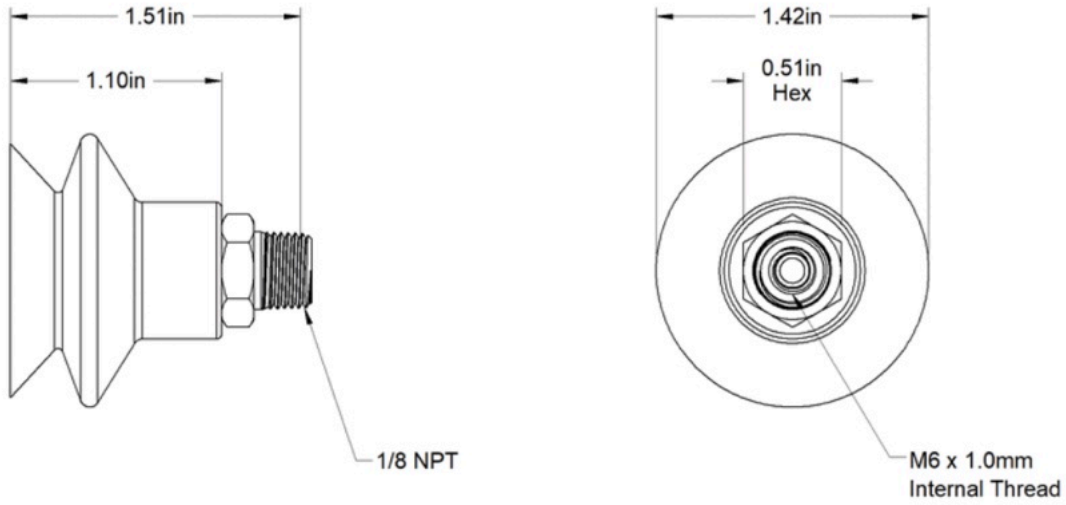


Figure C.5.1. Vacuum Suction Cup Design Drawing [39]

C.6. Adafruit APDS-9960 Documentation

Figure C.6.1 shows the electrical pinout for the Adafruit APDS-9960 sensor breakout board that is used as the distance sensor on the pneumatic suction gripper toolhead. Figure C.6.2, meanwhile, shows the experimental calibration curve obtained for this sensor, along with its corresponding first-order polynomial that is used for the practical distance conversion included in the gripper's Arduino code.

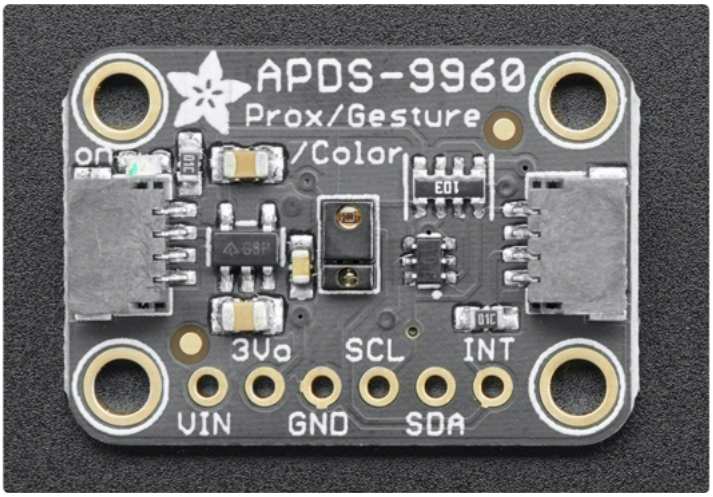


Figure C.6.1. Adafruit APDS-9960 Pinout [40]

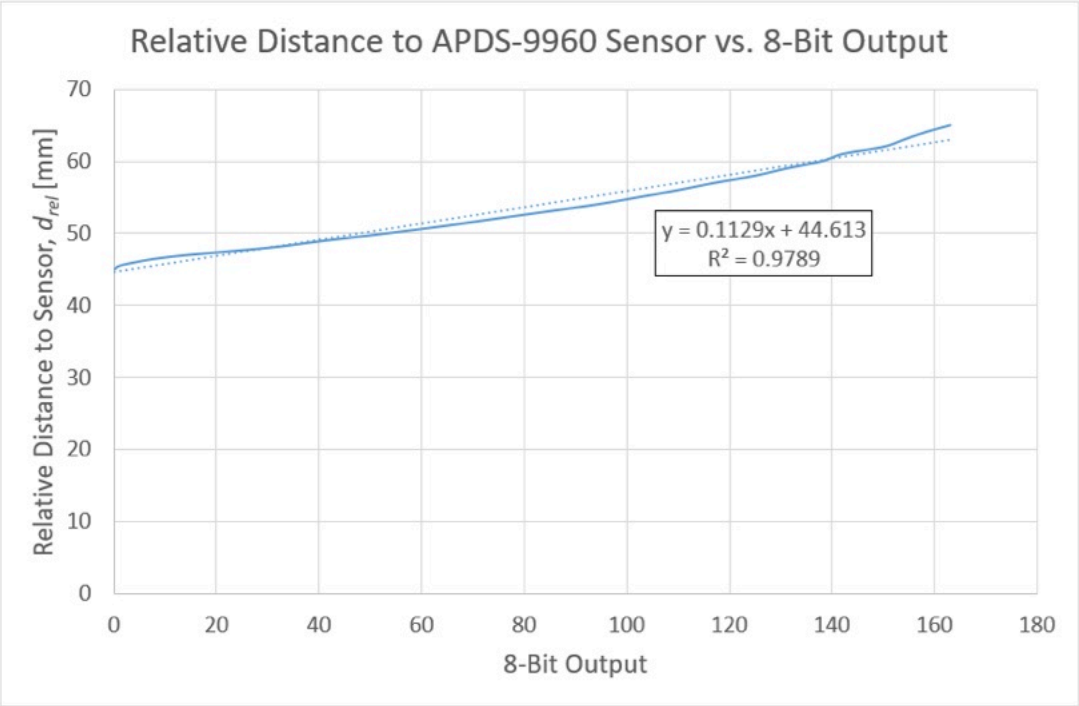


Figure C.6.2. Adafruit APDS-9960 Calibration Curve

C.7. Raspberry Pi 5 Pinout

Figure C.7.1 shows the pinout for the Raspberry Pi 5 microcontroller that is used as a part of the pump control unit for pneumatic suction gripper tool.

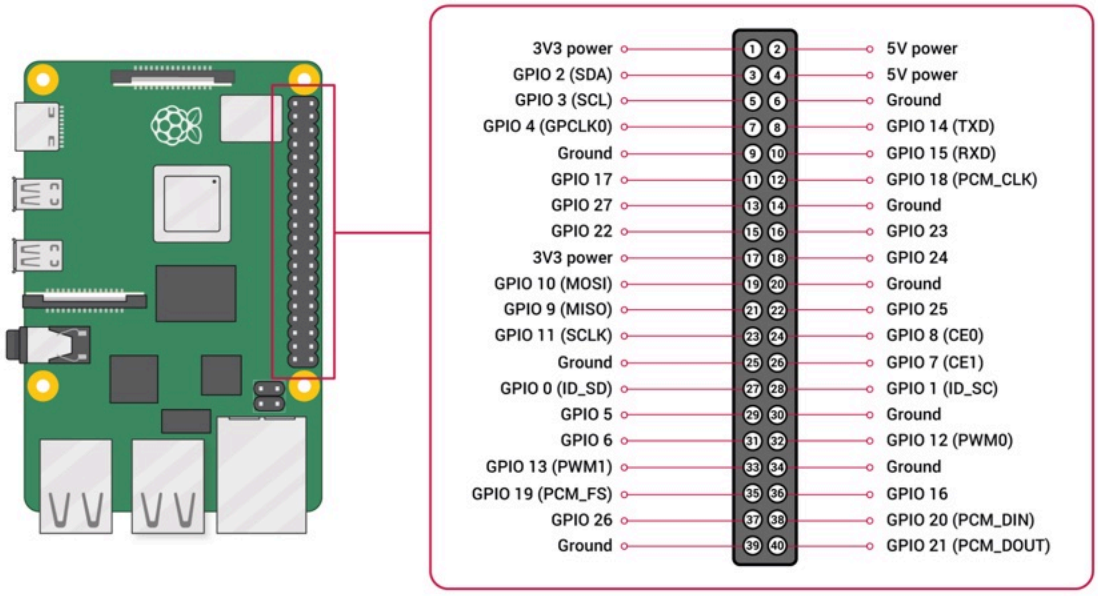


Figure C.7.1. Raspberry Pi 5 Pinout [37]

C.8. Raspberry Pi 5 DDL Relay Pinout

Figure C.8.1 shows the electrical pinout for the GeekPi-brand DDL relay shield that is mounted onto the Raspberry Pi 5 as the second part of the pneumatic suction pump's power control unit. The lower pin array corresponds directly to the GPIO pin array of the Raspberry Pi 5 onto which the shield is mounted, outlined in Appendix C.7 above.

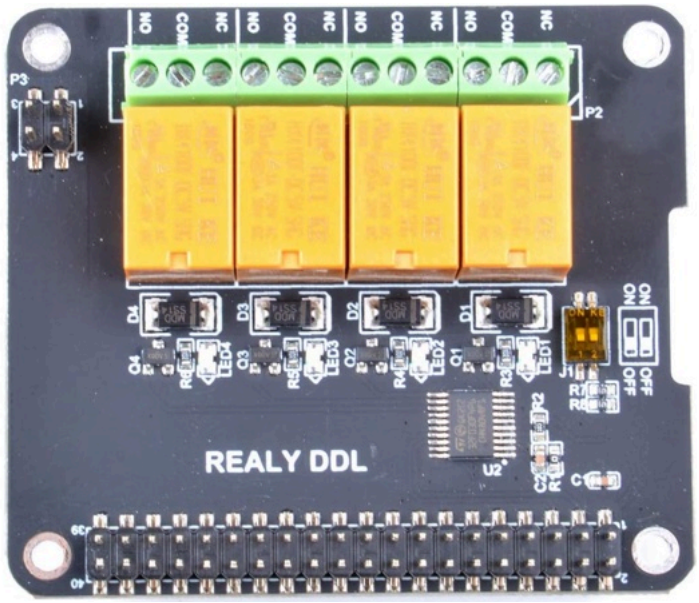


Figure C.8.1. GeekPi Relay Shield Pinout [41]

C.9. Loctite Air Line Regulator Documentation

Figure C.9.1 shows a summary of the key specifications for the Loctite air line regulator that forms the first stage of the adhesive applicator tool system. Figure C.9.2, meanwhile, is an isometric design drawing of the regulator unit with its key dimensions annotated.

Specifications:	
Port Size	1/8" NPT
Proof Pressure	1.5 MPa (217 PSI)
Max. Operating Pressure	1.0 Mpa (145 PSI)
Set Pressure Range	0.05 to 0.85 Mpa (7.25 to 123 PSI)
Ambient & Fluid Temperature	-5 to 60°C [no freezing] (23 to 140°F)
Filtration	5 micron
Drain Capacity	8 cm^3
Bowl Material	Polycarbonate
Construction	Relieving Style
Weight	.32 kg

Figure C.9.1. Loctite Air Line Regulator Key Specifications [42]

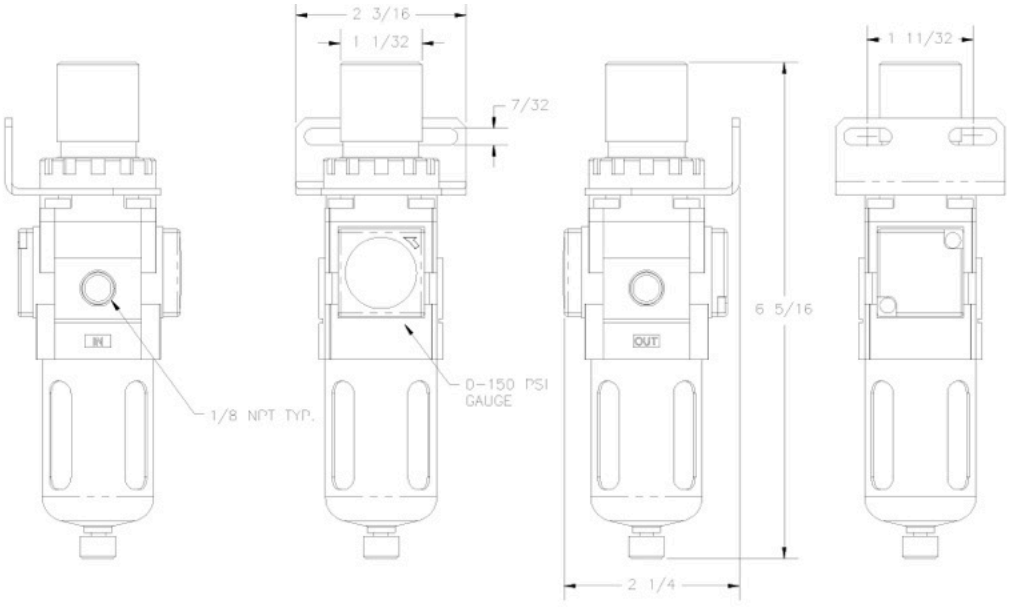


Figure C.9.2. Loctite Air Line Regulator Design Drawing [42]

C.10. Loctite RC50 Integrated Dispenser Documentation

Figure C.10.1 provides a summary of the key features and technical data pertaining to the Loctite RC50 integrated dispenser that is used as the second stage of the adhesive tool system.

Product Features

- Can be used to control two dispense valves at least.
- Two independent digital timing channels.
- Low level sensor alarm provides warning for adhesive bottle replacement.
- Interface connections for customer's PLC to indicate reservoir low level condition and cycle complete signal.
- Friendly HMI provided, easy-to-use big touch panel for entering values.
- Equipped with a 0 to 7 bar (0-100psi) digital pressure regulator (electronic proportional valve).
- Time range is adjustable from 0.01 to 99.99 seconds.
- Can be run in Continue or Time mode.
- Powerful I4.0 function, more convenient for user to interactive in real time via Wi-Fi, Bluetooth, or Ethernet.

Technical Data:

- Electrical:
  - Input voltage 100 -240 VAC, 50-60 Hz.
  - Internal voltage: 24 VDC
  - Power connection includes plug ends for North America, Japan, Continental Europe, UK, Australia and China.
- Dimensions:
  - Width: 348.84mm
  - Depth: 214mm
  - Height: 326.3mm
- Weight: 5 kg.
- Operation temperature: + 10 Deg. C to +40 Deg. C. (+50 Deg. F. to +104 Deg. F.)

Figure C.10.1. Loctite RC50 Integrated Dispenser Key Specifications [42]

C.11. Loctite Cyanoacrylate Adhesive Valve Documentation

The key specifications of the Loctite cyanoacrylate adhesive valve used in the final stage of the adhesive tool system are compiled into the bullet points of Figure C.11.1 below. The image below it, Figure C.11.2, is a design drawing of the valve which was crucial to sizing the applicator toolhead housing.

PRODUCT FEATURES

- Bubble free, no-drip dispensing
- Compact size and weight
- High resolution stroke adjustment
- Seal-less design
- Fail sage - air to open, spring closed
- Easy cleaning without valve demounting
- Field serviceable
- UHMW polyethylene valve chamber
- Easy integration with Loctite® reservoirs and controllers

Technical Data

- Dimensions: 3.1" length x 1" dia. (78.5mm x 26.9mm)
- Weight: 4.5 oz. (113g)
- Pneumatic Supply: 70 - 90 psi (4.8 - 6.2 bar)
- Max Fluid / Pressure: 70 psi (4.8 bar)

Figure C.11.1. Loctite Cyanoacrylate Adhesive Valve Key Specifications [42]

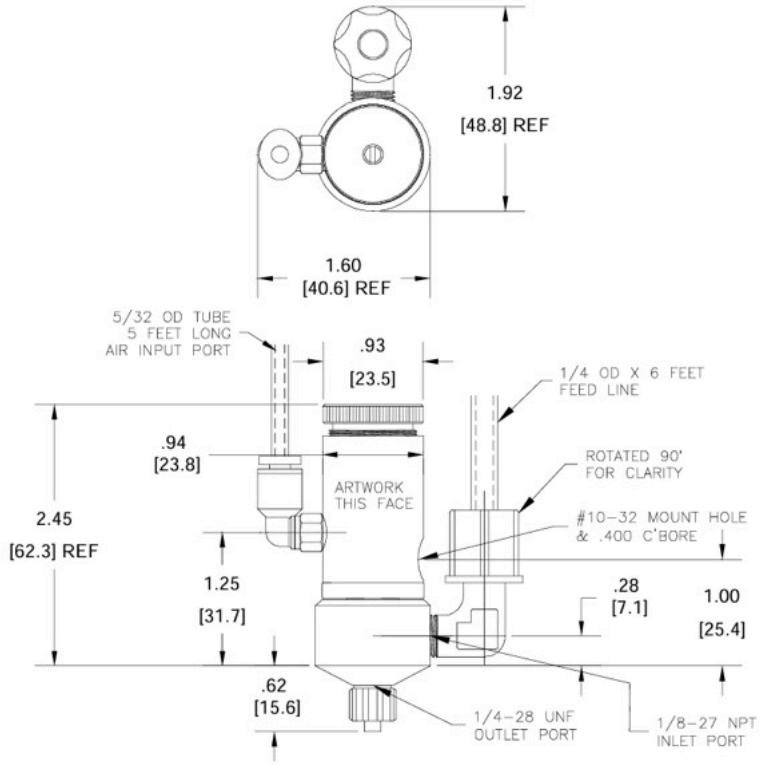


Figure C.11.2. Loctite Cyanoacrylate Adhesive Valve Design Drawing [42]

Appendix D: Bill of Materials

Table D.1 below is the comprehensive bill of materials (BoM) for the final MATRIX end effector portfolio, containing the components required to build one of each subsystem.

Subsystem	Part Description	Vendor	Part Number	Unit Cost	Qty.	Total Cost
Universal Mount / Tool Interlock	M5 Countersunk Screw, 14mm	McMaster-Carr	91420A324	$0.13	4	$0.51
	O-Ring, 10mm ID	McMaster-Carr	9262K104	$0.10	1	$0.10
	Pogo Connector (M+F), 5 Pins	Amazon	B0D44WD5C5	$5.99	1	$5.99
	19-Pin Stäubli Male Connector	Amazon	B0C1K21P96	$13.78	1	$13.78
	19-Pin Stäubli Female Connector	Elecbee	EB-400-3121-H	$9.86	1	$9.86
	PLA Filament, 110g	Amazon	B0DLJJFJHS	$1.70	1	$1.70
	18 AWG Wire, 5m	Amazon	B07T4SYVYG	$3.00	1	$3.00
	JST XH Connector, 5 Pins	DigiKey	1528-4875-ND	$0.95	1	$0.95
	5V DC Power Adapter	Amazon	B09W8X9VGK	$7.99	1	$7.99
	9V DC Power Adapter	Amazon	B087ZWS7CG	$8.99	1	$8.99
	6mm OD Quick Disconnect, M5 Base	DigiKey	2171-QSM-M5-6-ND	$2.65	2	$5.30
	6mm OD Pneumatic Tubing, 1m	DigiKey	2171-PUN-H-6X1-BL-500-DS-ND	$1.30	1	$1.30
	M5 Heat Set Insert, 9.6mm	McMaster-Carr	94459A436	$6.75	1	$6.75
	8mm OD Quick Disconnect, M10 Base	McMaster-Carr	5225K585	$10.07	1	$10.07
	8mm OD Pneumatic Tubing, 2ft	McMaster-Carr	3782N217	$1.42	1	$1.42
	Subsystem Total					**$77.71**
Tool Depot	PLA Filament, 160g	Amazon	B0DLJJFJHS	$2.48	1	$2.48
	20mm x 40mm T-Slot Extrusion, 5ft	McMaster-Carr	5537T111	$33.62	1	$33.62
	Corner Bracket for 20mm T-Slot	McMaster-Carr	5537T658	$9.30	2	$18.60
	M5 Csk. Screw, 18mm	McMaster-Carr	92010A793	$0.16	4	$0.65
	Subsystem Total					**$55.35**

Subsystem	Part Description	Vendor	Part Number	Unit Cost	Qty.	Total Cost
Electrical Parallel Gripper	Dynamixel XL430-W250-T Motor	Robotis	902-0135-000	$49.90	1	$49.90
	Formlabs Nylon 12 Powder, 150g	Formlabs	PD-FS-P12G-10-01	$14.99	1	$14.99
	PLA Filament, 50g	Amazon	B0DLJJFJHS	$0.77	1	$0.77
	M3 Hex Nut	McMaster-Carr	90591A250	$0.03	4	$0.11
	M3 Socket Head Screw, 10mm	McMaster-Carr	91292A113	$0.06	4	$0.24
	M2.5 Hex Nut	McMaster-Carr	90591A270	$0.02	12	$0.27
	M2.5 Socket Head Screw, 12mm	McMaster-Carr	91292A016	$6.54	4	$26.16
	M2.5 Socket Head Screw, 8mm	McMaster-Carr	91292A012	$6.54	8	$52.32
	M2.5 Socket Head Screw, 4mm	McMaster-Carr	91292A015	$7.43	4	$29.72
	M2 Socket Head Screw, 8mm	McMaster-Carr	91292A004	$15.76	4	$63.04
	Raspberry Pi Zero 2 W	PiShop	1799	$17.25	1	$17.25
	OpenRB-150 Motor Controller	Robotis	902-0183-000	$27.39	1	$27.39
	18 AWG Wire, 2m	Amazon	B07T4SYVYG	$1.20	1	$1.20
	M2.5 Countersunk Screw, 16mm	McMaster-Carr	91420A024	$0.06	4	$0.22
	Force Sensitive Resistor (0.3-10N)	Adafruit	166	$3.95	2	$7.90
	Neoprene Foam Pads, 1/8in Thick	McMaster	93375K428	$0.10	2	$0.20
	Subsystem Total					**$291.69**
Pneumatic Suction Gripper	12V Air Pump	Sparkfun	ROB-10398	$25.50	1	$25.50
	Raspberry Pi Zero 2 W	PiShop	1799	$17.25	1	$17.25
	Arduino Nano Every	Arduino	ABX00028	$13.70	1	$13.70
	PLA Filament, 68g	Amazon	B0DLJJFJHS	$1.05	1	$1.05
	APDS-9960 Distance Sensor	Adafruit	3595	$7.50	1	$7.50
	QWIIC to Dupont Cable (M), 4 Pins	Adafruit	4209	$0.95	1	$0.95

Subsystem	Part Description	Vendor	Part Number	Unit Cost	Qty.	Total Cost
Pneumatic Suction Gripper	Height-Adjusting Vacuum Cup, 1.32in OD	McMaster	5427A637	$13.38	1	$13.38
	M2.5 Hex Nut	McMaster-Carr	90591A270	$0.02	4	$0.09
	M2.5 Countersunk Screw, 16mm	McMaster-Carr	91420A024	$0.06	4	$0.22
	8mm OD Pneumatic Tubing, 2ft	DigiKey	4753-ZUSA-HT-3794-ND	$2.94	1	$2.94
	Raspberry Pi 5	Sparkfun	DEV-23550	$75.00	1	$75.00
	4-Channel Pi 5 Relay Shield	Amazon	B07Q2P9D7K	$15.99	1	$15.99
	Subsystem Total					**$173.57**
Adhesive Applicator	Loctite Air Line Regulator	Henkel	478603	$126.46	1	$126.46
	Loctite RC50 Integrated Dispenser	Henkel	2814024	$2,995.00	1	$2,995.00
	Loctite Cyanoacrylate Adhesive Valve	Henkel	318654	$937.35	1	$937.35
	Loctite 380 Black Max Adhesive, 1lb	Ellsworth	135424	$330.64	1	$330.64
	8mm OD Pneumatic Tubing, 25ft	McMaster-Carr	3782N217	$17.75	1	$17.75
	6mm Pneumatic Fitting, 1/8 NPT	McMaster	5350K129	$4.82	1	$4.82
	6mm Pneumatic Fitting, 1/4 NPT	McMaster	5350K123	$4.40	1	$4.40
	10/32 Socket Head Screw, 9/16in	McMaster	90128A316	$0.12	1	$0.12
	4mm OD Quick Disconnect, M5 Base	Digikey	2171-QSM-M5-4-ND	$2.55	1	$2.55
	8mm to 6mm Tubing Adapter	Digikey	2171-QS-8-6-ND	$3.88	1	$3.88
	PLA Filament, 64g	Amazon	B0DLJJFJHS	$0.99	1	$0.99
	Raspberry Pi Zero 2 W	PiShop	1799	$17.25	1	$17.25
	M2.5 Hex Nut	McMaster-Carr	90591A270	$0.02	4	$0.09
	M2.5 Countersunk Screw, 16mm	McMaster-Carr	91420A024	$0.06	4	$0.22
	Subsystem Total					**$4,441.53**
Total Cost - Taxes and Shipping NOT Included						**$5,039.85**

Appendix E: User Manual

E.1. Overview

The following guide contains the recommended operating procedures for the MATRIX end effector system, including the required equipment, safety disclaimers, installation procedure, and general operating workflow. The manual is written with the intended use environment being a robotic workcell running on ROS, similarly to the RSL workcell. The adhesive applicator system is not included here since its process engineering development is still in progress.

E.2. Required Equipment

- 6 DoF robot arm with control module, teach pendant, and computer running ROS
- Tool depot with mounting screws
- Universal mount section
- 4x M5 countersunk screws
- JST XH male-to-robot distal outlet male cable
- Robot base female connector with 9V and 5V adapters soldered on
- 8mm OD pneumatic tubing
- 6mm OD pneumatic tubing
- Quick disconnect for 8mm OD tubing (M10 threaded base)
- Quick disconnect for 6mm OD tubing (M5 threaded base)
- 2x tool interlock sections
- 8x M2.5 countersunk screws
- 8x M2.5 hex nuts
- Electrical parallel gripper
- Pneumatic suction gripper
- Pneumatic base unit

E.3. Safety Disclaimers

- Only perform installation or maintenance work on the robot arm when it is either fully turned off or the E-stop button is pressed.
- Maintain a radial distance of at least 4 ft from the robot arm while it is moving.
- Do not work on any electrical systems while they are under power.
- Turn off any pumps before connecting or disconnecting pneumatic tubing.

E.4. Installation Procedure

1. Attach the universal mount section to the distal end of the robot arm using the M5 countersunk screws.
    a. If the mount is not directly compatible with the arm, an adapter plate must be designed and 3D printed based on the arm's mounting point dimensions.

2. Install the M5 and M10 quick disconnects at the distal and base pneumatic ports of the arm, respectively.
3. Plug the JST and 6mm OD pneumatic tubing into their corresponding connectors on the universal mount and the robot arm's distal output panel.
4. Plug the custom passthrough cable with the adapters into the base of the arm.
5. Mount the pneumatic base unit on a surface near the robot arm's base.
    a. Mount the unit on the robot's cart for more flexibility, if available.
6. Connect the 8mm OD pneumatic tubing from the pump's suction outlet to the quick disconnect at the arm's base.
7. Plug in all four of the adapters needed for operation (9V and 5V for the electrical passthroughs, along with 5V and 12V for the pneumatic base unit).
8. Mount the tool depot such that all tool slots on it are reachable by the robot arm.
9. Place four M2.5 hex nuts in their respective cutouts in the electrical gripper's base plate, and use them with four M2.5 countersunk screws to fix the toolhead to a tool interlock plate. Solder any electrical connections between the two as needed.
10. Repeat Step 9 for the pneumatic gripper.
11. Place the assembled electrical and pneumatic gripper toolheads in their respective tool slots in the depot.

E.5. General Operation

1. Power on the robot using the ON switch located on its control module.
2. Power on the robot-side computer, spin up its ROS node, and connect it to the corresponding node on the workcell's central command computer.
3. Switch the teach pendant to manual mode and use the jog buttons on it to train the robot arm to the positions required to tool pick-up and drop-off operations. Use a software like RoboDK to log the waypoint positions.
4. Use the Python scripts in Appendices B.4 and B.5 as templates for the robot's tool pick-up and drop-off sequences, substituting the positions from Step 3 as relevant.
5. Verify both stages of the tool change process by running the scripts from Step 4 over the command PC-arm controller ROS node pair. Tune the waypoint positions as needed via the teach pendant.
6. Use the teach pendant to program any paths required for tool usage in the workspace, creating a ROS script like the one described in Step 4.
7. Merge the ROS scripts from Steps 4 and 6 and execute them to validate the full arm trajectory sequence for the process being developed.
8. Spin up the tools' ROS nodes (script in Appendix B.2 for the electrical gripper and B.3 for the pneumatic gripper) and add references to them in the main script from Step 7 anywhere tool actuations are required.
9. Run the full process loop using the combined pathing and tool usage script from Step 8.
10. Power off the robot and both computers when done using the system.